

GENERATORS OF STABILITY-PRESERVING SEMIGROUPS, SPECTRAL GAPS, AND CLASSICAL GROUND-STATE COMPUTATION

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ABSTRACT. We classify generators of complex stability-preserving semigroups on finite polynomial boxes and prove polynomial-time recognition from Gaussian-rational binary one-site and two-site inputs. Damping bounded below by $\mu > 0$ gives strict zero-freeness and a sharp, capacity-independent real-part spectral gap, extending the Hermitian 2μ gap of Bravyi, Gosset, Liu, and Wong. At bounded degree and local strength, deterministic classical algorithms approximate ground energies and relative amplitudes against stable product states for Suzuki–Fisher Hamiltonians with fixed positive longitudinal fields; field perturbation gives zero-field energy-value polynomial-time approximation schemes for Einstein–Podolsky–Rosen (EPR) and bipartite Quantum MaxCut on bounded-degree unweighted graphs. An all-temperature converse for full Gibbs tensors characterizes Hermitian qubit Hamiltonians, in a fixed basis, as precisely the edgewise phase-rotated Suzuki–Fisher family up to scalars. Together with strict positive-field Gibbs radii, this answers two questions of Wong, Bravyi, Gosset, and Liu. Weighted token-graph spectral concavity resolves the adjacency and signless-Laplacian conjectures of Apte, Parekh, and Sud. An exact certificate disproves the stronger proposed EPR ground-state radius.

2020 *Mathematics Subject Classification*. Primary 47D06, 30C15; Secondary 15A42, 81P45, 68Q25.

Key words and phrases. stable polynomial, generator classification, spectral gap, Lee–Yang theorem, quantum Hamiltonian, classical algorithm.

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1. INTRODUCTION

A polynomial with no zero in a prescribed domain imposes a global constraint on its coefficients. When the coefficients encode a matrix or a quantum state, that constraint can determine which interactions are possible and which spectral quantities can be computed. We study continuous linear evolutions that preserve all complex stable polynomials on a finite box of degrees, and use their infinitesimal structure to obtain spectral bounds and classical algorithms.

The classification of individual stability-preserving operators by their symbols gives a starting point for this question [7, 10, 8]. A semigroup imposes an additional condition: every operator e^{tA} , including those at arbitrarily small positive times, must be a preserver. This infinitesimal requirement is especially restrictive at collisions of roots. It is also useful. A generator description can identify all admissible interactions, distinguish physical positivity from complex stability, and supply a spectral structure that is not apparent from the entries of the operator.

The one-variable theory begins with the multiplier-sequence classification of Pólya and Schur [26]. Constant-coefficient differential operators and heat-type deformations preserving real zeros form another classical part of the subject; see Craven and Csordas [17]. Our question concerns arbitrary endomorphisms of finite multivariable boxes, whose variable coefficients and boundary degenerations must be controlled simultaneously.

Infinitesimal stability preservers have also been studied directly. Purbhoo constructs a family on multiaffine polynomials indexed by matrices with real diagonal and nonnegative off-diagonal entries [28, Proposition 4.5]. His result gives sufficient generators. The structural theorem below supplies necessary and sufficient conditions for arbitrary endomorphisms on every finite coordinate box.

This paper develops these consequences in the following chain:

$$\begin{aligned} &\text{root-collision constraints} \implies \text{generator structure} \\ &\implies \text{strict complex spectral control} \implies \text{classical computation.} \end{aligned}$$

The finite-box classification is the structural result. Strict damping turns it into a spectral and analytic tool, and a compact local input makes that tool algorithmic.

The all-temperature Hamiltonian converse is a structural application; comparisons between homogeneous sectors and the distinction between static preservers and generated flows give further consequences of the same classification.

1.1. The finite-dimensional structure theorem. Let V_κ be the space of polynomials with coordinatewise degree bounds $\kappa = (\kappa_1, \dots, \kappa_n)$. Every linear operator on this space has a canonical normally ordered differential representation. The first main theorem describes which such operators generate stability-preserving semigroups, without assuming positivity, locality, or a differential order in advance.

In real half-plane coordinates, the answer is second order. If

$$A = \sum_{\alpha \leq \kappa} P_\alpha(z) \partial^\alpha$$

is the canonical expression, preservation forces $P_\alpha = 0$ for $|\alpha| > 2$. Each diagonal second-order coefficient is a polynomial of its own variable and is nonpositive on the real line. Each mixed coefficient is a polynomial of the corresponding pair of variables and is nonpositive on the real plane. Together with the condition that A acts on V_κ , these conditions are also sufficient. This is Theorem 3.1. The finite-box constraint is substantial: it couples the highest terms of the lower-order coefficients to the second-order part.

For complex generators there is one further component. Every generator is the sum of such a real generator and

$$i \left(cI + \sum_i \left(q_i(z_i) \partial_i - \frac{\kappa_i}{2} q'_i(z_i) \right) \right), \quad c \in \mathbb{R}, \quad q_i \geq 0 \text{ on } \mathbb{R}, \quad (1)$$

where each q_i is a real polynomial of degree at most two. Conversely every such sum generates a complex stability-preserving semigroup (Theorem 3.2). The operators in (1) are the infinitesimal one-variable Möbius motions pointing into the half-plane. Thus arbitrary interactions in the imaginary part are excluded.

The proof has two different ingredients. Root-cluster asymptotics force the order bound and the signs. The local multiple-root mechanism has an antecedent in the variational analysis of Burke and Overton [16, Theorem 1.2 and Lemma 1.4]. A finite-dimensional algebraic decomposition then reduces sufficiency to explicit binary primitives. The required certificates use Gram matrices of orders at most four, independently of the numerical capacities. Polarization gives an exact binary-slot realization, with the binomial factors retained. These are generator statements; they are not obtained by differentiating a smooth boundary of the stable set. That boundary is singular, and its higher-multiplicity strata matter.

For a sum of one-site and two-site matrices on binary variables, the classification is effectively checkable. Corollary 3.13 decides preservation of the entire semigroup in polynomial bit time from this local description. It checks the sum of the local terms, allowing cancellations between them, and never constructs the full exponential-size matrix.

1.2. Strict damping and a complex spectral gap. Pass to the unit disk by a Cayley transform. Suppose A_0 generates a semigroup preserving all complex disk-stable polynomials in V_κ , and set

$$A = A_0 - 2 \sum_i \mu_i E_i, \quad E_i = z_i \partial_i, \quad \mu_i > 0. \quad (2)$$

The damping has a simple interpretation: it contracts each polynomial variable towards the center of the disk. The conclusion is stronger than preservation of the open disk. For every $t > 0$ the actual binomial-weighted kernel of e^{tA} is nonzero on the closed unit polydisk, and it is stable in a polydisk of some radius $r_t > 1$. For any fixed positive time, a common strict radius works at all later times (Theorem 4.3).

Consequently A has an algebraically simple principal eigenvalue a_* and

$$\Re a_* - \Re a \geq 2 \min_i \mu_i \quad \text{for every other eigenvalue } a. \quad (3)$$

The constant is sharp, and the bound is independent of the capacities and number of variables (Theorem 4.8). Neither self-adjointness nor coefficient positivity is assumed. The principal right and left objects are strictly stable.

The proof uses the finite-dimensional complex-cone theory of Rugh [33], together with a damping argument and contraction of stable kernels. The sharp radius-to-spectrum mechanism has a Hermitian Lee–Yang antecedent in [14]. Our extension concerns arbitrary complex finite-box stable generators and their actual kernels. The corresponding dynamical estimate concerns the oscillation of a quotient h/p over the polydisk. Its contraction, including time-dependent flows, is developed in Section 4. This strengthens the spectral conclusion; the algorithms below use the spectral gap and zero-free kernel, rather than this oscillation estimate. The projective-metric and oscillation constructions themselves have antecedents in [19, 21].

1.3. Comparing homogeneous sectors. Coefficient positivity gives another spectral consequence. Suppose a finite-box generator preserves real stability, has nonnegative off-diagonal entries, and preserves every homogeneous degree. The spectral bounds of its degree blocks form a concave sequence (Theorem 4.16). This also holds for nonself-adjoint, reducible blocks. The proof evolves a single polynomial with positive coefficients, applies Newton’s inequalities to its diagonal restriction, and takes exponential growth rates. The underlying growth-exponent principle is classical [11].

For a nonnegatively weighted graph G , let A_k, D_k be the adjacency and degree matrices of its k -token graph. The generator classification applies to the whole family

$$B_k(a) = A_k + aD_k, \quad -1 \leq a \leq 1.$$

Its largest eigenvalues satisfy

$$2\lambda_{\max}(B_k(a)) \geq \lambda_{\max}(B_{k-1}(a)) + \lambda_{\max}(B_{k+1}(a)),$$

and are symmetric under $k \leftrightarrow n - k$. They are therefore nondecreasing up to half filling. This proves the adjacency and signless-Laplacian monotonicity assertions of [2, Conjectures 5–6], with a stronger weighted concavity statement and a sharp universal interval for a (Theorem 4.17). The $a = 0$ semigroup is already contained in Purbhoo’s construction [28]; the new conclusion compares its particle-number sectors. The corresponding EPR interpretation uses the established token-graph correspondence of [2].

1.4. A classical algorithm beyond self-adjoint operators. Here is the algorithmic consequence in a form that specifies its input model. On n binary variables let A_0 be given as a sum of one-site and two-site matrices with Gaussian-rational entries. Suppose the interaction graph has degree at most d , each local matrix has

norm at most J , and e^{tA_0} preserves all complex disk-stable multiaffine polynomials. The preservation hypothesis concerns the whole generator; individual displayed summands need not be preservers. It is decidable in polynomial bit time by Corollary 3.13. For fixed d, J , and a fixed rational $\mu > 0$, the principal eigenvalue of

$$A_0 - 2\mu \sum_i E_i$$

can be approximated to absolute complex error δ by a deterministic classical algorithm in time

$$\text{poly}_{d,J,\mu}(n, \delta^{-1}, \text{input bit length}).$$

This is Theorem 5.1. The matrix has exponentially many entries, but the input is its sparse local description. The polynomial degree may depend on d, J, μ .

The same local input also gives relative access to the principal eigenvectors. Normalize the right and transpose-left vectors v, w to have vacuum coefficient one. For any tensor product $B = \bigotimes_i B_i$ of Gaussian-rational 2-by-2 matrices with $(B_i)_{00} = 1$ and disk-stable kernels, Theorem 5.11 computes

$$w^\top B v \quad \text{and} \quad \text{Tr}(B\Pi) = \frac{w^\top B v}{w^\top v}, \quad \Pi = \frac{vw^\top}{w^\top v},$$

to relative complex error ϵ in $\text{poly}_{d,J,\mu}(n, \epsilon^{-1}, L_{\text{in}})$ time. Both quantities are nonzero. This is scalar query access; no ordinary-norm conditioning bound for Π is required.

The algorithm uses two distinct analytic regions. A local expansion at a product vacuum has a size-independent convergence radius. The perturbation-coefficient algorithm of Bravyi, DiVincenzo, and Loss [12] supplies these coefficients. A complexification argument extends the coefficient bounds from Hermitian perturbations to arbitrary complex ones. The stability gap then continues the same principal branch through a right half-plane. An explicit disk map joining these regions turns logarithmically many local coefficients into a value estimate at the desired fixed damping. We give the branch-identification argument, the coefficient construction, and the algorithm in Section 5; the detailed rational bit bounds are collected in Appendix A.

For the state queries, the extra step is to obtain a nonvanishing scalar test on that same type of domain. The creation-coefficient bounds in [12] give a uniform vacuum disk for both eigenvectors. The half-plane continuation and strict disk contraction then show that $w^\top B v$ never vanishes. Its logarithm has a one-sided real-part bound of order n throughout the domain (Lemma 5.13). Connected coefficients of this logarithm can be computed from subsystems of logarithmic size, without requiring the induced subsystems to satisfy the global preservation promise.

Connected-cluster computation combined with analytic continuation is an established strategy [25, 42, 38]. Its point is that complex stability provides a global analytic domain for a class of sparse operators that need not be self-adjoint, positive, or Markovian. The computational statement uses that domain to go beyond the perturbative neighborhood where the coefficient expansion is initially justified.

1.5. Quantum consequences and their scope. For qubits, consider

$$H = - \sum_{ij} \left(J_{ij}^z Z_i Z_j + \sum_{a,b \in \{x,y\}} J_{ij}^{ab} \sigma_i^a \sigma_j^b \right) - \sum_i (h_i^x X_i + h_i^y Y_i + h_i^z Z_i), \quad (4)$$

$$J_{ij}^z \geq \|J_{ij}^\perp\|_{\text{op}}, \quad h_i^z \geq 0,$$

with real coefficients and $J_{ij}^\perp = (J_{ij}^{ab})_{a,b \in \{x,y\}}$. This is the phase-rotated Suzuki–Fisher family. Its Lee–Yang sufficiency, including the full transverse operator-norm condition, is classical: see [35, equation (2.47)] and [20, Theorem 2]. Asano’s contemporaneous account already uses a full matrix generating function and its composition law for the spin-one-half XXZ subfamily [4, equation (1.15), Theorem 1]. The tensor language of [41] is a modern formulation of this tradition. The generator theorem gives the converse: up to an additive scalar, these are precisely the finite-qubit Hamiltonians whose full Gibbs tensor is Lee–Yang at every temperature (Theorem 6.2). This answers negatively the question in [41, Section 6, “Beyond Suzuki–Fisher Hamiltonians”]: in the fixed computational basis, no genuine interaction supported on three or more qubits is compatible with the all-temperature full-tensor property, and the edgewise phase-rotated Suzuki–Fisher family is exhaustive. Ruelle’s all-temperature converse [32, Theorem 9] already forces pair interactions for classical coefficientwise Boltzmann Lee–Yang polynomials. The object here is the full matrix tensor of an arbitrary Hermitian Hamiltonian, including noncommuting interactions.

For rational Pauli coefficients, bounded degree and bounded local strength, and longitudinal fields bounded below by a fixed $\mu > 0$, Theorem 5.7 gives deterministic classical ground-energy approximation to arbitrary absolute accuracy. Theorem 5.16 additionally computes nonzero relative amplitudes against product states $(|0\rangle + s_i|1\rangle)/\sqrt{1+|s_i|^2}$, $s_i \in \mathbb{Q}(i)$ and $|s_i| \leq 1$, and relative probabilities of the corresponding product events. It gives a classical randomized sampler for adaptive equatorial measurements, with any prescribed total variation error. Each site is measured at most once; the strategy’s own running time and the bit lengths of its phases are included explicitly. At nonnegative fields, adding a fixed field of size proportional to ε gives an additive εn energy-value approximation scheme (Theorem 5.9). This covers bounded-degree EPR models on arbitrary graphs and, after a local unitary transformation, bounded-degree bipartite Quantum MaxCut. For both unweighted classes, the estimate gives a deterministic $(1 - \varepsilon)$ approximation to the optimum value, with polynomial running time for every fixed $\varepsilon > 0$.

Earlier classical algorithms cover quantum-ferromagnet subclasses [13, 22]; the broader classical computation question appears in [14], and related interpolation ideas in [39, 40]. Relative amplitudes and marginals were treated in [41, Section 4] using a fixed spatial zero-free margin, and thermal product measurements in [42, Theorem 15]; here a reciprocal-field analytic domain gives polynomial-time queries from the local Hamiltonian at any fixed positive field. For EPR and bipartite Quantum MaxCut on unweighted graphs, the zero-field conclusion concerns fixed-relative-error energy values, whereas [23] asks for inverse-polynomial absolute accuracy and [1] constructs approximating states.

Positive damping also proves strict zero-freeness of the actual Gibbs kernel and of its unique ground-state polynomial. Theorem 6.6 answers affirmatively the question in [41, Section 5.3]: every Suzuki–Fisher Hamiltonian with strictly positive

longitudinal fields has a full Gibbs tensor of some radius $r_\beta > 1$ at each inverse temperature $\beta > 0$. The stronger proposed radius $s^{-1/2}$ for a deformed EPR ground state is false. Appendix C gives an unweighted six-vertex tree at $s = 1/2$, together with an exact rational certificate of a zero strictly inside that radius. It leaves the qualitative radius-greater-than-one conclusion intact.

The converse also extends to arbitrary finite local spins (Section 7). The resulting Hamiltonian cone permits longitudinally dominated pair interactions and the corresponding on-site quadratic terms, with the coherent binomial normalization made explicit. This is an equilibrium classification, separate from classification of completely positive quantum channels.

1.6. Static preservers and the range of the theory. The generator cone does not exhaust the connected semigroup of static preservers. On the multiaffine space, every invertible complex disk-stability preserver can be joined to the identity by a path of preservers. Nevertheless, a nontrivial coordinate permutation is outside the closure of finite products of admissible one-parameter flows (Theorem B.1). This identifies a precise limitation of passing from an infinitesimal classification to static operators.

The coefficient-positive preservation problem and physical quantum Markov dynamics have their own generator cones and analytic hypotheses. These topics are outside the scope of the present paper, which is confined to finite boxes and the spectral and computational consequences of preserving the full complex stable class.

1.7. Organization. Section 2 fixes the normalizations. Section 3 proves the generator classification and its local-input recognition corollary. Section 4 develops strictification, the spectral gap, dynamical contraction, and sector concavity. Section 5 proves the principal-eigenvalue, state-query, and measurement-sampling algorithms, including the zero-field value scheme. Sections 6 and 7 classify all-temperature Hamiltonians for qubits and higher spins. The appendices give the exact-arithmetic details, the obstruction to flow generation, and the certified radius counterexample.

2. CONVENTIONS AND ZERO-FREE KERNELS

2.1. Regions, degrees, and preservation. Write $\mathbb{H} = \{z \in \mathbb{C} : \Im z > 0\}$ and $\mathbb{D}_r = \{z \in \mathbb{C} : |z| < r\}$, with $\mathbb{D} = \mathbb{D}_1$.

Definition 2.1 (Stability and preservation). A nonzero polynomial is Ω -stable if it has no zero when all its variables lie in Ω . For a product of different regions the corresponding meaning is used. A polynomial is *real stable* if its coefficients are real and it is $\mathbb{H}$ -stable. In one variable this is equivalent to real-rootedness. A *positive stable polynomial* has nonnegative real coefficients and is real stable. Positivity throughout refers to coefficients, unless an operator on a Hilbert space is explicitly said to be positive semidefinite. An operator *preserves* one of these classes when it sends every member to another member or to zero.

Let $\mathbb{N} = \{0, 1, 2, \dots\}$. For $\kappa \in \mathbb{N}^n$ put

$$V_\kappa = \{f \in \mathbb{C}[z_1, \dots, z_n] : \deg_{z_i} f \leq \kappa_i\}, \quad \binom{\kappa}{\alpha} = \prod_i \binom{\kappa_i}{\alpha_i}.$$

Variables of capacity zero can be deleted. A degree bound is not an assertion that the polynomial has that degree. All finite-box statements include degree drops and

constants. For a real operator, its action on complex polynomials always means its complex-linear extension. Every operator on V_κ has a unique canonical expression

$$A = \sum_{\alpha \leq \kappa} P_\alpha(z) \partial^\alpha, \quad (5)$$

as an operator on V_κ . Uniqueness follows by applying the expression successively to monomials, starting with 1: the equation for z^α determines $\alpha!P_\alpha$ after all lower terms are known. The phrase *differential order* refers to this expression.

A finite-dimensional generator A preserves a class when e^{tA} preserves it for all $t \geq 0$. All generator and spectral statements here act on finite-dimensional polynomial boxes. Positive dynamics at infinite capacities and their analytic generation are outside the scope of this paper.

Definition 2.2 (Complete stability preservation). An operator is *completely stable* if its tensor product with the identity in arbitrarily many extra polynomial variables preserves the same complex stable class, for every finite choice of the extra degree bounds.

This is a statement about polynomial stability, distinct from complete positivity of a quantum channel. Positive stability with nonnegative coefficients and complete complex stability must also be distinguished.

2.2. Polarization and the disk kernel. The normalized polarization of z_i^a into κ_i binary slots is

$$z_i^a \mapsto \binom{\kappa_i}{a}^{-1} e_a(z_{i1}, \dots, z_{i\kappa_i}).$$

It is symmetric in the slots and specializes to z_i^a on the diagonal. Polarization preserves stability in the convex circular regions used here, including open disks and half-planes; diagonalization preserves it directly. We use the Grace–Walsh–Szegő coincidence theorem in this form; see [7, 8, 9]. The binomial normalization is part of the definition, not an optional choice of basis.

For $T : V_\kappa \rightarrow V_\lambda$ define its disk kernel by

$$K_T(z, w) = T_z \prod_i (1 + z_i w_i)^{\kappa_i} = \sum_{\alpha \leq \kappa} \binom{\kappa}{\alpha} (Tz^\alpha)(z) w^\alpha. \quad (6)$$

The notation z_i in the first expression denotes the input variables on which T acts; its result is a polynomial in the output variables. The kernel has output degree bound λ and input degree bound κ . In particular $K_{\text{Id}}(z, w) = \prod_i (1 + z_i w_i)^{\kappa_i}$.

For two polynomials of degree at most κ in paired variables define their bilinear contraction by

$$[f(u), g(v)]_\kappa = \sum_{\alpha \leq \kappa} \binom{\kappa}{\alpha}^{-1} f_\alpha g_\alpha, \quad f(u) = \sum_\alpha f_\alpha u^\alpha, \quad g(v) = \sum_\alpha g_\alpha v^\alpha. \quad (7)$$

There is no complex conjugation in this formula. Matrix composition is exactly contraction of the corresponding kernels:

$$K_{ST}(z, w) = [K_S(z, u), K_T(v, w)]_\kappa,$$

where κ is the intermediate capacity.

Lemma 2.3 (Disk contraction). *Suppose $f(z, u)$ and $g(v, w)$ are stable in products of disks and $\deg_{u_i} f, \deg_{v_i} g \leq \kappa_i$. If their paired radii r_i, s_i satisfy $r_i s_i \geq 1$, then their contraction (7) is stable in the remaining disks or is identically zero. If all $r_i s_i > 1$, contraction at any point in the remaining disks is nonzero.*

Proof. First consider a multiaffine polynomial $F(u, v) = a + bu + cv + duv$ that is nonzero on $\mathbb{D}_r \times \mathbb{D}_s$. Then $a \neq 0$ and $F(rt, st)$ is nonzero for $|t| < 1$. If $d \neq 0$, the product of its two roots has modulus at least one, so $|d|rs \leq |a|$; if $d = 0$, this inequality is immediate. Thus $a + d \neq 0$ when $rs > 1$. The same argument applies after fixing any uncontracted variables in their respective disks. Consequently the operation $a + bu + cv + duv \mapsto a + d$ preserves stability under strict separation of the paired radii. It may be iterated on a general multiaffine polynomial, without a factorization assumption.

Polarize the paired variables of f and g into κ_i slots for coordinate i , and apply this operation to each pair of corresponding slots in their product. For a fixed multi-index α , the matching slot subsets have cardinality $\binom{\kappa}{\alpha}$, while the two normalized polarizations contribute the factor $\binom{\kappa}{\alpha}^{-2}$. The result is therefore exactly (7). This proves strict nonvanishing, including degree deficiencies. It is the weighted form of circular-domain contraction; see [4, 7, 8]. For equality of paired radii, replace every paired variable in each input polynomial by $(1 - \varepsilon)$ times that variable, apply the strict statement, and let $\varepsilon \downarrow 0$. Hurwitz's theorem gives the stable-or-zero conclusion. The argument is valid after fixing any values of the unpaired variables in their respective disks. $\square$

The finite-box symbol theorem [7, 8] identifies stability preservers of rank at least two by the appropriate stable kernel; the low-rank alternatives are retained whenever they arise. In particular an invertible disk-stability preserver has a stable kernel (6). We do not infer a kernel criterion for arbitrary positive-coefficient preservers from this theorem.

The Cayley isomorphism is

$$(\mathcal{C}_\kappa f)(x) = \prod_i (x_i + i)^{\kappa_i} f\left(\frac{x_1 - i}{x_1 + i}, \dots, \frac{x_n - i}{x_n + i}\right). \quad (8)$$

It is a linear isomorphism between the degree boxes, and f is $\mathbb{D}$ -stable if and only if $\mathcal{C}_\kappa f$ is $\mathbb{H}$ -stable. Thus the generator classifications in half-plane coordinates and the spectral results in disk coordinates concern conjugate finite-dimensional semigroups.

2.3. Coherent coordinates and physical tensors. Equip V_κ with the inner product for which $\binom{\kappa}{\alpha}^{1/2} z^\alpha$ is an orthonormal basis. Equivalently, identify the standard basis $|\alpha\rangle$ of $\bigotimes_i \mathbb{C}^{\kappa_i+1}$ with $\binom{\kappa}{\alpha}^{1/2} z^\alpha$. The polynomial of a vector ψ is

$$f_\psi(z) = \sum_{\alpha \leq \kappa} \binom{\kappa}{\alpha}^{1/2} \psi_\alpha z^\alpha.$$

An operator M on this Hilbert space has coherent tensor

$$F_M(z, w) = \sum_{\alpha, \beta \leq \kappa} \binom{\kappa}{\alpha}^{1/2} \binom{\kappa}{\beta}^{1/2} M_{\alpha\beta} z^\alpha w^\beta. \quad (9)$$

This is precisely (6) for its action in coherent coordinates. Bra variables are independent complex variables; they are not conjugated when testing zero-freeness. For a positive semidefinite density matrix ρ , the assertion that F_ρ is $\mathbb{D}$ -stable is called its *Lee–Yang property*.

For a linear map Φ between matrix algebras the full coherent Choi tensor is defined by replacing $M_{\alpha\beta}$ in (9) with the coefficients $\Phi(|\gamma\rangle\langle\delta|)_{\alpha\beta}$ and adding input monomials $u^\gamma v^\delta$ with the corresponding square-root binomial weights. Its stability is called *full-Choi Lee–Yang preservation*. For qubits all capacities are one, so these weights are one. They cannot be omitted for higher spin. Ordinary physical preservation tests only positive semidefinite Lee–Yang inputs; it is a weaker condition for isolated maps.

We use the Pauli matrices

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Thus $E_i = z_i \partial_i$ represents $|1\rangle\langle 1|_i$ and $Z_i = I - 2E_i$ on a qubit. For general capacity $E_i = z_i \partial_i$ has eigenvalues $0, \dots, \kappa_i$. A *principal eigenvalue* below means an algebraically simple eigenvalue whose real part is strictly greater than the real parts of all other eigenvalues; for a discrete operator it means the corresponding strict modulus dominance.

3. GENERATORS ON A FINITE POLYNOMIAL BOX

We first classify semigroups acting on the whole stable class, with no assumption on the signs of coefficients of the polynomials or of the operators. The distinction from preservation of the coefficient-positive stable class is essential: the two problems have different generator cones.

Let $\kappa = (\kappa_1, \dots, \kappa_d)$ have finite positive integer coordinates, and write

$$V_\kappa^{\mathbb{F}} = \mathbb{F}[z_1, \dots, z_d]_{\deg_{z_i} \leq \kappa_i}, \quad \mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}.$$

Coordinates of capacity zero are omitted. If no coordinate remains, the space consists of constants and every scalar generator has the requisite preservation property. Stability and complete preservation have the meanings of Definitions 2.1 and 2.2; in this section the domain of stability is $\mathbb{H}^d$. A real operator acts on $V_\kappa^{\mathbb{C}}$ by complexification.

For a polynomial q in one variable of degree at most two, define

$$J_{i, \kappa_i}(q) = q(z_i) \partial_i - \frac{\kappa_i}{2} q'(z_i). \quad (10)$$

These are endomorphisms of the box. The three choices $q = 1, z_i, z_i^2$ give

$$J_i^- = \partial_i, \quad J_i^0 = z_i \partial_i - \frac{\kappa_i}{2}, \quad J_i^+ = z_i^2 \partial_i - \kappa_i z_i.$$

Their span is the sitewise $\mathfrak{sl}_2$ algebra.

Every endomorphism has a unique canonical normal representation

$$A = \sum_{\alpha \leq \kappa} P_\alpha(z) \partial^\alpha. \quad (11)$$

Here the coefficient polynomials are not assumed to belong to the original box. The order of A means the largest $|\alpha|$ for which its canonical coefficient P_α is nonzero. The uniqueness of this representation, proved below, is important at small capacities: in particular ∂_i^2 is absent when $\kappa_i = 1$.

Theorem 3.1 (Real generators). *Let $A \in \text{End}_{\mathbb{R}}(V_{\kappa}^{\mathbb{R}})$. The following conditions are equivalent.*

- (1) *For every $t \geq 0$, e^{tA} preserves real stability.*
- (2) *For every $t \geq 0$, its complexification preserves complex stability, also after adjoining arbitrary inert polynomial variables.*
- (3) *The canonical representation of A has order at most two, and its second-order coefficients satisfy*

$$P_{2e_i}(s) \leq 0 \quad \text{for all } s \in \mathbb{R}, \quad \kappa_i \geq 2, \quad (12)$$

$$P_{e_i+e_j}(s, t) \leq 0 \quad \text{for all } (s, t) \in \mathbb{R}^2, \quad i < j. \quad (13)$$

In the third condition, box preservation forces P_{2e_i} to depend only on z_i , with degree at most four, and $P_{e_i+e_j}$ to depend only on z_i, z_j , with bidegree at most $(2, 2)$. There are no further inequalities on the zeroth- and first-order coefficients beyond the identities already forced by $A(V_{\kappa}^{\mathbb{R}}) \subseteq V_{\kappa}^{\mathbb{R}}$.

The coefficient $P_{e_i+e_j}$ in this statement is the full coefficient of $\partial_i \partial_j$, with $i < j$. Thus a convention writing the second-order part as $\sum_{i,j} Q_{ij} \partial_i \partial_j$, with Q symmetric, has $P_{e_i+e_j} = 2Q_{ij}$. We use the full-coefficient convention throughout this section.

Theorem 3.2 (Complex generators). *Let $A \in \text{End}_{\mathbb{C}}(V_{\kappa}^{\mathbb{C}})$. The following conditions are equivalent.*

- (1) *For every $t \geq 0$, e^{tA} preserves complex stability.*
- (2) *For every $t \geq 0$, e^{tA} preserves complex stability after adjoining arbitrary inert polynomial variables.*
- (3) *There are a real generator R satisfying Theorem 3.1, a real scalar c , and real polynomials q_i of degree at most two, nonnegative on $\mathbb{R}$, such that*

$$A = R + i \left(c \text{Id} + \sum_{i=1}^d J_{i, \kappa_i}(q_i) \right). \quad (14)$$

Equivalently, the canonical representation has order at most two, all its second-order coefficients are real and satisfy (12)–(13), and every $\Im P_{e_i}$ is nonnegative on $\mathbb{R}^d$. In this formulation box preservation forces $\Im P_{e_i}(z) = q_i(z_i)$ with $\deg q_i \leq 2$, and forces the corresponding imaginary zeroth-order term in (14).

The two theorems identify ordinary and complete preservation for *semigroups starting at the identity*. We will prove this by constructing the generators. It is not an assertion that an arbitrary isolated real stability preserver preserves complex stability or admits every tensor extension.

3.1. Canonical differential operators and box preservation. We record the algebraic restrictions separately from the stability inequalities. All statements in this subsection hold over either $\mathbb{R}$ or $\mathbb{C}$.

Lemma 3.3 (Canonical normal representation). *Every linear endomorphism of $V_{\kappa}^{\mathbb{F}}$ has a unique representation (11).*

Proof. Apply the proposed representation successively to monomials in increasing total degree. The constant monomial determines $P_0 = A1$. Once P_{β} is known for

all $|\beta| < |\alpha|$, the equation on z^α determines

$$P_\alpha = \frac{1}{\alpha!} \left(Az^\alpha - \sum_{\substack{\beta \leq \alpha \\ \beta \neq \alpha}} P_\beta(z) (\alpha)_\beta z^{\alpha-\beta} \right).$$

Here $(\alpha)_\beta = \prod_i \alpha_i! / (\alpha_i - \beta_i)!$. No derivative whose index is not bounded by α acts nontrivially on z^α . The recursion therefore proves existence and uniqueness on the monomial basis. $\square$

Descriptions of differential operators on finite polynomial modules by $\mathfrak{sl}_2$ enveloping algebras are classical in quasi-exact solvability [36, 37]; we include the following decomposition to make the box conventions and small-capacity degeneracies explicit.

Proposition 3.4 (Operators of order at most two on a box). *Every box endomorphism of canonical order at most two is a linear combination of Id, the operators J_i^-, J_i^0, J_i^+ , and their products of length two, including products at the same site. Every endomorphism of order at most one has a unique expression*

$$c \text{Id} + \sum_{i=1}^d J_{i, \kappa_i}(q_i), \quad c \in \mathbb{F}, \quad q_i \in \mathbb{F}[z_i]_{\leq 2}. \quad (15)$$

Consequently the second-order coefficients have the variable dependence and degree bounds stated in Theorem 3.1. The vector space of endomorphisms of order at most two has dimension

$$1 + 3d + 5\#\{i : \kappa_i \geq 2\} + 9\binom{d}{2}. \quad (16)$$

Proof. Write $p_{\alpha, \beta}$ for the coefficient of z^β in P_α , and group terms according to the integer monomial shift $\nu = \beta - \alpha$. For input exponent $x \in \prod_i \{0, \dots, \kappa_i\}$, the coefficient of $z^{x+\nu}$ is the restriction to this grid of

$$F_\nu(x) = \sum_{\substack{|\alpha| \leq 2, \alpha \leq \kappa \\ \alpha + \nu \geq 0}} p_{\alpha, \alpha + \nu}(x)_\alpha. \quad (17)$$

This polynomial has total degree at most two and degree at most κ_i in each x_i . We use the polynomial (17) also when $x + \nu$ is outside the exponent grid.

Suppose first that $\nu_i < 0$. Every summand has $\alpha_i \geq -\nu_i$, so F_ν is divisible by $(x_i)_{-\nu_i}$. If $-\nu_i > \kappa_i$, there are no summands and $F_\nu = 0$. In particular, whenever a grid point would give a negative output exponent, this falling factorial makes F_ν vanish there.

If $\nu_i > 0$, box preservation gives vanishing on every overflow face

$$x_i \in \{\kappa_i - \nu_i + 1, \dots, \kappa_i\} \cap \{0, \dots, \kappa_i\}.$$

To justify the divisibility as a polynomial identity, fix one such value of x_i . On the remaining tensor grid the value of F_ν is zero: either the output has a negative coordinate, in which case the preceding falling-factorial argument applies, or it is a genuine monomial outside the box, whose coefficient must vanish. Tensor-product interpolation, using the coordinate degree bounds in (17), shows that the restriction to this face is identically zero as a polynomial in all remaining variables. Thus the

corresponding linear factor in x_i divides F_ν . If $\nu_i > \kappa_i$, the same argument gives vanishing on all $\kappa_i + 1$ coordinate values and hence $F_\nu = 0$.

For every remaining nonzero shift polynomial, these factors give

$$F_\nu = D_\nu G_\nu, \quad D_\nu(x) = \prod_i (x_i)_{\nu_i^-} (\kappa_i - x_i)_{\nu_i^+}, \quad \deg G_\nu \leq 2 - |\nu|_1. \quad (18)$$

The factors belonging to distinct coordinate values are coprime. Thus $F_\nu \neq 0$ implies $|\nu|_1 \leq 2$.

Put $E_i = z_i \partial_i$. For $\nu = 0$, every possible F_ν is realized by a combination of $\text{Id}, E_i, E_i^2, E_i E_j$. For $\nu = \pm e_i$, the required constant or affine factor G_ν is realized by $J_i^\pm$ and $J_i^\pm E_j$. For $\nu = \pm 2e_i$ use $(J_i^\pm)^2$; these shifts can be nonzero only if $\kappa_i \geq 2$. For $\nu = \pm e_i \pm e_j$, with $i \neq j$, use the corresponding products $J_i^\pm J_j^\pm$. The sign of the raising coefficient $x_i - \kappa_i = -(\kappa_i - x_i)$ is absorbed in the scalar multiplier. This accounts for every possibility in (18) on the input grid, and hence for every matrix entry of A .

This argument also covers binary sites. For example, if $\kappa_i = 1$, the canonical coordinate degree of F_ν in x_i is at most one; any inadmissible quadratic dependence has already been excluded before the realization step. Some displayed products may then be linearly dependent or zero, which does not affect the spanning assertion.

Repeating the same argument with total degree at most one leaves only a scalar and the three sitewise operators at each coordinate. They are linearly independent: the off-diagonal shifts separate the raising and lowering terms, while an affine polynomial in the diagonal exponents vanishing on the full tensor grid has every coefficient zero. This proves (15), including uniqueness.

The first-order coefficients of J_i^-, J_i^0, J_i^+ span $1, z_i, z_i^2$. Products at one site therefore have principal coefficients spanning all polynomials of degree at most four. Products at two different sites span all biforms of bidegree at most $(2, 2)$. At a site with $\kappa_i \geq 2$, these are canonical second-order coefficients; at a binary site that derivative is absent. The kernel of this principal-coefficient map is exactly the order-at-most-one space. Its dimension is $1 + 3d$, and the respective principal-coefficient spaces have dimensions five and nine. Surjectivity gives (16). $\square$

The proposition says that the operators of canonical order at most two form the image of the degree-at-most-two part of the enveloping algebra of the sitewise $\mathfrak{sl}_2$'s. This is an algebraic statement about the whole box. Stability will impose precisely the signs of its principal coefficients and, in the complex case, an inward sign on the imaginary first-order fields.

3.2. Root clusters and infinitesimal necessity. The relevant localization uses an isolated root cluster and is valid even when the ambient polynomial degree changes. No full-degree tangent formula is applied to a lower-degree polynomial. The underlying multiple-root contact principle is classical: after rotation of the half-plane, it is contained in the polynomial abscissa tangent theorem of Burke and Overton [16, Theorem 1.2 and Lemma 1.4]. We give a direct moment proof to record the signs and the local factorization needed for the operator argument.

Lemma 3.5 (Lower-half-plane root clusters). *Let $p_t(s)$, $t \geq 0$, be a curve of complex polynomials of uniformly bounded degree whose coefficients are right differentiable at zero. Suppose*

$$p_0(s) = s^m g(s), \quad m \geq 1, \quad g(0) \neq 0,$$

and that p_t has no zero in $\mathbb{H}$ for all sufficiently small t . Let C_t be the monic factor consisting of the m roots converging to zero. Its coefficients are right differentiable, and

$$\left. \frac{d}{dt} C_t(s) \right|_{t=0} = c_1 s^{m-1} + c_2 s^{m-2}, \quad \Im c_1 \geq 0, \quad c_2 \in \mathbb{R}, \quad c_2 \leq 0. \quad (19)$$

When $m = 1$, the c_2 term is absent. If every p_t is real rooted, then c_1 is real as well.

Proof. Choose a small circle about zero containing no other zero of p_0 . Coefficient continuity and Rouché's theorem isolate exactly m roots inside it for small t . If these roots are $\lambda_1(t), \dots, \lambda_m(t)$, their power sums are

$$M_k(t) = \sum_{j=1}^m \lambda_j(t)^k = \frac{1}{2\pi i} \int s^k \frac{p'_t(s)}{p_t(s)} ds, \quad k \geq 1,$$

where the contour is the chosen circle. The integrands are right differentiable with a denominator bounded away from zero. Thus each M_k is right differentiable and $M_k(t) = O(t)$. Newton's identities give the same differentiability for the coefficients of C_t .

Write $\lambda_j = a_j + ib_j$. The roots lie in the closed lower half-plane, so $b_j \leq 0$, and

$$\sum_j |b_j| = -\Im M_1(t) = O(t), \quad \sum_j b_j^2 = O(t^2).$$

Moreover,

$$\sum_j a_j^2 = \Re M_2(t) + \sum_j b_j^2 = O(t).$$

Consequently

$$\Im M_2(t) = 2 \sum_j a_j b_j = O(t^{3/2}) = o(t),$$

and, for $k \geq 3$,

$$|M_k(t)| \leq \sum_j |\lambda_j(t)|^k \leq \left(\sum_j |\lambda_j(t)|^2 \right)^{k/2} = O(t^{k/2}) = o(t).$$

The second inequality uses the ordinary finite counting sum over the roots.

Write $C_t = s^m + \gamma_1(t)s^{m-1} + \dots + \gamma_m(t)$. At $t = 0$, all γ_j and M_j vanish. Differentiating Newton's identities gives $\gamma'_j(0) = -M'_j(0)/j$. Hence every derivative below the coefficient of s^{m-2} vanishes. The first coefficient has nonnegative imaginary part because $\Im M_1(t) \leq 0$. For the second coefficient, $M'_2(0)$ is real, and

$$\Re M'_2(0) = \lim_{t \downarrow 0} \frac{\sum_j a_j(t)^2 - \sum_j b_j(t)^2}{t} \geq 0.$$

This proves (19). Real roots make M_1 real. □

Proposition 3.6 (Order and sign from product contacts). *Suppose $A \in \text{End}_{\mathbb{C}}(V_{\kappa}^{\mathbb{C}})$ generates a complex stability-preserving semigroup. Then A has canonical order at most two, every second-order canonical coefficient is real and nonpositive on $\mathbb{R}^d$, and $\Im P_{e_i} \geq 0$ on $\mathbb{R}^d$. If A is real and its semigroup is only assumed to preserve real stability, the order bound and the nonpositivity of its second-order coefficients still follow.*

Proof. For $0 \neq \alpha \leq \kappa$ and $y \in \mathbb{R}^d$, use the stable product

$$f_{\alpha,y}(z) = \prod_i (z_i - y_i)^{\alpha_i}.$$

In the canonical representation, every derivative at $z = y$ vanishes except for ∂^α , and therefore

$$(Af_{\alpha,y})(y) = \alpha! P_\alpha(y). \quad (20)$$

Choose $v \in \mathbb{R}_{>0}^d$ and restrict the polynomial curve to $z = y + sv$. Its initial value is

$$\left(\prod_i v_i^{\alpha_i} \right) s^{|\alpha|}.$$

Its coefficients are differentiable and its degree is bounded by $|\kappa|$. It has no zero for $s \in \mathbb{H}$, because such an s puts every active coordinate in $\mathbb{H}$. The restricted polynomial cannot be identically zero, since that would contradict multivariate stability. Invertibility of e^{tA} also excludes the zero multivariate output.

Apply Lemma 3.5. Factoring the restriction as $C_t(s)G_t(s)$, its constant first variation is

$$G_0(0) \left. \frac{d}{dt} C_t(0) \right|_{t=0}.$$

The term involving the derivative of G_t vanishes because $C_0(0) = 0$. Here $G_0(0) = \prod_i v_i^{\alpha_i} > 0$. For $|\alpha| \geq 3$, the constant variation is zero, so (20) gives $P_\alpha(y) = 0$. For $|\alpha| = 2$, it is real and nonpositive, giving the corresponding assertion for $P_\alpha(y)$. For $\alpha = e_i$, its imaginary part is nonnegative.

Since y is arbitrary, polynomial identity proves the vanishing of all coefficients of order greater than two. A complex polynomial real-valued on all of $\mathbb{R}^d$ has real coefficients; this proves reality of the principal coefficients. For a real semigroup, the same inputs and restrictions are real rooted, so the real case follows from the same lemma. Proposition 3.4 then supplies the asserted variable dependence. $\square$

The moment calculation identifies the obstruction to higher differential order: one-sided control of the imaginary root depths, together with the second root moment, eliminates every higher first-order cluster moment. Positivity of coefficients of the input polynomial is not used; every real contact point is available.

3.3. Binary apolar generators and sums of squares. The sufficiency proof starts in two binary variables. For $f = f_{00} + f_{10}s + f_{01}t + f_{11}st \in V_{(1,1)}^{\mathbb{R}}$, put

$$\Delta(f) = f_{10}f_{01} - f_{00}f_{11}.$$

A nonzero real f is stable if and only if $\Delta(f) \geq 0$. Indeed, if $f = a + bs + ct + dst$ and $b + dt \neq 0$, its zero in s is $-(a + ct)/(b + dt)$, whose imaginary part equals $-\Delta(f)\Im t/|b + dt|^2$. The cases in which f is independent of s follow directly.

Lemma 3.7 (Low-degree nonnegative polynomials). *Every nonnegative real univariate polynomial of degree at most four is a sum of squares of real polynomials of degree at most two. Every nonnegative real polynomial of bidegree at most $(2, 2)$ is a sum of squares of real bilinear polynomials.*

Proof. For a nonzero nonnegative univariate polynomial, every real root has even multiplicity and its nonreal roots occur in conjugate pairs. A positive real quadratic is a sum of two squares of linear polynomials. Repeated use of

$$(a^2 + b^2)(c^2 + d^2) = (ac - bd)^2 + (ad + bc)^2$$

gives the first assertion, with square degrees at most two. The zero polynomial presents no exception.

For the second assertion let $R(s, t) \geq 0$ and homogenize it to

$$\tilde{R}(x, y, u) = u^4 R(x/u, y/u).$$

The bidegree bound makes this a ternary quartic. It is nonnegative when $u \neq 0$, and hence everywhere by continuity. Hilbert's theorem on nonnegative ternary quartics writes it as a sum of squares of real quadratic forms; we use the version including all degenerate nonnegative forms [27]. The coefficients of x^4 and y^4 in $\tilde{R}$ are zero. They are sums of the squares of the x^2 and y^2 coefficients of the quadratic forms, respectively. Thus every form is a linear combination of xy, xu, yu, u^2 . Setting $u = 1$ yields bilinear squares. $\square$

Lemma 3.8 (Apolar rank-one primitive). *For a real bilinear polynomial $F = a + bs + ct + dst$, define*

$$\ell_F(f) = df_{00} - cf_{10} - bf_{01} + af_{11}, \quad B_F f = -F \ell_F(f).$$

The full mixed canonical coefficient of B_F is $-F^2$. Moreover

$$B_F^2 = 2\Delta(F)B_F, \tag{21}$$

$$\Delta(f + kB_F f) = \Delta(f) + k(1 + k\Delta(F))\ell_F(f)^2 \quad (k \in \mathbb{R}). \tag{22}$$

For all $t \geq 0$, e^{tB_F} preserves real stability.

Proof. For any endomorphism B of the binary box, its mixed coefficient is

$$P_B(s, t) = B(st) - sB(t) - tB(s) + stB(1). \tag{23}$$

The four evaluations of B_F give $P_{B_F} = -F^2$. Also $\ell_F(F) = 2ad - 2bc = -2\Delta(F)$, which gives (21). The polarization of the quadratic form Δ satisfies $D\Delta_f[F] = -\ell_F(f)$. Expanding $\Delta(f - kF\ell_F(f))$ proves (22).

If $D_F = \Delta(F)$, equation (21) gives

$$e^{tB_F} = \text{Id} + k_t B_F, \quad k_t = \begin{cases} (e^{2tD_F} - 1)/(2D_F), & D_F \neq 0, \\ t, & D_F = 0. \end{cases}$$

For $t \geq 0$, $k_t \geq 0$ and

$$1 + k_t D_F = \frac{1 + e^{2tD_F}}{2} > 0.$$

Thus (22) preserves the inequality $\Delta(f) \geq 0$. The exponential is invertible, so a nonzero input does not become the zero polynomial. These formulas also include $F = 0$ and $D_F = 0$. $\square$

For use below, we make explicit the passage from real to complex preservation in this binary semigroup.

Lemma 3.9 (Orientation at the identity). *Let B be a real endomorphism of $V_{(1,1)}^{\mathbb{R}}$. Then e^{tB} preserves real stability for all $t \geq 0$ if and only if its mixed coefficient P_B is nonpositive on $\mathbb{R}^2$. In this case every e^{tB} is a complete complex stability preserver.*

Proof. For necessity, take $f = (s - x)(t - y)$. It has $\Delta(f) = 0$. Differentiating $\Delta(e^{\tau B} f)$ at $\tau = 0$, and using (23), gives $-P_B(x, y)$. Real preservation implies that this derivative is nonnegative.

Conversely, Lemma 3.7 gives $-P_B = \sum_{\ell} F_{\ell}^2$. The operator $B - \sum_{\ell} B_{F_{\ell}}$ has zero mixed coefficient and hence has canonical order at most one. Proposition 3.4 writes it as a real scalar plus two sitewise operators $J_{i,1}(q_i)$, with real quadratic q_i , without a sign restriction on these two quadratics.

For clarity, the real sitewise flows preserve complex stability in both time directions. If $q(z) = q_0 + q_1 z + q_2 z^2$, let

$$M_{\tau} = \exp \left[\tau \begin{pmatrix} q_1/2 & q_0 \\ -q_2 & -q_1/2 \end{pmatrix} \right] = \begin{pmatrix} a_{\tau} & b_{\tau} \\ c_{\tau} & d_{\tau} \end{pmatrix}.$$

This is a real matrix of determinant one. The polynomial action

$$U_{\tau} f(z) = (c_{\tau} z + d_{\tau})^{\kappa} f \left(\frac{a_{\tau} z + b_{\tau}}{c_{\tau} z + d_{\tau}} \right) \quad (24)$$

has infinitesimal generator $J_{\kappa}(q)$. The quotient maps $\mathbb{H}$ to itself, since its imaginary part is $\Im z / |c_{\tau} z + d_{\tau}|^2$, and the denominator has no zero there. Formula (24) is polynomial after cancellation term by term and holds for every real τ . It also works with inert variables.

Lemma 3.8 and the finite-dimensional Lie product formula now show that $e^{\tau B}$ preserves real stability. Products of preserving factors converge coefficientwise; Hurwitz's theorem and invertibility of the limiting exponential exclude a zero output.

It remains to justify complex preservation of the apolar factors and of the resulting binary semigroup. We use the finite-box classification of Borcea and Brändén [7, Theorems 1.1 and 1.2]. A real stability preserver of rank greater than two has a stable plus symbol or a stable reflected-minus symbol. For the binary box these are

$$G_T^+(s, t; u, v) = T((s + u)(t + v)), \quad G_T^-(s, t; u, v) = T((s - u)(t - v)),$$

where T acts only in s, t . Every $e^{\tau B}$ has rank four. At $\tau = 0$ the minus symbol is $(s - u)(t - v)$, which is not stable: it vanishes when $s = u \in \mathbb{H}$. The cone consisting of stable polynomials and zero is closed in this fixed coefficient space. Continuity in τ therefore excludes the minus alternative for all sufficiently small nonnegative τ . The plus symbol is then stable, and the complex classification gives complex preservation.

More explicitly, adjoining an identity operator on an auxiliary box of capacity λ multiplies the plus symbol by $\prod_j (\xi_j + \eta_j)^{\lambda_j}$. This remains stable. The same complex classification proves preservation on every enlarged box. Finally, divide an arbitrary time into sufficiently small equal pieces and compose. This proves complete preservation for every time and completes the lemma. $\square$

This use of the symbol theorem is limited and precise. The real rank-two exceptional case is not invoked: the binary pair operator has rank four at every time. A single binary site is handled directly by (24).

3.4. Polarization and the proof of the real classification. Let $\mathcal{P}_\kappa$ be normalized polarization into $N = |\kappa|$ binary variables, grouped into κ_i slots at site i :

$$\mathcal{P}_\kappa(z_i^r) = \frac{e_r(z_{i,1}, \dots, z_{i,\kappa_i})}{\binom{\kappa_i}{r}}.$$

Let $\mathcal{D}_\kappa$ identify all slots belonging to each site. The two maps are inverse between $V_\kappa^\mathbb{C}$ and the subspace symmetric within each group of slots. Normalized polarization preserves upper-half-plane stability, and diagonalization preserves it in the reverse direction [7, Proposition 2.4], with the normalization fixed in the preliminaries.

We will need the exact derivative factors

$$\mathcal{D}_\kappa \partial_{i,a} \mathcal{P}_\kappa f = \frac{\partial_i f}{\kappa_i}, \quad (25)$$

$$\mathcal{D}_\kappa \partial_{i,a} \partial_{j,b} \mathcal{P}_\kappa f = \frac{\partial_i \partial_j f}{\kappa_i \kappa_j}, \quad i \neq j, \quad (26)$$

$$\mathcal{D}_\kappa \partial_{i,a} \partial_{i,b} \mathcal{P}_\kappa f = \frac{\partial_i^2 f}{\kappa_i(\kappa_i - 1)}, \quad a \neq b. \quad (27)$$

The last formula is used only for $\kappa_i \geq 2$. They follow by differentiating the elementary symmetric polynomial on a monomial and then diagonalizing.

Proof of Theorem 3.1. The implication from complete complex preservation to real preservation is immediate, since the operator is real. Necessity of the order and sign conditions is Proposition 3.6. We prove sufficiency constructively.

Write $a_i = P_{2e_i}$ and $b_{ij} = P_{e_i + e_j}$. For each $i < j$, apply Lemma 3.7 to obtain

$$-b_{ij}(s, t) = \sum_{\ell} F_{ij,\ell}(s, t)^2$$

with real bilinear $F_{ij,\ell}$. The binary operator $B_{ij} = \sum_{\ell} B_{F_{ij,\ell}}$ has mixed coefficient b_{ij} and generates a complete preserving semigroup. On the full slot space, place a copy of B_{ij} on each pair consisting of one slot at site i and one at site j , and sum these $\kappa_i \kappa_j$ copies. Every summand is a complete generator, so the sum is one by the Lie product formula. It commutes with permutations of slots at either site and therefore preserves the symmetric subspace.

Descend this operator by $\mathcal{D}_\kappa$ and $\mathcal{P}_\kappa$. Equation (26) shows that its full mixed coefficient is exactly $b_{ij}(z_i, z_j)$. No other second-order coefficient is created. Indeed, after writing each binary summand in normal order, zeroth- and first-order terms remain of those respective orders after diagonalization, and the only second-order derivative involves its two selected slots.

For a site with $\kappa_i \geq 2$, write

$$-a_i(z) = \sum_{\ell} p_{i,\ell}(z)^2, \quad p_{i,\ell}(z) = a_{\ell} z^2 + b_{\ell} z + c_{\ell},$$

using the univariate part of Lemma 3.7. Set

$$F_{i,\ell}(s, t) = a_{\ell} s t + \frac{b_{\ell}}{2}(s + t) + c_{\ell}.$$

These forms are symmetric in their two variables and satisfy $F_{i,\ell}(z, z) = p_{i,\ell}(z)$. The symmetric binary operator $2 \sum_{\ell} B_{F_{i,\ell}}$ has mixed coefficient

$$P_i(s, t) = -2 \sum_{\ell} F_{i,\ell}(s, t)^2, \quad P_i(z, z) = 2a_i(z).$$

Symmetry of the operator follows directly from symmetry of both $F_{i,\ell}$ and its apolar functional. Sum copies of this operator over all unordered pairs of distinct slots at site i . It is again a complete generator and commutes with all slot permutations at that site. Its descended same-site principal coefficient, by (27), is

$$\binom{\kappa_i}{2} \frac{P_i(z_i, z_i)}{\kappa_i(\kappa_i - 1)} = a_i(z_i).$$

This is the reason for the factor two in the binary operator. There is no same-site construction when $\kappa_i = 1$.

Let A_{pr} be the sum of all descended generators. For each summand, permutation invariance implies that its exponential on the symmetric subspace is intertwined by polarization with the exponential of the descended operator. Polarization, complete preservation on the full slot space, and diagonalization therefore show that A_{pr} generates a complete preserving semigroup on the original box. It has exactly the same second-order canonical coefficients as A .

The difference $A - A_{\text{pr}}$ has order at most one. By Proposition 3.4, it is a real scalar plus a sum of real sitewise $J_{i,\kappa_i}(q_i)$. Formula (24) proves complete preservation for each of their flows, with no sign restriction on q_i . A final Lie product formula proves complete preservation for e^{tA} . The argument takes place on any fixed enlarged finite box, and the limiting exponential is invertible there, so Hurwitz's zero alternative never occurs for a nonzero stable input. This proves the theorem. $\square$

3.5. Imaginary inward fields and the complex classification. The remaining complex generators have a particularly explicit one-site realization.

Lemma 3.10 (Imaginary parabolic primitive). *Let $a, b \in \mathbb{R}$ and $\kappa \geq 1$. The generator $iJ_\kappa((az + b)^2)$ has the semigroup*

$$(U_t f)(z) = d_t(z)^\kappa f(\phi_t(z)), \quad d_t(z) = 1 - iat(az + b), \quad \phi_t(z) = \frac{z + ibt(az + b)}{1 - iat(az + b)}. \quad (28)$$

For every $t \geq 0$, this is a complete complex stability preserver.

Proof. The corresponding fractional-linear matrix is

$$M_t = \text{Id} + it \begin{pmatrix} ab & b^2 \\ -a^2 & -ab \end{pmatrix}.$$

The displayed nonidentity matrix has square zero, so $M_{t+s} = M_t M_s$ and $\det M_t = 1$. Denominators in (28) cancel on each monomial of degree at most κ , giving a polynomial operator and an exact semigroup. Differentiating at zero gives

$$\left. \frac{d}{dt} U_t f \right|_{t=0} = i((az + b)^2 f' - \kappa a(az + b)f) = iJ_\kappa((az + b)^2)f.$$

For $z \in \mathbb{H}$ and $t \geq 0$, direct multiplication by the conjugate denominator gives

$$\Im \phi_t(z) = \frac{\Im z + t|az + b|^2}{|1 - iat(az + b)|^2} > 0. \quad (29)$$

When $a \neq 0$ and $t > 0$, the denominator has its only zero at $-b/a - i/(a^2 t)$, below the real axis. When $a = 0$, it is identically one and $\phi_t(z) = z + ib^2 t$. Thus the substitution maps $\mathbb{H}$ to itself and the prefactor does not vanish there. The same argument with all other variables held inert proves complete preservation. The cases $t = 0$ and $a = b = 0$ give the identity. $\square$

Proof of Theorem 3.2. Assume ordinary complex preservation and decompose $A = R + iS$ by entrywise real and imaginary parts in the ordinary monomial basis. Both R and S are real box endomorphisms. The canonical recursion in Lemma 3.3 is real linear, so it commutes with this decomposition.

By Proposition 3.6, A has order at most two and its second-order coefficients are real and nonpositive. The same coefficients are therefore the principal coefficients of R , which satisfies Theorem 3.1. Meanwhile S has order at most one. Proposition 3.4 gives

$$S = c\text{Id} + \sum_i J_{i,\kappa_i}(q_i), \quad c \in \mathbb{R}, \quad q_i \in \mathbb{R}[z_i]_{\leq 2}.$$

Its first-order coefficients are exactly $q_i = \Im P_{e_i}$; the product-contact test makes them nonnegative on $\mathbb{R}$. This proves the necessity of (14).

Conversely, every real quadratic nonnegative on $\mathbb{R}$ is a sum of squares of real linear polynomials. For $q(z) = az^2 + bz + c$ with $a > 0$, complete the square:

$$q(z) = \left(\sqrt{a}z + \frac{b}{2\sqrt{a}} \right)^2 + c - \frac{b^2}{4a}.$$

The last constant is nonnegative. If $a = 0$, nonnegativity forces $b = 0$ and $c \geq 0$. Lemma 3.10 and the Lie product formula therefore give complete preservation for each $iJ(q_i)$. The generator R has the same property by the real theorem, and $ic\text{Id}$ contributes a nonzero scalar phase. One further product formula proves complete preservation for their sum. Invertibility of the finite-dimensional exponential again excludes the zero limit. Complete preservation implies ordinary preservation, finishing the equivalence. $\square$

In particular, every complex second-order coefficient in an admissible generator is forced to be real. The imaginary part has no pair term: it consists entirely of sitewise inward quadratic fields and a scalar. The sign in (28) is fixed by the upper-half-plane convention: $i\partial$ shifts the argument upward and moves polynomial zeros downward.

3.6. Finite certificates, quotient cones, and exact lifts. The construction records both a finite membership certificate and a representation by local binary generators.

Corollary 3.11 (Gram certificates and lineality). *In Theorem 3.1, the principal inequalities are equivalent to the existence of real positive semidefinite matrices G_i of size three and G_{ij} of size four such that*

$$-P_{2e_i}(s) = (1, s, s^2)G_i(1, s, s^2)^\top, \quad \kappa_i \geq 2, \quad (30)$$

$$-P_{e_i+e_j}(s, t) = (1, s, t, st)G_{ij}(1, s, t, st)^\top, \quad i < j. \quad (31)$$

The extra inequalities in Theorem 3.2 have size-two certificates

$$q_i(s) = (1, s)K_i(1, s)^\top, \quad K_i \succeq 0. \quad (32)$$

These matrix sizes do not depend on the numerical capacities.

Let $\mathcal{N}_4, \mathcal{N}_{2,2}, \mathcal{N}_2$ denote, respectively, the cones of nonnegative real univariate quartics of degree at most four, nonnegative real biforms of bidegree at most $(2, 2)$, and nonnegative real univariate quadratics. The real generator cone has lineality

$$\mathfrak{g}_\kappa = \mathbb{R}\text{Id} + \sum_i J_{i,\kappa_i}(\mathbb{R}[z_i]_{\leq 2}),$$

and its quotient by this lineality is naturally isomorphic to

$$\prod_{i:\kappa_i \geq 2} \mathcal{N}_4 \times \prod_{i < j} \mathcal{N}_{2,2}. \quad (33)$$

The complex generator cone, regarded as a real cone, has lineality

$$\mathbb{C} \text{Id} + \sum_i J_{i,\kappa_i}(\mathbb{R}[z_i]_{\leq 2}),$$

and its quotient is the product in (33) times $\prod_i \mathcal{N}_2$.

Proof. The square decompositions in Lemma 3.7 give (30) and (31) by taking sums of outer products of the coefficient vectors. Conversely a real positive semidefinite matrix is a sum of rank-one positive semidefinite matrices, yielding the required square decomposition. The quadratic case is the same argument with the vector $(1, s)$.

The negative-principal-coefficient map has kernel precisely the order-at-most-one space, by Proposition 3.4. The polarization construction realizes every tuple of nonnegative principal polynomials, so it is onto the product in (33). A real A and its negative are both admissible if and only if all their second-order coefficients vanish. This gives exactly the stated real lineality and quotient.

For complex generators also record the imaginary first-order polynomials q_i . Their nonnegativity is independent of the real principal coefficients, as (14) shows. Both signs of a generator can be admissible only if all these q_i and all second-order coefficients vanish. The remaining imaginary scalar is unrestricted. This proves the complex assertion. $\square$

Corollary 3.12 (Exact binary-slot lift). *For every generator A in Theorem 3.2, there is a complete stability-preserving generator $\tilde{A}$ on the full multiaffine space with $|\kappa|$ variables such that*

$$\tilde{A}\mathcal{P}_\kappa = \mathcal{P}_\kappa A, \quad e^{t\tilde{A}}\mathcal{P}_\kappa = \mathcal{P}_\kappa e^{tA} \quad (t \geq 0). \quad (34)$$

It commutes with every permutation of slots within a site. It can be chosen as a sum of a scalar, one-slot real Möbius generators, one-slot imaginary parabolic generators, and nonnegative multiples of the rank-one apolar generators B_F on pairs of slots.

Proof. For the real second-order part use exactly the slot sums constructed in the proof of Theorem 3.1. They commute with within-site permutations and descend to the chosen principal-coefficient realization. Lift each remaining sitewise drift by

$$J_{i,\kappa_i}(q) \mapsto \sum_{a=1}^{\kappa_i} \left(q(z_{i,a}) \partial_{i,a} - \frac{1}{2} q'(z_{i,a}) \right).$$

Equation (25) shows that its diagonalization on a polarized input is precisely $J_{i,\kappa_i}(q)$. The lifted sum is permutation invariant, so this equality upgrades to intertwining with $\mathcal{P}_\kappa$. The same lift applies to the imaginary quadratics, which are sums of the parabolic primitives of Lemma 3.10. The scalar lifts as the same scalar identity.

All summands generate complete preserving flows on the full slot space, hence so does their sum. This proves the first relation in (34); exponentiation gives the second. $\square$

The lift is a generator intertwining, not merely a static polarization followed by an unrelated preserving map. It is not asserted to have a nonnegative coefficient matrix. Consequently it imposes no coefficient-positive lifting principle on a different preservation problem. Likewise, the signed generation statements proved above are finite dimensional; they contain no infinite-capacity domain assertion.

3.7. Exact recognition from local matrices. The small nonnegativity certificates also give an exact recognition algorithm when the operator is supplied by its local terms.

Corollary 3.13 (Polynomial-time local recognition). *Let A be specified as a finite sum of one-site and two-site operators on $(\mathbb{C}^2)^{\otimes n}$, with Gaussian-rational entries in the coefficient basis $1, z_i$ at each site. There is a deterministic algorithm, polynomial in n and the total input bit length, deciding whether e^{tA} preserves every nonzero complex disk-stable multiaffine polynomial for all $t \geq 0$. The test is on the sum A ; the displayed summands need not individually generate preserving semigroups. No degree bound on the interaction graph is required.*

Proof. In the basis $1, z$, the Cayley map from disk to half-plane coordinates is

$$C = \begin{pmatrix} i & -i \\ 1 & 1 \end{pmatrix}, \quad (Cf)(x) = (x+i)f\left(\frac{x-i}{x+i}\right).$$

Thus $B = C^{\otimes n} A (C^{-1})^{\otimes n}$ is the conjugated half-plane generator. Conjugating a local term requires only a matrix of order two or four, and preserves its support. All entries remain Gaussian rational with polynomial bit length.

For a transformed two-site term U , with variables x, y , set

$$\begin{aligned} p_0 &= U1, & p_x &= U(x) - xU1, & p_y &= U(y) - yU1, \\ p_{xy} &= U(xy) - yU(x) - xU(y) + xyU1. \end{aligned}$$

Checking the four monomials $1, x, y, xy$ proves $U = p_0 + p_x \partial_x + p_y \partial_y + p_{xy} \partial_x \partial_y$. The one-site formula is the same with the y -terms omitted. Add these expressions and merge equal monomials. Canonical uniqueness shows that the result is exactly

$$B = P_0 + \sum_i P_i \partial_i + \sum_{i < j} P_{ij} \partial_i \partial_j.$$

It has order at most two automatically. Each P_{ij} depends only on its two variables and has bidegree at most $(2, 2)$; all remaining coefficients also have sparse polynomial representations with only a constant number of contributions per input term.

By Theorems 3.1 and 3.2, acceptance is equivalent to the following finite tests on the merged coefficients:

- (1) each P_{ij} is real and $P_{ij}(s, t) \leq 0$ for every $(s, t) \in \mathbb{R}^2$;
- (2) each $\Im P_i$ is a polynomial $q_i(z_i)$ of degree at most two, and $q_i(s) \geq 0$ for every $s \in \mathbb{R}$;
- (3) $\Im P_0 + \frac{1}{2} \sum_i q'_i$ is a real constant.

Indeed the real part of B is a real box endomorphism with the required principal signs, while the last two conditions give exactly its allowed imaginary part. Conversely the complex classification forces all three conditions. The Cayley transformation is invertible, and so is e^{tA} ; stable-or-zero preservation therefore gives preservation of the nonzero stable class here.

The identity tests require rational arithmetic and merging only polynomially many sparse records. Forming common denominators by products shows that the bit lengths of all merged coefficients are polynomial in the input length. After clearing denominators by a positive multiplier, each remaining sign test is a real sentence in at most two variables, with one polynomial of total degree at most four. Fixed-dimensional, fixed-degree real quantifier elimination has polynomial bit complexity [6, Chapter 14]. There are only polynomially many such sentences. Weak inequalities include zero coefficients, repeated roots, and degree drops. This proves the claimed bit bound without expanding the global matrix. $\square$

Efficient tests for bivariate real stability and static univariate real-rootedness preservers were obtained by Raghavendra, Ryder, and Srivastava [29]. Here the finite-box generator classification reduces recognition from a compact many-site local description to fixed-dimensional nonnegativity tests.

4. SPECTRAL CONTRACTION AND SECTOR GROWTH

We first study arbitrary complex stability-preserving semigroups on a finite polynomial space. No self-adjointness, positivity of coefficients, or differential representation of the generator is assumed. Positive damping of polynomial degree has two consequences: it improves the zero-free domain of the actual evolution, and it gives a sharp contraction estimate in a norm determined by stable polynomials. The last subsection returns to real half-plane stability and assumes coefficient positivity and preservation of homogeneous degree. It compares spectral bounds between sectors without damping.

4.1. Weighted kernels and the stable cone. Zero-capacity coordinates may be omitted. If no coordinate remains, the kernel and eigenvector assertions reduce to nonzero constants. For the gap statements assume that at least one coordinate remains. Fix $d \geq 1$ and $\kappa \in \mathbb{N}_{>0}^d$, write

$$V_\kappa = \{p \in \mathbb{C}[z_1, \dots, z_d] : \deg_{z_i} p \leq \kappa_i\}, \quad b_\alpha = \binom{\kappa}{\alpha},$$

and let $\mathbb{D}_r^d = \{z : |z_i| < r \text{ for all } i\}$. Let $\mathcal{C}_\kappa$ consist of zero and all polynomials in V_κ that do not vanish on $\mathbb{D}_1^d$. We call a polynomial *strictly stable* if it does not vanish on $\overline{\mathbb{D}_1^d}$. For $T \in \text{End}_{\mathbb{C}}(V_\kappa)$, use the kernel of (6), namely

$$K_T(z, w) = T_z \prod_{i=1}^d (1 + z_i w_i)^{\kappa_i} = \sum_{\alpha \leq \kappa} b_\alpha (T z^\alpha)(z) w^\alpha. \quad (35)$$

The corresponding coefficient pairing is bilinear:

$$[f, g]_\kappa = \sum_{\alpha \leq \kappa} b_\alpha^{-1} f_\alpha g_\alpha. \quad (36)$$

In particular, $[K_T(z, \cdot), p]_\kappa = T p(z)$. There is no complex conjugation in this identity. In the orthonormal coherent basis $\sqrt{b_\alpha} z^\alpha$, the matrix kernel has one square-root binomial weight on each of its two legs.

We shall use the classical disk contraction theorem, Lemma 2.3, in the following form. If two nonzero stable coefficient tensors have matched variables of the same capacities, their contraction by (36) is stable or zero whenever each pair of contracted radii has product at least one. If every such product is greater than

one, the contraction is nonzero. The radii of all uncontracted variables are retained. This is the Grace–Asano contraction principle, with the finite-capacity formulation obtained by polarization; see [7, 8, 41]. Indeed, a coefficient of multidegree α is divided by b_α under normalized polarization, and summation over its b_α equal subsets leaves exactly the reciprocal weight in (36). Applied to two kernels, this pairing gives the kernel of their operator product.

Lemma 4.1. *An invertible operator T preserves $\mathcal{C}_\kappa$ if and only if K_T is zero-free on $\mathbb{D}_1^d \times \mathbb{D}_1^d$. If K_T is zero-free on $\mathbb{D}_r^d \times \mathbb{D}_r^d$, where $r > 1$, then Tp is nonzero and zero-free on $\mathbb{D}_r^d$ for every $p \in \mathcal{C}_\kappa \setminus \{0\}$, without an invertibility assumption on T in this last assertion.*

Proof. For fixed $w \in \mathbb{D}_1^d$, the polynomial $\prod_i (1 + z_i w_i)^{\kappa_i}$ is stable and nonzero. Its image under an invertible stability preserver is stable and nonzero. Evaluating at any $z \in \mathbb{D}_1^d$ proves the forward implication. Conversely, contract the input variables of K_T against a stable polynomial. The contraction theorem gives a stable polynomial or zero; invertibility excludes zero on a nonzero input. For the final assertion the contracted radii have product $r > 1$, so the strict contraction theorem itself excludes zero and retains output radius r . $\square$

Lemma 4.2. *The set $\mathcal{C}_\kappa$ is a closed, complex-scalar-invariant cone containing no complex two-dimensional subspace. Its interior consists exactly of the strictly stable polynomials. Every invertible operator preserving this cone maps its interior into its interior.*

Proof. Closedness follows from Hurwitz’s theorem on the connected polydisk. Every complex two-dimensional subspace contains a nonzero polynomial annihilated by evaluation at zero, and such a polynomial is not stable. A polynomial nonzero on the closed polydisk remains so under sufficiently small coefficient perturbations, by compactness. Conversely, a boundary zero ζ of a stable polynomial moves into the open polydisk for the arbitrarily small perturbation $p((1 + \varepsilon)z)$, at $z = \zeta/(1 + \varepsilon)$. Zero is not an interior point, since arbitrarily small nonzero multiples of z_1 do not belong to the cone. Finally, an invertible linear operator is open, so its image of the cone interior is an open subset of the cone. $\square$

4.2. Strictification by degree damping. Put $E_i = z_i \partial_i$, $E = \sum_i E_i$, and $P_0 p = p(0)$. The following theorem improves the zero-free domain of the exponential itself, rather than that of an auxiliary splitting operator.

Theorem 4.3 (Degree-damping strictification). *Suppose $A_0 \in \text{End}_{\mathbb{C}}(V_\kappa)$ satisfies $e^{tA_0} \mathcal{C}_\kappa \subseteq \mathcal{C}_\kappa$ for $t \geq 0$. Let $\mu_i > 0$ and*

$$A = A_0 - 2 \sum_{i=1}^d \mu_i E_i.$$

For every $t > 0$, the kernel $K_{e^{tA}}$ is nonzero on the closed unit polydisk in all $2d$ variables. Consequently it has a zero-free radius $r_t > 1$. The same radius works for the image of every nonzero unit-disk-stable input polynomial.

Moreover, for each $t_0 > 0$, one radius $r > 1$ works simultaneously for the kernels $K_{e^{tA}}$, $t \geq t_0$, and for their action on all nonzero unit-disk-stable inputs.

Proof. First consider uniform damping $A = A_0 - 2\mu E$. Fix $t > 0$ and allow μ to vary in the complex half-plane $\Re \mu > 0$. For $s \geq 0$,

$$e^{-2s\mu E} p(z) = p(e^{-2s\mu} z)$$

preserves disk stability. Thus every Lie product

$$\left(e^{tA_0/m} e^{-2t\mu E/m} \right)^m$$

preserves stability and is invertible. The finite-dimensional product formula, followed by Hurwitz and invertibility of the limiting exponential, proves preservation by $T_t(\mu) = e^{t(A_0 - 2\mu E)}$. Lemma 4.1 gives

$$K_{T_t(\mu)}(z, w) \neq 0 \quad \text{if } z, w \in \mathbb{D}_1^d, \quad \Re \mu > 0. \quad (37)$$

Each kernel coefficient is an entire function of μ .

We next establish a real large-damping limit, valid without normality. Let $a_{00} = (A_0 1)(0)$. In the Euclidean norm on monomial coefficients, E is diagonal, nonnegative, and has the constants as its zero eigenspace. Hence, for real $\mu > 0$,

$$\|e^{-2s\mu E}\| \leq 1, \quad e^{-2s\mu E} \longrightarrow P_0 \quad (s > 0).$$

Expand $T_t(\mu)$ in its Dyson series about $-2\mu E$. The term of order $k \geq 1$ is the integral over $0 < s_1 < \dots < s_k < t$ of

$$e^{-2\mu(t-s_k)E} A_0 e^{-2\mu(s_k-s_{k-1})E} \dots A_0 e^{-2\mu s_1 E}.$$

Every time interval is positive almost everywhere. The integrand therefore converges to $a_{00}^k P_0$, and its norm is bounded by $\|A_0\|^k$, independently of μ . Dominated convergence on the simplex gives the limit $t^k a_{00}^k P_0 / k!$. These integrated terms are uniformly bounded by the summable sequence $t^k \|A_0\|^k / k!$. Including the order-zero term and passing the limit through the sum proves

$$T_t(\mu) \longrightarrow e^{ta_{00}} P_0 \quad \text{in operator norm as real } \mu \longrightarrow +\infty. \quad (38)$$

This is a finite-dimensional instance of strong-coupling limits such as [15, Theorem 1]; the displayed argument fixes its normalization directly. Since $K_{P_0} = 1$, the limiting kernel is the nonzero constant $e^{ta_{00}}$.

Fix any $(z, w) \in \overline{\mathbb{D}}_1^d \times \overline{\mathbb{D}}_1^d$. For $0 < \rho < 1$, the functions

$$f_\rho(\mu) = K_{T_t(\mu)}(\rho z, \rho w)$$

are holomorphic and nonvanishing on $\Re \mu > 0$. Their limit as $\rho \uparrow 1$ is locally uniform in that half-plane, because the kernel has finitely many entire coefficient functions. Hurwitz therefore says that $K_{T_t(\mu)}(z, w)$, as a function of μ , is identically zero or nowhere zero. The limit (38) excludes the first alternative. This proves closed-polydisk nonvanishing at the desired real $\mu > 0$. Compactness then supplies a radius $r_t > 1$. Lemma 4.1 gives the common input-output assertion.

For nonuniform rates, let $\mu = \min_i \mu_i > 0$. The operator $A_0 - 2 \sum_i (\mu_i - \mu) E_i$ generates a stable semigroup by the same product-formula argument. Apply the uniform result to this operator and the remaining damping $-2\mu E$.

Finally fix $t_0 > 0$, and choose $r > 1$ for $B = e^{t_0 A/2}$. For every $t \geq t_0$,

$$e^{tA} = B e^{(t-t_0)A} B.$$

The middle kernel has radius one, including at $t = t_0$, and the outer kernels have radius r . The two kernel contractions pair radii whose products are $r > 1$, retaining

radius r on both external legs. A further contraction with any stable input proves the uniform input assertion. $\square$

Remark 4.4. The radius can depend on the capacity, generator, and initial positive time. The theorem gives neither a dimension-independent lower bound for $r - 1$ nor an improvement on inert ancillary variables. Complex scalar shifts of the generator do not affect its conclusions. No normalization by a trace is used.

4.3. Dominant eigenvectors and their dual certificates. We use the following finite-dimensional complex-cone theorem of Rugh [33, Theorem 8.4]: if a nontrivial closed complex-scalar-invariant cone contains no complex plane and a linear map sends its nonzero part into its interior, then the map has an algebraically simple nonzero eigenvalue of strictly largest modulus. Convexity of the cone and self-adjointness of the map are not hypotheses.

Proposition 4.5. *Under the hypotheses of Theorem 4.3, A has an algebraically simple eigenvalue a_* uniquely largest in real part. Choose its right eigenvector v and left eigenfunctional ℓ so that $\ell(v) = 1$. Both v and*

$$Q_\ell(w) = \sum_{\alpha \leq \kappa} b_\alpha \ell(z^\alpha) w^\alpha \quad (39)$$

are zero-free on a common polydisk of radius greater than one. Furthermore $\ell(p) \neq 0$ for every nonzero stable p , and $e^{-ta_} e^{tA} p \rightarrow \ell(p)v$.*

Proof. For each fixed $t > 0$, Theorem 4.3 and Lemma 4.2 verify all the hypotheses of Rugh's theorem for $T = e^{tA}$; the cone contains all constant polynomials and is therefore nontrivial. Its simple dominant eigenspace is invariant under A , because A commutes with T . Thus it is an A -eigenline, say for a_* . Spectral mapping gives strict real-part dominance: another generator eigenvalue with equal real part would give the same modulus for the exponential. If two distinct generator eigenvalues exponentiated to the same value, that value would have algebraic multiplicity at least two. A nontrivial Jordan block at a_* would also give a nontrivial exponential Jordan block, since $te^{ta_*} \neq 0$. Both alternatives contradict algebraic simplicity for T .

Fix a kernel radius $r > 1$ for T , and put $\Lambda = e^{ta_*}$. Finite-dimensional spectral theory gives

$$\Lambda^{-m} T^m \longrightarrow P = v \otimes \ell, \quad \ell(v) = 1.$$

Every power has kernel radius r , because its internal contraction radii have product $r^2 > 1$. The kernel transform is injective, so K_P is not the zero polynomial. Hurwitz yields

$$K_P(z, w) = v(z) Q_\ell(w) \neq 0 \quad (z, w \in \mathbb{D}_r^d).$$

Both factors consequently have radius r . The strict weighted contraction of Q_ℓ with a nonzero unit-stable polynomial is $\ell(p)$, so it is nonzero. The strict real-part separation and the finite-dimensional spectral expansion then give the asserted convergence, including when the remaining spectrum has Jordan blocks. $\square$

4.4. Sharp spectral bounds. We first record the scalar estimate used in both the spectral and dynamical arguments. It is the diameter form of the Schwarz lemma used in [14, Theorem 2].

Lemma 4.6. *If g is holomorphic on $\mathbb{D}_1^d$, continuous on its closure, and $0 \leq q \leq 1$, then*

$$\text{diam } g(q\overline{\mathbb{D}_1^d}) \leq q \text{diam } g(\overline{\mathbb{D}_1^d}).$$

The same conclusion holds for a coordinatewise dilation all of whose moduli are at most q .

Proof. The cases $q = 1$, zero diameter, and $q = 0$ are immediate. For $x, y \in q\overline{\mathbb{D}_1^d}$, put $\rho = \max(\|x\|_\infty, \|y\|_\infty)$. If $\rho > 0$,

$$f(\zeta) = \frac{g(\zeta x/\rho) - g(\zeta y/\rho)}{\text{diam } g(\overline{\mathbb{D}_1^d})}$$

is holomorphic on the unit disk, bounded in modulus by one, and vanishes at zero. Schwarz at $\zeta = \rho < 1$ bounds $|g(x) - g(y)|$ by ρ times the denominator. Take the supremum over x, y . A coordinatewise dilation has its image in $q\overline{\mathbb{D}_1^d}$. $\square$

Theorem 4.7 (Complex radius bound). *Suppose $r > 1$ and K_T is zero-free on $\mathbb{D}_r^d \times \mathbb{D}_r^d$. Then T has an algebraically simple nonzero eigenvalue $\Lambda_\star$ of strictly largest modulus, with an eigenvector of radius r . Every other eigenvalue satisfies*

$$|\Lambda/\Lambda_\star| \leq r^{-2}. \quad (40)$$

No invertibility, Hermiticity, or diagonalizability is assumed.

Proof. The kernel maps the nonzero stable cone into its interior by Lemma 4.1. Rugh's theorem therefore supplies the simple dominant eigenvalue. Its rank-one spectral projection is nonzero on some interior cone vector, since a proper linear hyperplane cannot contain an open set. Normalized iterates of that vector converge to a nonzero eigenvector ψ in the closed cone. Applying T once improves ψ to radius r .

Let ϕ be an eigenvector of a different nonzero eigenvalue, and put $g = \phi/\psi$, $\alpha = \Lambda/\Lambda_\star$. The function g is nonconstant and holomorphic on $\mathbb{D}_r^d$. For $1/r < s < r$, we claim

$$\alpha g(\mathbb{D}_r^d) \subseteq g(\mathbb{D}_s^d).$$

Otherwise, for some $v \in \mathbb{D}_r^d$, the nonzero polynomial $f = \phi - \alpha g(v)\psi$ would be stable on $\mathbb{D}_s^d$, but $Tf = \Lambda(\phi - g(v)\psi)$ would vanish at v . This contradicts strict contraction, since $rs > 1$.

For $1/r < s < t < r$, let $\Delta_u = \text{diam } g(\overline{\mathbb{D}_u^d})$. These diameters are finite and positive. The inclusion gives $|\alpha|\Delta_t \leq \Delta_s$. Applying Lemma 4.6 after rescaling radius t gives $\Delta_s \leq (s/t)\Delta_t$. Letting $s \downarrow 1/r$ and $t \uparrow r$ proves (40). Zero eigenvalues satisfy the inequality trivially. $\square$

The quantitative mechanism in this proof is the ratio-squeezing argument of [14]; the complex-cone theorem supplies the eigenvector without a positive-semidefinite hypothesis.

Theorem 4.8 (Sharp degree-damping gap). *Under the hypotheses of Theorem 4.3, set $\mu = \min_i \mu_i$. The dominant generator eigenvalue satisfies*

$$\Re a_\star - \Re a \geq 2\mu \quad (a \in \text{spec}(A), a \neq a_\star). \quad (41)$$

The constant is sharp for every positive finite capacity vector.

Proof. For small $h > 0$, put

$$D_h = e^{-h \sum_i \mu_i E_i}, \quad T_h = D_h e^{h A_0} D_h.$$

The kernel of $e^{h A_0}$ has radius one by Lemma 4.1. If $q_i = e^{-h \mu_i}$, direct coefficient calculation gives

$$K_{T_h}(z, w) = K_{e^{h A_0}}(q_1 z_1, \dots, q_d z_d, q_1 w_1, \dots, q_d w_d).$$

It therefore has radius $e^{h \mu}$. Theorem 4.7 gives a simple dominant eigenvalue $\Lambda_\star(h)$ and bounds every other eigenvalue by $e^{-2h \mu} |\Lambda_\star(h)|$.

Now $B_h = (T_h - I)/h \rightarrow A$ in norm. Its eigenvalues are exactly $b(h) = (\Lambda(h) - 1)/h$, with algebraic multiplicities. These multisets are bounded and converge to the spectrum of A . Uniformly for bounded complex b ,

$$h^{-1} \log |1 + hb| = \Re b + O(h).$$

The modulus inequality consequently implies $\Re b_\star(h) - \Re b(h) \geq 2\mu + O(h)$. Choose a sequence along which $b_\star(h)$ converges. After removing this one eigenvalue from each multiset, the remaining eigenvalues converge, with multiplicity, to the remaining copies of the spectrum of A . The limiting inequality separates all those copies by 2μ . In particular, the distinguished limit is algebraically simple and is the unique maximal-real-part eigenvalue, hence is $a_\star$. This proves (41) without differentiable eigenvalue labels or a normal-matrix perturbation bound. For $A_0 = 0$, the eigenvalues are $-2 \sum_i \mu_i \alpha_i$; a degree-one monomial at a site of minimum rate attains equality. $\square$

4.5. Contraction along the actual evolution. For strictly stable p and arbitrary $h \in V_\kappa$, define

$$\text{Osc}_p(h) = \text{diam}\{h(z)/p(z) : z \in \overline{\mathbb{D}_1^d}\}. \quad (42)$$

It is a complex seminorm with kernel $\mathbb{C}p$: homogeneity and the triangle inequality follow from the diameter definition, and zero diameter means that the rational function is constant.

Lemma 4.9. *If T is an invertible stability preserver, p is strictly stable, and $h \in V_\kappa$, then Th is strictly stable and*

$$(Th/Thp)(\mathbb{D}_1^d) \subseteq (h/p)(\mathbb{D}_1^d).$$

In particular $\text{Osc}_{Thp}(Th) \leq \text{Osc}_p(h)$.

Proof. Strictness follows from Lemma 4.2. If w does not belong to the input ratio image, then $h - wp$ is a nonzero stable polynomial. Its image $Th - wThp$ is nonzero and stable, excluding w from the output ratio image. Taking diameters proves the assertion; the open and closed polydisks give the same diameter by continuity and density. $\square$

Theorem 4.10 (Moving-denominator contraction). *Let A_0 generate a stability-preserving semigroup, and let $A = A_0 - 2 \sum_i \mu_i E_i$, with $\mu_i \geq \mu \geq 0$. For every strictly stable p , every $h \in V_\kappa$, and $t \geq 0$,*

$$\text{Osc}_{e^{tA}p}(e^{tA}h) \leq e^{-2\mu t} \text{Osc}_p(h). \quad (43)$$

The constant is sharp.

Proof. The damping operator $D_\tau = e^{-2\tau \sum_i \mu_i E_i}$ acts by a coordinatewise dilation of maximum modulus $e^{-2\mu\tau}$. Lemma 4.6, applied to h/p , gives the corresponding oscillation contraction. Each $e^{\tau A_0}$ is nonexpansive by Lemma 4.9. Iterating along $(e^{tA_0/m} D_{t/m})^m$ therefore gives exactly the factor $e^{-2\mu t}$, with all intermediate denominators strictly stable.

The products converge to e^{tA} , which is invertible and preserves stability by the product formula and Hurwitz. Its image of p is strictly stable. The minimum modulus of that limiting polynomial on the closed polydisk is positive, so coefficient convergence gives uniform convergence of the ratios there. Diameters are continuous under uniform convergence, proving (43). Equality holds for pure damping with $p = 1$ and $h = z_j$ at a minimum-rate site. $\square$

Corollary 4.11 (Nonautonomous evolution). *On each compact time interval, suppose $A_0(t)$ and $\mu_i(t)$ are bounded and piecewise continuous, with finitely many continuity pieces, each $A_0(t)$ a stability generator and $\mu_i(t) \geq 0$. Let $U(t, s)$ solve the linear evolution equation with generator $A_0(t) - 2 \sum_i \mu_i(t) E_i$. Then, for strictly stable p ,*

$$\text{Osc}_{U(t,s)p}(U(t, s)h) \leq \exp\left(-2 \int_s^t \min_i \mu_i(u) du\right) \text{Osc}_p(h).$$

Proof. Approximate on each continuity piece by step functions taking actual values of the coefficients. They converge in L^1 . Their frozen evolutions satisfy Theorem 4.10, including when the minimum rate is zero. Multiplying the bounds gives the exponential of the integral of the stepwise minimum rates. These integrals converge to the stated integral, since the minimum of finitely many real coordinates is Lipschitz. The integral-equation estimate for finite matrix ODEs gives uniform convergence of the propagators. The limiting fundamental matrix is invertible and preserves stability by Hurwitz; Lemma 4.2 gives a strictly stable limiting denominator. Uniform convergence of the ratios passes the estimate to the limit, as in the autonomous proof. $\square$

Corollary 4.12 (Adapted norm and resolvent). *Assume positive damping, and normalize v, ℓ as in Proposition 4.5. On $W = \ker \ell$, set $\|h\|_v = \text{Osc}_v(h)$ and $B = (A - a_\star I)|_W$. Then $\|\cdot\|_v$ is a norm and*

$$\|e^{tB}\|_{v \rightarrow v} \leq e^{-2\mu t}, \quad \|(\zeta I - B)^{-1}\|_{v \rightarrow v} \leq \frac{1}{\Re \zeta + 2\mu} \quad (\Re \zeta > -2\mu).$$

Every eigenvalue of B with real part -2μ is semisimple.

Proof. The only possible null vectors of the seminorm are multiples of v , and $\ell(v) = 1$. Thus it is a norm on W . The left eigenfunctional identity makes W invariant. Since $e^{tA}v = e^{ta_\star}v$, Theorem 4.10 gives the first estimate, including the complex phase of the scalar. For $\Re \zeta > -2\mu$, the absolutely convergent integral $\int_0^\infty e^{-t\zeta} e^{tB} dt$ is the inverse of $\zeta I - B$, by integration of its derivative. Integrating the norm estimate gives the stated bound. A nontrivial Jordan block at a boundary eigenvalue would produce unbounded polynomial growth after multiplication by $e^{2\mu t}$, contradicting the first estimate and norm equivalence on its finite-dimensional generalized eigenspace. $\square$

Example 4.13 (No uniform strict radius). Fix $\mu, t > 0$. On V_1 , in the coefficient basis $(1, z)$, let X and Z be the usual Pauli matrices and put

$$A_{0,R} = RX, \quad A_R = A_{0,R} - 2\mu E = \begin{pmatrix} 0 & R \\ R & -2\mu \end{pmatrix}, \quad R > 0.$$

The undamped flow preserves disk stability, also with arbitrary inert variables. Indeed, writing $c = \cosh(Rs)$, $b = \sinh(Rs)$, its action is the explicit weighted substitution

$$(e^{sA_{0,R}}p)(z) = (c + bz)p\left(\frac{b + cz}{c + bz}\right).$$

The prefactor is nonzero on the disk and the fraction is a disk automorphism, since $c^2 - b^2 = 1$. Put $\omega_R = \sqrt{R^2 + \mu^2}$. The identity $(A_R + \mu I)^2 = \omega_R^2 I$ gives

$$e^{tA_R} = e^{-\mu t} \left[\cosh(\omega_R t) I + \frac{\sinh(\omega_R t)}{\omega_R} (A_R + \mu I) \right].$$

Thus the image of the fixed input 1 has its unique zero at

$$z_R = -\frac{\omega_R \coth(\omega_R t) + \mu}{R}.$$

For every finite R , $|z_R| > 1$, but $|z_R| \rightarrow 1$ as $R \rightarrow \infty$. Since $K_{e^{tA_R}}(z, 0) = e^{tA_R}1(z)$, the same zero obstructs a common kernel radius depending only on μ, t . This failure already occurs for Hermitian generators on one binary variable. The dominant eigenvector has its zero at modulus $(\omega_R + \mu)/R$, which also tends to one.

Example 4.14 (Unbounded mixing remainder at a fixed gap). Fix $\mu > 0$, and on V_1 put

$$G_R = R(X + iZ), \quad B_R = G_R - 2\mu E = \begin{pmatrix} iR & R \\ R & -iR - 2\mu \end{pmatrix}, \quad R > 0.$$

Here $G_R^2 = 0$ and $G_R^*Z + ZG_R = 0$. Its stability-preserving action is explicit: with $a = 1 + iRs$ and $b = Rs$,

$$(e^{sG_R}p)(z) = (a + bz)p\left(\frac{b + \bar{a}z}{a + bz}\right), \quad |a + bz|^2 - |b + \bar{a}z|^2 = 1 - |z|^2.$$

The denominator is nonzero in the disk, and the fraction maps the disk to itself. The formula also proves preservation with inert variables.

Choose the square root $\omega_R = \sqrt{\mu^2 + 2i\mu R}$ with $x_R = \Re \omega_R > 0$. The two eigenvalues of B_R are $a_{\pm} = -\mu \pm \omega_R$, and

$$x_R = \sqrt{\frac{\mu\sqrt{\mu^2 + 4R^2} + \mu^2}{2}} \geq \mu.$$

The dominant spectral projection and normalized exponential are

$$P_R = \frac{1}{2} \left(I + \frac{B_R + \mu I}{\omega_R} \right), \quad S_R(t) = e^{-ta_+} e^{tB_R} = P_R + e^{-2\omega_R t} (I - P_R).$$

Their off-diagonal entries give

$$\|P_R\|_2 \geq \frac{R}{2|\omega_R|}, \quad |[S_R(t)]_{10}| = \frac{R}{2|\omega_R|} |1 - e^{-2\omega_R t}| \geq \frac{R}{2|\omega_R|} (1 - e^{-2x_R t}).$$

Since $|\omega_R| = (\mu^4 + 4\mu^2 R^2)^{1/4}$, both quantities diverge as $R \rightarrow \infty$, the second for every fixed $t > 0$.

To keep the actual spectral gap fixed, pass to $V_{(1,1)}$ and set

$$\mathcal{A}_R = G_R \otimes I - 2\mu(E_1 + E_2) = B_R \otimes I + I \otimes (-2\mu E).$$

The undamped part is still a joint stability generator by the displayed weighted substitution. Its four eigenvalues are $a_+, a_-, a_+ - 2\mu, a_- - 2\mu$. Hence its dominant eigenvalue is a_+ , its real-part gap is exactly 2μ , and its dominant projector is $\mathcal{P}_R = P_R \otimes P_0$. For the fixed strictly stable input $p(z_1, z_2) = 1 + z_2/2$, define $\mathcal{R}_R(t) = e^{-ta_+} e^{t\mathcal{A}_R} - \mathcal{P}_R$. The coefficient of $z_1 z_2$ is exactly

$$[z_1 z_2] \mathcal{R}_R(t) p = \frac{1}{2} e^{-2\mu t} \frac{R}{2\omega_R} (1 - e^{-2\omega_R t}).$$

The projector term contains no z_2 , which proves this identity directly from the tensor exponential. Its modulus tends to infinity for every fixed $\mu, t > 0$. Thus even after subtraction of the dominant projector there is no coefficient-norm mixing bound depending only on the damping, time, and exact spectral gap, in a fixed two-variable box and on one fixed strictly stable input. Equivalence of fixed finite-dimensional norms gives the same obstruction for every fixed coefficient norm. This is consistent with Corollary 4.12, whose norm depends on the dominant polynomial. The example uses linear normalization by e^{ta_+} ; it makes no claim about nonlinear normalization by an evolving coefficient.

Remark 4.15. For the stable cone, the forbidden-pencil projective distance of [19] is $\text{osc}_{\mathbb{D}_1^d} \log |q/p|$. Its real infinitesimal form is $\text{osc } \Re(h/p)$; maximizing over a scalar phase of h gives (42). Thus the metric and the general complex-cone contraction principle are inherited. The estimates above specify the exact degree-damping rate for arbitrary tangent vectors along the actual, possibly nonautonomous evolution.

4.6. Sector growth and token-graph spectral concavity. We now return to real upper-half-plane stability. There is a second spectral consequence when coefficient positivity and total degree are preserved. It does not require degree damping or self-adjointness. A single stable evolution compares the exponential growth rates of all homogeneous sectors.

Theorem 4.16 (Concavity of sector growth). *Let V_κ be a finite real coordinate box and $N = \sum_i \kappa_i$. Suppose that A has nonnegative off-diagonal entries in the monomial basis, preserves every homogeneous subspace $V_{\kappa,k}$, and that e^{tA} preserves nonnegative real stability for every $t \geq 0$. Write*

$$s_k = \max\{\Re \lambda : \lambda \in \text{spec}(A|_{V_{\kappa,k}})\}, \quad 0 \leq k \leq N.$$

Then

$$2s_k \geq s_{k-1} + s_{k+1} \quad (1 \leq k < N).$$

Neither irreducibility nor symmetry of A is assumed.

Proof. Start at the stable polynomial $f_0(z) = \prod_i (1 + z_i)^{\kappa_i}$. Every allowed monomial has a strictly positive coefficient. Choose α such that $A + \alpha I$ is entrywise nonnegative. Then $e^{tA} = e^{-\alpha t} e^{t(A+\alpha I)}$ is entrywise nonnegative and has a strictly positive diagonal. Consequently $f_t = e^{tA} f_0$ still has every allowed coefficient strictly positive. Its diagonal restriction is a degree- N polynomial

$$P_t(x) = f_t(x, \dots, x) = \sum_{k=0}^N c_k(t) x^k, \quad c_k(t) > 0,$$

whose roots are all strictly negative.

Newton's inequalities give

$$\left(\frac{c_k(t)}{\binom{N}{k}} \right)^2 \geq \frac{c_{k-1}(t)}{\binom{N}{k-1}} \frac{c_{k+1}(t)}{\binom{N}{k+1}}. \quad (44)$$

Here is the particular form needed. The derivative $R = P_t^{(k-1)}$ has $m = N - k + 1$ negative roots, so $R(x) = R(0) \prod_{j=1}^m (1 + u_j x)$ with $u_j > 0$. Cauchy's inequality gives

$$\sum_{i < j} u_i u_j \leq \frac{m-1}{2m} \left(\sum_i u_i \right)^2.$$

The first three coefficients of R are $(k-1)!c_{k-1}$, $k!c_k$, and $(k+1)!c_{k+1}/2$. Substitution gives (44), including its binomial normalization.

Let $A_k = A|_{V_{\kappa,k}}$, and let v_k be the coefficient vector of the degree- k part of f_0 . Degree preservation gives

$$c_k(t) = \mathbf{1}^\top e^{tA_k} v_k.$$

Since every entry of v_k is positive, this quantity is bounded above and below by fixed positive multiples of the sum of all entries of e^{tA_k} . For an entrywise nonnegative m_k -by- m_k matrix M , that sum lies between $\|M\|_1$ and $m_k\|M\|_1$, where $\|\cdot\|_1$ is the induced maximum column-sum norm. Spectral mapping gives $\|e^{tA_k}\|_1 \geq e^{ts_k}$, and Jordan normal form gives

$$\|e^{tA_k}\|_1 \leq C_k(1 + t^{m_k-1})e^{ts_k} \quad (t \geq 0)$$

for a fixed constant C_k . Therefore

$$\lim_{t \rightarrow \infty} t^{-1} \log c_k(t) = s_k.$$

Take logarithms in (44), divide by t , and let $t \rightarrow \infty$. The fixed binomial factors disappear, proving the assertion. For $N \leq 1$ there is no inequality to check. $\square$

The proof uses coefficient sums, not $\text{Tr}(e^{tA_k})$. No real-rooted generating function for the latter traces is asserted. The passage from coefficient log-concavity to concavity of growth exponents is the one-variable tropical-stability principle [11, Theorem 5 and the discussion following it]. The proof above establishes the needed exponential version directly, including nonsymmetric blocks and their Jordan factors. The preservation theorem supplies the input needed to apply it.

Token-graph spectral theory includes the nesting of Laplacian spectra proved by Dalfó and coauthors [18], and the study of adjacency spectral radii and new Laplacian spectral layers by Reyes, Dalfó, and Fiol [31]. The monotonicity assertions addressed below are those for adjacency and signless-Laplacian largest eigenvalues in [2, Conjectures 5–6]. The new-layer Laplacian quantities in [31, Definition 3.1 and Conjecture 3.3] are different spectral invariants.

Let G be a finite simple graph on $[n]$, with nonnegative edge weights w_{ij} . Its k -token graph has the k -subsets as vertices. The edge from S to $S - \{i\} + \{j\}$ has weight w_{ij} whenever $i \in S$, $j \notin S$, and ij is an edge of G . Denote its adjacency and weighted degree matrices by $A_k(G)$ and $D_k(G)$. The sectors $k = 0, n$ are singletons with zero matrices.

Theorem 4.17 (Token-graph spectral concavity). *For every $a \in [-1, 1]$, put*

$$\Lambda_k(a) = \lambda_{\max}(A_k(G) + aD_k(G)).$$

For every nonnegatively weighted G ,

$$2\Lambda_k(a) \geq \Lambda_{k-1}(a) + \Lambda_{k+1}(a) \quad (1 \leq k < n), \quad \Lambda_k(a) = \Lambda_{n-k}(a).$$

In particular the sequence is nondecreasing up to half filling. The interval $[-1, 1]$ is the exact interval of real parameters for which this concavity holds for every such graph.

Proof. On the binary polynomial box define, for every unordered edge,

$$\mathcal{B}_{ij} = w_{ij}[(z_j + az_i)\partial_i + (z_i + az_j)\partial_j - (z_i^2 + 2az_iz_j + z_j^2)\partial_i\partial_j].$$

It annihilates 1, z_iz_j and sends z_i to $w_{ij}(az_i + z_j)$ and z_j to $w_{ij}(z_i + az_j)$. Thus $\mathcal{B} = \sum_{ij} \mathcal{B}_{ij}$ preserves total degree, is a real symmetric Metzler matrix, and restricts on the degree- k sector to $A_k(G) + aD_k(G)$. Its mixed second-order coefficient is

$$-w_{ij}[(s + at)^2 + (1 - a^2)t^2] \leq 0.$$

Theorem 3.1 therefore gives stability preservation. Theorem 4.16 yields concavity, since for a real symmetric matrix its spectral bound is its largest eigenvalue.

Complementation of subsets gives a weight-preserving isomorphism between the k - and $(n-k)$ -token graphs, proving symmetry. Set $\delta_k = \Lambda_k - \Lambda_{k-1}$. Concavity says that the δ_k are nonincreasing, while symmetry gives $\delta_{n-k+1} = -\delta_k$. If $k \leq (n+1)/2$, it follows that $\delta_k \geq -\delta_k$, proving monotonicity.

For sharpness, take the unweighted star $K_{1,3}$. Its two-token graph is a six-cycle. On the equal-leaf subspace, the one-token matrix is $\begin{pmatrix} 3a & \sqrt{3} \\ \sqrt{3} & a \end{pmatrix}$; its orthogonal complement has eigenvalue a . Thus

$$\Lambda_1(a) = 2a + \sqrt{a^2 + 3}, \quad \Lambda_2(a) = 2a + 2.$$

For $a > 1$, $\Lambda_1(a) > \Lambda_2(a)$; symmetry then contradicts concavity at $k = 2$. For $a < -1$, a single edge has $\Lambda_0 = \Lambda_2 = 0$ and $\Lambda_1 = a + 1 < 0$, again violating concavity. At $a = 1$, the star has $\Lambda_1 = \Lambda_2 = 4$, so connectedness does not imply strict monotonicity. $\square$

For completeness, the local preservation used here has a short verification independent of the generator classification. For $\tau \geq 0$, set $b = e^{a\tau} \sinh \tau$ and $c = e^{a\tau} \cosh \tau$. The unweighted edge exponential fixes $1, xy$ and sends x, y to $cx + by, bx + cy$. Its algebraic symbol is

$$xy + uv + b(xu + yv) + c(xv + yu).$$

Writing this symbol as $\zeta^\top M \zeta / 2$ in the order $\zeta = (x, y, u, v)^\top$, one has

$$M = \begin{pmatrix} 0 & 1 & b & c \\ 1 & 0 & c & b \\ b & c & 0 & 1 \\ c & b & 1 & 0 \end{pmatrix}.$$

Its eigenvalues are

$$1 + e^{(a+1)\tau}, \quad 1 - e^{(a+1)\tau}, \quad -1 - e^{(a-1)\tau}, \quad -1 + e^{(a-1)\tau}.$$

Exactly one is positive when $-1 \leq a \leq 1$. The quadratic form is positive on every strictly positive real vector Y . A zero at $X + iY$ would imply $X^\top MY = 0$ and

$X^\top MX = Y^\top MY > 0$, contradicting the one-positive-eigenvalue property. Hence the symbol is stable. The classical symbol theorem [7] gives complete preservation for each edge. The finite-dimensional product formula and coefficientwise closure give preservation for their sum; invertibility of the limiting exponential excludes zero outputs.

Taking $a = 1$ and $a = 0$ gives, respectively, monotonicity of the largest eigenvalues of the signless Laplacian and the adjacency matrix, as conjectured in [2, Conjectures 5–6]. Theorem 4.17 proves the stronger weighted concavity throughout the displayed interpolation interval. The adjacency-preserving generator at $a = 0$ is already contained in Purbhoo’s construction [28, Proposition 4.5]. The contribution here is the cross-sector spectral conclusion. The bounds in Conjectures 1–4 of [2] were settled by [5] and are not claimed here.

In the occupation basis, the same block matrices arise from

$$B_a(G) = \frac{1}{2} \sum_{ij} w_{ij} [X_i X_j + Y_i Y_j + a(I - Z_i Z_j)].$$

Equivalently, for the XXZ Hamiltonian

$$H_\Delta(G) = \frac{1}{2} \sum_{ij} w_{ij} (X_i X_j + Y_i Y_j + \Delta Z_i Z_j) = B_{-\Delta}(G) + \frac{\Delta}{2} \sum_{ij} w_{ij} I,$$

the maximum energies in the magnetization sectors $\sum_i Z_i = n - 2k$ form a concave sequence for $|\Delta| \leq 1$. Indeed, the sector maxima are $\Lambda_k(-\Delta) + (\Delta/2) \sum_{ij} w_{ij}$, with an additive constant independent of k . The negative Hamiltonian has convex sector ground-state energies. At $a = 1$, the common one-qubit rotation $R = e^{-i\pi X/4}$, with $RYR^* = Z$ and $RZR^* = -Y$, identifies $B_1(G)$ with the EPR Hamiltonian in [2]. There is a maximum-energy state in a half-filled sector in the rotated basis. This locates a maximizing sector; it does not give an efficient algorithm within that exponentially large sector. For $a < 0$ the quantity in Theorem 4.17 is the largest eigenvalue, not in general the spectral radius.

5. CLASSICAL COMPUTATION OF PRINCIPAL EIGENVALUES AND STATES

The spectral theorem of the preceding section has an algorithmic consequence for operators with a bounded-degree local description. A positive damping rate selects a holomorphic eigenvalue throughout a complex half-plane. This domain can be joined to a disk on which perturbation coefficients are efficiently computable. An explicit change of variable then reduces evaluation at any fixed positive damping rate to logarithmically many coefficients.

The coefficient algorithm is due to Bravyi, DiVincenzo, and Loss [12]. Our use of it includes a complexification argument, so that every invocation of their algorithm remains within its Hermitian hypotheses. Neither an extension of Hermitian eigenvalue ordering to nonnormal matrices nor numerical continuation of eigenvectors is needed. We first estimate the principal eigenvalue. Starting in Section 5.7, we then compute stable product tests of its eigenvectors, including relative ground-state amplitudes, product-event probabilities, and adaptive equatorial measurement records. These scalar queries do not require a bound on the condition number of the spectral projection.

5.1. Local input and the approximation theorem. Identify the coefficient space

$$\mathcal{P}_n = \{p \in \mathbb{C}[z_1, \dots, z_n] : \deg_{z_i} p \leq 1\} \cong (\mathbb{C}^2)^{\otimes n}, \quad z^\alpha \longleftrightarrow |\alpha\rangle.$$

We use the Euclidean norm in this coefficient basis and the induced operator norm. Let $\mathcal{S}_n$ be the set of nonzero polynomials in $\mathcal{P}_n$ that do not vanish in $\mathbb{D}^n$, where $\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\}$. The degree operator is

$$E = \sum_{i=1}^n |1\rangle\langle 1|_i = \sum_{i=1}^n z_i \partial_{z_i}. \quad (45)$$

All local matrices below are supplied in the displayed tensor-product basis. A Gaussian rational means an element of $\mathbb{Q}(i)$.

Theorem 5.1 (Principal-eigenvalue approximation). *Fix $d \in \mathbb{N} \cup \{0\}$ and positive rational numbers J, μ . For $n \geq 1$, suppose that the input specifies*

$$A_0 = \sum_{i=1}^n K_i + \sum_{\{i,j\} \in \mathcal{E}} K_{ij}, \quad (46)$$

where the graph $(\{1, \dots, n\}, \mathcal{E})$ has maximum degree at most d , every local matrix has Gaussian-rational entries and operator norm at most J , and

$$e^{tA_0} \mathcal{S}_n \subseteq \mathcal{S}_n \quad (t \geq 0). \quad (47)$$

Let $a_\star$ be the eigenvalue of $A = A_0 - 2\mu E$ with largest real part. This eigenvalue is unique and algebraically simple. Given a positive rational δ , a deterministic classical algorithm returns $\hat{a} \in \mathbb{Q}(i)$ such that

$$|\hat{a} - a_\star| \leq \delta.$$

For fixed d, J, μ , its bit complexity is polynomial in n , the input bit length, and δ^{-1} .

Only the sum A_0 is assumed to satisfy (47); the displayed local terms need not individually generate preserving semigroups. The theorem uses the supplied local representation. Corollary 3.13 verifies its preservation hypothesis in polynomial bit time. The local representation itself is part of the input. No positivity, Hermiticity, stochastic normalization, or diagonalizability is assumed.

We will give explicit choices of all continuation parameters. In particular, put

$$N = 2n, \quad D = d + 1, \quad \rho = \frac{2^{-18}}{DJ}, \quad c = \frac{1}{\mu}. \quad (48)$$

The number p of perturbation coefficients required by the proof satisfies

$$p = O\left(\left(1 + \frac{c}{\rho}\right)^2 \log\left(2 + \frac{C_{d,J,\mu} n}{\delta}\right)\right), \quad (49)$$

for a constant $C_{d,J,\mu}$ independent of n, δ . The arithmetic operation count is $N \exp(O_D(p))$, with a polynomial bit overhead. In particular, after substitution the coefficient computation takes

$$N \left(2 + \frac{C_{d,J,\mu} n}{\delta}\right)^{O_D((1+2^{18}DJ/\mu)^2)}$$

arithmetic operations, up to polynomial factors in the coefficient order. The constant implicit in O_D retains the graph-degree dependence of the perturbation algorithm. In compact form, the same arithmetic bound is

$$n \left(2 + \frac{C_{d,J,\mu} n}{\delta} \right)^{O_d(1+(J/\mu)^2)},$$

up to polynomial factors in the coefficient order, with the bit overhead specified in Appendix A.1. For fixed inverse-polynomial accuracy $\delta = n^{-b}$, this is $n^{O_{d,b}(1+(J/\mu)^2)}$, up to a parameter-dependent factor and polynomial factors in the coefficient order. The large constants in (48) are inherited from the perturbation estimate. They are sufficient for a complexity statement, not a claim of practical efficiency. In particular, Theorem 5.1 does not assert polynomial dependence on μ^{-1} .

5.2. Perturbation coefficients and complexification. We first state precisely the published perturbation input. For a graph on N qubits with maximum degree at most $D \geq 1$, set

$$H_0 = 2 \sum_{u=1}^N |1\rangle\langle 1|_u, \quad \Omega = |0\rangle^{\otimes N}. \quad (50)$$

For any matrix V , the eigenvalue of $H_0 + xV$ issuing from the simple eigenvalue 0 has a convergent germ at $x = 0$, which we write

$$F_V(x) = \sum_{k \geq 1} e_k(V) x^k. \quad (51)$$

Proposition 5.2 (Bravyi–DiVincenzo–Loss). *Let $W = \sum_{\{u,v\}} W_{uv}$ be Hermitian, with each W_{uv} Hermitian and of norm at most $L > 0$. For H_0 as in (50), put*

$$R_L = \frac{2^{-16}}{DL}.$$

Then

$$|e_k(W)| \leq 2^{-15} N R_L^{-k} \quad (k \geq 1). \quad (52)$$

For every $p \geq 1$, the coefficients $e_1(W), \dots, e_p(W)$ can be computed in $N \exp(O_D(p))$ arithmetic operations.

These are Lemma 6 and Theorem 3 of [12], specialized to unperturbed gap 2. Their Theorem 2 also gives convergence of the energy expansion throughout the corresponding high-field disk. The coefficient algorithm is formal: computing the first p coefficients does not require the eventual evaluation point to lie in that disk. We use the coefficient bound and algorithm exactly in this Hermitian form. The required rational-input bit implementation is specified below.

Lemma 5.3 (Complex perturbation coefficients). *Let $V = \sum_{\{u,v\}} V_{uv}$, where the local matrices are arbitrary complex matrices with $\|V_{uv}\| \leq J$. The germ (51) extends to a holomorphic eigenvalue function on*

$$|x| < \rho, \quad \rho = \frac{2^{-18}}{DJ},$$

and its coefficients satisfy

$$|e_k(V)| \leq 2^{-15} N \rho^{-k} \quad (k \geq 1). \quad (53)$$

If the entries of the local matrices are Gaussian rational, for each integer $p \geq 1$ its first p coefficients can be computed exactly using $N \exp(O_D(p))$ arithmetic operations on Gaussian rationals.

Proof. Let $Q = \text{Id} - |\Omega\rangle\langle\Omega|$, and let S be the inverse of H_0 on $Q(\mathbb{C}^2)^{\otimes N}$, extended by zero on Ω . Normalize the eigenvector germ by

$$v(x) = \Omega + \sum_{k \geq 1} v_k x^k, \quad \langle \Omega, v_k \rangle = 0.$$

Writing $v_0 = \Omega$, coefficient comparison in $(H_0 + xV)v(x) = F_V(x)v(x)$ gives

$$e_k(V) = \langle \Omega, V v_{k-1} \rangle, \quad (54)$$

$$v_k = -SQ \left(V v_{k-1} - \sum_{j=1}^{k-1} e_j(V) v_{k-j} \right). \quad (55)$$

The vacuum coordinate functional in these formulas is fixed; it does not conjugate any variable matrix entry of V . Induction shows that v_k and $e_k(V)$ are homogeneous polynomials of degree k in the entries of V , with no conjugate variables. The same recursion also proves uniqueness of the formal normalized eigenpair. Thus the coefficients returned by Proposition 5.2 are the restrictions of these polynomials to Hermitian inputs.

Decompose the local terms as

$$B_{uv} = \frac{V_{uv} + V_{uv}^*}{2}, \quad C_{uv} = \frac{V_{uv} - V_{uv}^*}{2i},$$

and write $V = B + iC$ for the resulting sums. Both B_{uv} and C_{uv} are Hermitian of norm at most J . For real θ , the local perturbation

$$W_\theta = B \cos \theta + C \sin \theta = \frac{1}{2} e^{-i\theta} V + \frac{1}{2} e^{i\theta} V^*$$

is Hermitian, with each local norm at most $2J$. Since e_k is homogeneous of degree k , the Fourier coefficient of frequency $-k$ in $e_k(W_\theta)$ is $2^{-k} e_k(V)$. Indeed, a monomial of degree k contributes that frequency only when its V -term is selected in all k factors. Consequently

$$e_k(V) = \frac{2^k}{2\pi} \int_0^{2\pi} e^{ik\theta} e_k(W_\theta) d\theta. \quad (56)$$

Apply (52) with $L = 2J$. The corresponding radius is $2^{-17}/(DJ)$, so (56) gives

$$|e_k(V)| \leq 2^k 2^{-15} N \left(\frac{2^{-17}}{DJ} \right)^{-k} = 2^{-15} N \rho^{-k}.$$

This proves normal convergence on compact subsets of $|x| < \rho$. Near zero the sum is the actual eigenvalue germ, by (54)–(55) and the simple eigenvalue of H_0 . The identity theorem applied to

$$\det(H_0 + xV - F_V(x) \text{Id})$$

therefore proves the eigenvalue identity on the entire disk, providing an analytic eigenvalue branch there.

For the coefficient algorithm, take the distinct rational nodes

$$s_\ell = -1 + \frac{2\ell}{p}, \quad 0 \leq \ell \leq p.$$

At each node, $B + s_k C$ is a Hermitian local perturbation with norm bound $2J$. Apply Proposition 5.2 at each of these $p + 1$ inputs. For $k \leq p$, the function

$$s \mapsto e_k(B + sC)$$

is a polynomial of degree at most k . Its values at the nodes determine its value at $s = i$, namely $e_k(V)$, by exact rational interpolation. The factor $p + 1$ in the number of calls and the polynomial interpolation cost are absorbed in $N \exp(O_D(p))$. Thus the integral in (56) is used only for a bound; the algorithm uses no numerical quadrature. $\square$

Remark 5.4. The factor four between R_J and ρ has two sources: the Hermitian family has local norm bound $2J$, and extracting the extremal Fourier coefficient contributes 2^k . The argument transfers the coefficient estimate, not a Hermitian spectral-ordering or spectral-simplicity theorem. This distinction is necessary for arbitrary nonnormal inputs.

One-site perturbations can be incorporated without altering this published input. Attach a private auxiliary qubit i' to each physical site i , and represent a one-site term V_i by $V_i \otimes \text{Id}_{i'}$ on the edge $\{i, i'\}$. Give every auxiliary qubit the unperturbed term $2|1\rangle\langle 1|_{i'}$. The resulting operator is exactly

$$\tilde{H}(x) = (2E + xV) \otimes \text{Id}_{\text{aux}} + \text{Id}_{\text{phys}} \otimes 2 \sum_{i'} |1\rangle\langle 1|_{i'}. \quad (57)$$

Its vacuum eigenvalue germ and all its coefficients agree with those of the physical operator. The number of qubits is $N = 2n$, the maximum degree is at most $D = d + 1$, and the local norm bound is unchanged. The identity on the auxiliary factor is important: the auxiliary qubit is not coupled to the physical perturbation. This construction is only a device for using the coefficient algorithm; it imposes no stability assumption on the auxiliary description.

5.3. A global eigenvalue branch.

Lemma 5.5 (Half-plane continuation). *Under (47), the matrix*

$$A(h) = A_0 - 2hE$$

has an algebraically simple eigenvalue $a(h)$, uniquely maximizing real part, for every h with $\Re h > 0$. The function a is holomorphic on that half-plane. For $V = -A_0$, the function

$$F(x) = -x a(1/x), \quad \Re x > 0, \quad (58)$$

is a holomorphic eigenvalue of $2E + xV$. It agrees with the vacuum germ of Lemma 5.3 near the positive real origin.

Proof. Write $h = u + iv$, where $u > 0$. The imaginary Euler flow acts on polynomials by

$$e^{-2itvE} p(z_1, \dots, z_n) = p(e^{-2itv} z_1, \dots, e^{-2itv} z_n),$$

and therefore preserves $\mathcal{S}_n$. The finite-dimensional product formula gives

$$e^{t(A_0 - 2ivE)} = \lim_{m \rightarrow \infty} (e^{tA_0/m} e^{-2itvE/m})^m.$$

For each stable input, every approximating output is stable. The limit is stable or zero by Hurwitz's theorem, and it cannot be zero on a nonzero input because the limiting exponential is invertible. Hence $A_0 - 2ivE$ also generates a preserving semigroup.

Theorem 4.8, applied with damping u , supplies an algebraically simple eigenvalue of $(A_0 - 2ivE) - 2uE = A(h)$ whose real part exceeds those of all the others by at least $2u$. At each point, the implicit function theorem for the characteristic polynomial makes this eigenvalue locally holomorphic. Spectral continuity preserves its strict real-part inequality in a sufficiently small neighborhood. Thus the locally defined functions agree wherever their neighborhoods overlap: they select the same uniquely distinguished eigenvalue. This proves the global holomorphic assertion.

The reciprocal map takes the right half-plane into itself, so (58) is holomorphic there. The matrix identity

$$-xA(1/x) = 2E + xV$$

proves the eigenvalue assertion. For positive real x , the scalar $-x$ reverses real-part order, so $F(x)$ is the unique eigenvalue of $2E + xV$ with smallest real part. We do not make this ordering assertion for nonreal x .

It remains to identify the high-field germ without assuming Hermiticity of V . Choose positive x so small that $x\|V\| < 1/4$. Normality of $2E$ implies that every eigenvalue of $2E + xV$ is within distance $1/4$ of $\{0, 2, \dots, 2n\}$. To see the inclusion, outside these disks the resolvent of $2E$ has norm at most 4, so its product with xV has norm less than 1. The Neumann series therefore makes the perturbed resolvent invertible, including when $V = 0$. The Riesz projection inside $|z| = 1$ has rank one: the contour stays in the resolvent set throughout $2E + \tau xV$, $0 \leq \tau \leq 1$, and its rank at $\tau = 0$ is one. Hence the eigenvalue near zero is precisely the vacuum branch. All other eigenvalues have real part greater than one. The vacuum branch is therefore the uniquely minimal real-part eigenvalue, proving its agreement with $F(x)$ on a positive real interval. $\square$

Apply the identity theorem on the connected overlap $\{\Re x > 0\} \cap \{|x| < \rho\}$. Lemma 5.5 and Lemma 5.3, with the auxiliary construction when needed, yield one holomorphic function on

$$\Omega_\rho = \{x \in \mathbb{C} : \Re x > 0\} \cup \{x \in \mathbb{C} : |x| < \rho\}. \quad (59)$$

It remains an eigenvalue of $\tilde{H}(x)$ throughout this domain, by (57) and the characteristic-polynomial identity. In particular,

$$|F(x)| \leq \|\tilde{H}(x)\| \leq 2N + |x|W, \quad W = NDJ. \quad (60)$$

The displayed W is a convenient rational upper bound on the sum of all local norms. The scalar estimate $|\lambda| \leq \|T\|$ for an eigenvalue of a matrix T does not require normality. Finally,

$$a_\star = -\mu F(1/\mu). \quad (61)$$

5.4. Effective continuation and exact arithmetic. The next lemma gives the continuation step independently of its operator-theoretic application.

Lemma 5.6 (An explicit continuation map). *Let $\rho, c > 0$, and suppose that F is holomorphic on Ω_ρ . Define*

$$\eta = \frac{\rho^2}{4c + 2\rho}, \quad R = c + \eta, \quad q = \frac{c}{R}, \quad b = \frac{R^2 - c^2}{R}, \quad \phi(w) = \frac{bw}{1 - qw}. \quad (62)$$

Then $0 < q < 1$, $\phi(\mathbb{D}) \subset \Omega_\rho$, $\phi(0) = 0$, and $\phi(q) = c$. If $|F(x)| \leq L_0 + L_1|x|$ on Ω_ρ , the function $G = F \circ \phi$ satisfies

$$|G(w)| \leq M := L_0 + L_1(c + R) \quad (w \in \mathbb{D}).$$

Writing $F(x) = \sum_{j \geq 0} f_j x^j$ at zero and $G(w) = \sum_{k \geq 0} g_k w^k$, one has $g_0 = f_0$ and

$$g_k = \sum_{j=1}^k f_j b^j q^{k-j} \binom{k-1}{j-1} \quad (k \geq 1). \quad (63)$$

For every $p \geq 0$,

$$\left| F(c) - \sum_{k=0}^p g_k q^k \right| \leq \frac{M q^{p+1}}{1-q}. \quad (64)$$

Proof. The definitions give $q \in (0, 1)$, $\eta \leq \rho/2$, and

$$0 < R^2 - c^2 = 2c\eta + \eta^2 < \rho^2.$$

If $|x - c| < R$ and $\Re x \leq 0$, then

$$|x|^2 = |x - c|^2 + 2c\Re x - c^2 < R^2 - c^2 < \rho^2.$$

Thus the disk of center c and radius R lies in Ω_ρ . The identity

$$\phi(w) = c + R \frac{w - q}{1 - qw}$$

shows that ϕ maps $\mathbb{D}$ biholomorphically onto this disk. Direct substitution gives its values at 0 and q . Since $|\phi(w)| < c + R$, the bound on G follows.

For $j \geq 1$, expansion at zero gives

$$\phi(w)^j = b^j w^j (1 - qw)^{-j} = b^j \sum_{k \geq j} \binom{k-1}{j-1} q^{k-j} w^k.$$

Only $j \leq k$ contributes to the coefficient of w^k , which proves (63). Applying Cauchy's estimate on circles with radii increasing to one gives $|g_k| \leq M$. The geometric tail at $w = q$ now proves (64). No holomorphic extension to the boundary of $\mathbb{D}$ is used. $\square$

Proof of Theorem 5.1. The existence, uniqueness, and simplicity of a_* follow from Theorem 4.8. Use $V = -A_0$, the auxiliary construction (57), and the parameters (48). Lemmas 5.3 and 5.5 give the function F on Ω_ρ , with (60). Apply Lemma 5.6 with

$$L_0 = 2N, \quad L_1 = W, \quad M = 2N + (c + R)W.$$

All the parameters in (62) are rational.

Choose the smallest integer $p \geq 0$ for which

$$\mu M q^{p+1} \leq \delta(1 - q). \quad (65)$$

The algorithm finds this integer by rational multiplication and comparison; evaluation of logarithms is unnecessary. If $p > 0$, compute $e_1(V), \dots, e_p(V)$ by Lemma 5.3, and then compute $g_0, \dots, g_p$ by (63). Here $g_0 = F(0) = 0$. Return the Gaussian rational

$$\hat{a} = -\mu \sum_{k=0}^p g_k q^k. \quad (66)$$

For $p = 0$, no coefficient call is made and the same formula returns zero. Equations (61), (64), and (65) give $|\hat{a} - a_*| \leq \delta$.

We first bound the number of arithmetic operations. A useful exact identity is

$$\frac{1}{1-q} = \frac{R}{\eta} = 1 + \frac{4c^2}{\rho^2} + \frac{2c}{\rho}. \quad (67)$$

Since $-\log q \geq 1 - q$, the stopping rule implies

$$p = O\left(\frac{1}{1-q} \log\left(2 + \frac{\mu M}{\delta(1-q)}\right)\right).$$

The bound (49) follows, because $M = O_{d,J,\mu}(n)$. For fixed d, J, μ , this is $p = O_{d,J,\mu}(\log(2 + n/\delta))$. The coefficient algorithm uses $N \exp(O_D(p))$ arithmetic operations. Interpolation, the triangular transform, and evaluation in (66) add polynomially many operations in p .

Appendix A.1 proves that the exact coefficient computation, interpolation, and evaluation have polynomial bit overhead. Together with (49), this completes the proof. $\square$

5.5. Suzuki–Fisher ground energies. The preceding result applies to the Suzuki–Fisher class of Theorem 6.2. We keep its sign convention: the longitudinal term is $-\sum_i \mu_i^z Z_i$, with $\mu_i^z \geq 0$. The Lee–Yang theorem gives disk stability preservation by the Gibbs semigroup e^{-tH} ; see [35, 41, 14].

Theorem 5.7 (Fixed positive longitudinal field). *Fix a maximum degree d , a positive rational bound J on the absolute values of the Pauli coefficients, and a positive rational μ . Let H be a Suzuki–Fisher Hamiltonian with rational Pauli coefficients on an n -vertex graph of maximum degree at most d , and suppose*

$$\mu_i^z \geq \mu \quad (1 \leq i \leq n).$$

For every positive rational δ , its ground energy $E_0(H)$ can be approximated to absolute error at most δ by a deterministic classical algorithm polynomial in n , the input bit length, and δ^{-1} . The polynomial may depend on d, J, μ .

Proof. Define

$$V = H + \mu \sum_i Z_i.$$

The longitudinal fields of V are $\mu_i^z - \mu \geq 0$; all its interaction inequalities are unchanged. Thus V is again Suzuki–Fisher, and $A_0 = -V$ satisfies (47). Its local matrices have Gaussian-rational entries. Combining the finitely many Pauli terms on each edge or site gives a local norm bound depending only on J, μ , and leaves the degree bound unchanged. For example, sums of the absolute Pauli coefficients give explicit rational norm upper bounds, so the algorithm need not compute exact matrix norms.

Since $2E = n \text{Id} - \sum_i Z_i$,

$$A_0 - 2\mu E = -H - \mu n \text{Id}. \quad (68)$$

The matrix on the right is Hermitian. Its eigenvalue with largest real part is therefore $-E_0(H) - \mu n$. Apply Theorem 5.1 with error δ . If it returns $\hat{a}$, return the real rational

$$\hat{E} = -\Re \hat{a} - \mu n.$$

Equation (68) gives the required error bound and the asserted complexity. $\square$

Remark 5.8 (Hermitian coefficient bound). For this application, the complexification step can be omitted: the perturbation V is Hermitian. Applying Proposition 5.2 directly gives the larger guaranteed disk $2^{-16}/(DJ')$, where J' is the local norm bound after the auxiliary construction. The same continuation proof then applies to the ground-energy branch of $2E + xV$. The factor-four improvement in this disk changes constants, not the fixed-parameter complexity conclusion.

Theorem 5.7 includes nonstoquastic interactions and does not require conservation of total degree. Its computational output and parameter dependence differ from several earlier results. Bravyi and Gosset [13] give randomized algorithms for a two-axis stoquastic family on arbitrary graphs. Harrow, Mehraban, and Soleimanifar [22, Section 7 of the full version] give an all-temperature quasipolynomial partition-function algorithm for a number-conserving XXZ subclass. Their fugacity-polynomial reduction uses the commutation of the interaction with total magnetization. Bravyi, Gosset, Liu, and Wong [14] give a polynomial-time quantum ground-energy algorithm for the full Suzuki–Fisher family, including zero fields and inverse-polynomial absolute accuracy. Their sharp Hermitian gap and quantum algorithm are not consequences newly claimed here. Theorem 5.7 gives a deterministic classical conclusion on bounded-degree inputs at every fixed positive field. It does not replace the unrestricted quantum theorem.

5.6. Energy-value approximation at zero field. A fixed positive field can also be used as a controlled perturbation when the requested error is proportional to the number of sites.

Theorem 5.9 (Additive energy-value approximation). *Fix a maximum degree d , a bound J on the absolute Pauli coefficients, and a positive rational ε . For every rational Suzuki–Fisher Hamiltonian on an n -vertex graph satisfying these bounds, a deterministic classical algorithm polynomial in n and the input bit length returns $\hat{E} \in \mathbb{Q}$ such that*

$$|\hat{E} - E_0(H)| \leq \varepsilon n.$$

The longitudinal fields may vanish. The polynomial may depend on d, J, ε . The output is an energy value; no approximating state is asserted.

Proof. It suffices to consider $0 < \varepsilon \leq 1$, since a smaller accuracy parameter also meets a larger requested tolerance. Put $\nu = \varepsilon/4$ and

$$H_\nu = H - \nu \sum_i Z_i.$$

This is a Suzuki–Fisher Hamiltonian whose longitudinal fields are all at least the fixed positive number ν . Its local coefficients are bounded in terms of J, ε . The variational characterization of the lowest eigenvalue gives

$$|E_0(H_\nu) - E_0(H)| \leq \|H_\nu - H\| \leq \nu n.$$

Use Theorem 5.7 to estimate $E_0(H_\nu)$ with absolute error at most $\varepsilon n/2$. The returned value differs from $E_0(H)$ by at most

$$\frac{\varepsilon n}{4} + \frac{\varepsilon n}{2} = \frac{3\varepsilon n}{4} \leq \varepsilon n.$$

For fixed d, J, ε , the field ν and all local bounds are fixed. The runtime in Theorem 5.7 is therefore polynomial in the input size and n . $\square$

We spell out the consequence for Quantum MaxCut and EPR Hamiltonians, keeping value approximation separate from state construction. For a graph $G = (V, \mathcal{E})$ with nonnegative rational weights, let

$$Q_G = \sum_{\{i,j\} \in \mathcal{E}} w_{ij} |\Psi^-\rangle \langle \Psi^-|_{ij}, \quad |\Psi^-\rangle = \frac{|01\rangle - |10\rangle}{\sqrt{2}},$$

and write $\text{OPT}(G) = \lambda_{\max}(Q_G)$. Similarly, define the EPR objective

$$P_G = \sum_{\{i,j\} \in \mathcal{E}} w_{ij} |\Phi^+\rangle \langle \Phi^+|_{ij}, \quad |\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}.$$

Corollary 5.10 (EPR and bipartite Quantum MaxCut values). *For fixed degree and upper weight bounds and fixed rational $\varepsilon > 0$, the optimal values of P_G on arbitrary graphs and of Q_G on bipartite graphs admit deterministic classical additive- $\varepsilon|V|$ approximation in polynomial time. For both unweighted classes and each fixed rational $0 < \varepsilon < 1$, it returns a rational value L satisfying*

$$(1 - \varepsilon) \text{OPT} \leq L \leq \text{OPT},$$

where OPT denotes the largest eigenvalue of the respective objective. The output is a value, rather than a state attaining it.

Proof. The negative EPR objective is Suzuki–Fisher up to a scalar:

$$-P_G = -\frac{1}{4} \sum_{\{i,j\} \in \mathcal{E}} w_{ij} (\text{Id} + X_i X_j - Y_i Y_j + Z_i Z_j). \quad (69)$$

On each edge the transverse matrix has operator norm $w_{ij}/4$, equal to its longitudinal interaction coefficient. Subtract the known scalar $-\frac{1}{4} \sum_{\{i,j\}} w_{ij}$ when applying Theorem 5.9, and restore it afterward. This proves the assertion for P_G .

If G is bipartite with parts A, B , let $U = \prod_{i \in A} Y_i$. On every edge, U maps the singlet vector to the EPR vector up to a scalar of modulus one. Hence

$$U Q_G U^* = P_G.$$

The spectra agree, proving the additive assertion for Q_G . This is the same local change of basis used in [14, Corollary 2].

For the unweighted assertions, remove isolated vertices and let N, m be the remaining vertex and edge counts. If $m = 0$, return zero. Otherwise $N \leq 2m$. For EPR, the product vector $|0^N\rangle$ has expectation $m/2$. For bipartite Quantum MaxCut, the product vector which is 0 on one part and 1 on the other has the same expectation. In either case $\text{OPT} \geq m/2 \geq N/4$. Run the additive algorithm with error $\eta = \varepsilon N/8$, obtaining $\hat{v}$, and return $L = \max(0, \hat{v} - \eta)$. Then

$$L \leq \text{OPT}, \quad L \geq \text{OPT} - 2\eta \geq (1 - \varepsilon) \text{OPT}.$$

For each fixed ε , the running time is polynomial. $\square$

The additive statement for bounded weights does not automatically give a relative guarantee when positive weights may be arbitrarily small. Nor is the value scheme a state-producing approximation algorithm of the type studied, for example, in [1]. The inverse-polynomial absolute-accuracy conjectures for general EPR problems in [23] remain outside its scope. Indeed, introducing a field of order δ/n would make the factor c/ρ in (49) grow with n . The resulting coefficient count

need no longer be logarithmic, and the algorithm proved here need no longer run in polynomial time.

5.7. Stable tests of principal eigenvectors. We now pass from one eigenvalue to queries about its principal eigenvectors. The input remains (46)–(47). For the binary coefficient space, the transpose exchanges the two groups of variables in a matrix kernel. Proposition 4.5 therefore gives strictly disk-stable right eigenvectors of both $A = A_0 - 2\mu E$ and $A^\top$. Normalize them by

$$v_\emptyset = w_\emptyset = 1.$$

The transpose, rather than the adjoint, makes the dependence on a complex parameter holomorphic. A *stable product test* is $B = \bigotimes_{i=1}^n B_i$, where $B_i \in M_2(\mathbb{C})$, $(B_i)_{00} = 1$, and

$$K_{B_i}(z, w) = (B_i)_{00} + (B_i)_{10}z + (B_i)_{01}w + (B_i)_{11}zw$$

does not vanish in $\mathbb{D}^2$. The matrices B_i need not be Hermitian, positive, or invertible.

Theorem 5.11 (Classical product tests of principal states). *Fix d, J, μ as in Theorem 5.1, and suppose that A_0 satisfies that theorem's input assumptions. Given a stable product test with Gaussian-rational entries, both*

$$G_B = w^\top B v, \quad \text{Tr}(B\Pi) = \frac{w^\top B v}{w^\top v}, \quad \Pi = \frac{v w^\top}{w^\top v},$$

are nonzero. For every rational $0 < \epsilon < 1$, a deterministic classical algorithm returns Gaussian-rational estimates of these two quantities with relative complex error at most ϵ . Its bit complexity is

$$\text{poly}_{d,J,\mu}(n, \epsilon^{-1}, L_{\text{in}}),$$

where L_{in} includes the local generator, the test matrices, and the accuracy. Relative complex error means $|\hat{z}/z - 1| \leq \epsilon$.

This is query access to specified scalar tests, not a listing of an exponentially long vector. No polynomial bound on $\|\Pi\|$ is assumed or obtained. The proof instead constructs a nonvanishing scalar function with a controlled logarithm in the reciprocal-field parameter.

5.8. A uniform domain for normalized eigenvectors. Write $V = -A_0$, $H(x) = 2E + xV$, and set

$$r_0 = \frac{2^{-18}}{(d+1)J}, \quad \rho_s = \frac{r_0}{8}. \quad (70)$$

The following use of the Kirkwood–Thomas expansion concerns its creation coefficients, not just the energy estimate of Proposition 5.2. For a nonempty set $M \subseteq [n]$, put

$$a_M^\dagger = \prod_{i \in M} |1\rangle\langle 0|_i.$$

These operators commute, and products with overlapping supports vanish. The vacuum-normalized eigenvector germ has a unique expression

$$v(x) = \exp \left(- \sum_{\emptyset \neq M \subseteq [n]} C(M; x) a_M^\dagger \right) \Omega, \quad C(M; x) = \sum_{k \geq 1} C_k(M) x^k.$$

Uniqueness follows by taking the finite logarithm in the nilpotent commutative algebra generated by the $a_i^\dagger$.

Lemma 5.12 (Creation coefficients and a zero-free vacuum disk). *For every local complex V with the bounds above,*

$$\max_i \sum_{M \ni i} |C_k(M)| \leq 2^{-15} r_0^{-k}, \quad C_k(M) = 0 \quad \text{if } |M| > k+1. \quad (71)$$

The normalized right eigenvector germs of $H(x)$ and $H(x)^\top$ extend holomorphically to $|x| < \rho_s$. Their coefficient polynomials are nonzero on the closed polydisk $\{|z_i| \leq 2\}$ throughout this parameter disk.

Proof. For Hermitian perturbations with local norm bound L on a graph of maximum degree D , Lemma 5, equation (32), and Lemma 7, equation (47), of [12], with unperturbed gap 2, give

$$\max_i \sum_{M \ni i} |C_k(M)| \leq 2^{-15} R_L^{-k}, \quad R_L = 2^{-16}/(DL),$$

and say that a nonzero $C_k(M)$ is supported inside a connected set of at most $k+1$ vertices. In particular $|M| \leq k+1$. Use the private auxiliary qubits in (57) for one-site terms, with $D = d+1$. The normalized eigenvector germ of the augmented operator is exactly the physical germ tensored with the auxiliary vacuum. Its creation coefficients involving an auxiliary vertex vanish; all other coefficients agree with the physical ones.

The recursion (54)–(55) and the finite logarithm show that $C_k(M)$ is a homogeneous polynomial of degree k in the local entries of V , without their conjugates. With W_θ from the proof of Lemma 5.3,

$$C_k(M; V) = \frac{2^k}{2\pi} \int_0^{2\pi} e^{ik\theta} C_k(M; W_\theta) d\theta.$$

Taking absolute values, summing over $M \ni i$, and applying the Hermitian estimate with $L = 2J$ proves (71). The support conclusion is inherited by the same coefficient identity. Transposition gives the identical estimates for the left germ. For $|x| < r_0/8$, it follows that

$$\max_i \sum_{M \ni i} 4^{|M|} |C(M; x)| \leq 2^{-15} \sum_{k \geq 1} 4^{k+1} (|x|/r_0)^k < 2^{-13}. \quad (72)$$

All these series converge locally uniformly.

We give the elementary polymer argument that turns this estimate into zero-freeness. For weights u_M on nonempty subsets of a finite set, define

$$Z(S) = \sum_{\substack{\mathcal{A} \text{ a collection of} \\ \text{pairwise disjoint nonempty subsets of } S}} \prod_{M \in \mathcal{A}} u_M.$$

Suppose that $\sup_i \sum_{M \ni i} |u_M| 2^{|M|-1} \leq 1/2$. Induction on $|S|$ proves

$$Z(S) \neq 0, \quad \left| \frac{Z(S)}{Z(S \setminus \{i\})} - 1 \right| \leq \frac{1}{2} \quad (i \in S). \quad (73)$$

Indeed,

$$Z(S) = Z(S \setminus \{i\}) + \sum_{\substack{M \subseteq S \\ i \in M}} u_M Z(S \setminus M).$$

By the induction hypothesis, deleting the $|M| - 1$ additional vertices gives $|Z(S \setminus M)/Z(S \setminus \{i\})| \leq 2^{|M|-1}$. The claimed ratio estimate follows, and its disk does not contain zero. The empty set starts the induction.

The polynomial of $v(x)$ is this polymer sum with $u_M = -C(M; x)z^M$. The factorials in the exponential cancel the orderings of pairwise disjoint sets. For $|z_i| \leq 2$, the left side of the polymer condition is at most half the left side of (72), hence is less than $1/2$. This proves the asserted zero-freeness.

The convergent creation series and its finite exponential give holomorphic vectors on $|x| < \rho_s$. Their formal eigenvalue is the convergent energy series of Lemma 5.3. The eigenvector equations hold near zero and then throughout the disk by the identity theorem. No assertion of simplicity on the entire disk is needed. $\square$

Lemma 5.13 (Nonvanishing principal tests). *The vectors in Lemma 5.12 and the normalized principal vectors of $A(1/x)$ glue to holomorphic vectors $v(x), w(x)$ on*

$$\Omega_{\rho_s} = \{\Re x > 0\} \cup \{|x| < \rho_s\}.$$

For every stable product test,

$$G_B(x) = w(x)^\top Bv(x) \quad (74)$$

is nonvanishing on this domain, $G_B(0) = 1$, and

$$|G_B(x)| \leq 4^n, \quad \Re F_B(x) \leq n \log 4, \quad F_B = \log G_B, \quad F_B(0) = 0. \quad (75)$$

Proof. The half-plane argument in Lemma 5.5 applies to the principal eigenvectors as well. Local analytic eigenvectors exist for a simple eigenvalue; their vacuum coordinates are nonzero because their coefficient polynomials are strictly stable. Vacuum normalization makes these local choices agree on overlaps. Use $A(h)^\top$ for the left vectors. On a sufficiently small positive real interval they coincide with the vacuum vectors, by the same Riesz-projection argument as in Lemma 5.5. That interval may depend on n ; it is used only to identify the branches, not to choose ρ_s . The identity theorem on the connected overlap gives the required gluing.

In the half-plane, $v(x)$ and $w(x)$ each have a strict polydisk radius greater than one. In the small disk they have radius at least two by Lemma 5.12. Contract their legs against the two legs of every K_{B_i} . The strict version of the disk contraction rule, equivalently polarized apolarity, applies because each contracted pair has product of radii greater than one; see Lemma 2.3. The resulting scalar is (74), hence is nonzero. This argument allows singular tests and needs no uniform lower bound for the strict radius.

For completeness, a multiaffine stable polynomial $f(z) = \sum_S f_S z^S$ with $f_\emptyset = 1$ satisfies $|f_S| \leq 1$. Set variables outside S to zero and all remaining variables to t . If $f_S \neq 0$, the resulting polynomial has degree $|S|$, constant term one, leading coefficient f_S , and all roots of modulus at least one. Their product proves the bound. The case $f_S = 0$ is immediate. Apply this observation to v, w and each K_{B_i} . The expansion of $w^\top Bv$ has 4^n terms of modulus at most one. Also $G_B(0) = \prod_i (B_i)_{00} = 1$. The domain is star-shaped about zero, so its nonvanishing holomorphic function has a unique logarithm vanishing there. Taking its real part gives (75). $\square$

Lemma 5.14 (Continuation with a one-sided logarithmic bound). *Use the parameters (62) with $\rho = \rho_s$ and $c = 1/\mu$. Write $F_B(x) = \sum_{j \geq 1} f_j x^j$ and $F_B(\phi(u)) =$*

$\sum_{k \geq 1} g_k u^k$. The transform (63) remains valid, and

$$|g_k| \leq 4n, \quad \left| F_B(c) - \sum_{k=1}^p g_k q^k \right| \leq \frac{4nq^{p+1}}{1-q}. \quad (76)$$

In particular $|F_B(c)| \leq Kn$, where $K = 4q/(1-q)$.

Proof. The map lemma puts $\phi(\mathbb{D})$ in Ω_{ρ_s} . The holomorphic function $U(u) = 2n - F_B(\phi(u))$ has nonnegative real part and $U(0) = 2n$. On a circle of radius $r < 1$, the positive function $\Re U(re^{it})$ has average $2n$. Its k -th Fourier coefficient, for $k \geq 1$, equals $-g_k r^k/2$, so $|g_k| r^k \leq 4n$. Let $r \uparrow 1$ and sum the geometric tail. The coefficient transform is a formal identity already proved in Lemma 5.6. $\square$

5.9. Connected coefficients and finite bit complexity. We compute the coefficients of F_B without expanding the full eigenvectors. This is an instance of the connected-logarithm method for multiplicative analytic graph functions; compare [25, 42]. The needed multiplicativity is formal at the vacuum and does not require a preservation promise on induced subgraphs.

For $U \subseteq [n]$, keep just the local terms supported inside U , obtaining V_U , and let $B_U = \bigotimes_{i \in U} B_i$. Define the vacuum-normalized formal vectors v_U, w_U for $2E_U + xV_U$ and its transpose. Put

$$F_{B,U}(x) = \log(w_U(x)^T B_U v_U(x)), \quad \Phi_{B,U}(x) = \sum_{W \subseteq U} (-1)^{|U|-|W|} F_{B,W}(x).$$

For the empty subsystem all logarithms are zero.

Lemma 5.15 (Local computation of test logarithms). *For $k \geq 1$,*

$$[x^k] F_B(x) = \sum_{\substack{U \subseteq [n] \text{ connected} \\ |U| \leq k+1}} [x^k] \Phi_{B,U}(x).$$

For Gaussian-rational input, the first p coefficients can be computed exactly with $n \exp(O_d(p)) \text{poly}(L_{\text{in}}, n, p)$ bit operations.

Proof. If the induced graph on U splits into components, the normalized formal vectors tensor across components and the test does likewise. Thus $F_{B,U}$ is the sum of the component logarithms. Subset inversion shows that $\Phi_{B,U} = 0$ when U is disconnected.

To obtain the order bound as well, give each local perturbation term its own parameter. An order- k coefficient is a homogeneous polynomial of degree k in these parameters, by the unique formal eigenvector recursion. If a parameter monomial has disconnected union of supports, set all unused parameters to zero. Tensor factorization makes the logarithm additive across those components, so that monomial has coefficient zero. A connected union of at most k one- or two-site supports has at most $k+1$ vertices. Finally the coefficient of a monomial with support union T is unchanged in every induced subsystem containing T . The alternating sum over $W \subseteq U$ retains it only if $T = U$. This proves the formula, including both vanishings.

Here is a direct finite-input implementation. In each subsystem of size $m \leq p+1$, use the exact recursions (54)–(55), and their transposes, through order p . If

$$a_k = \sum_{i+j=k} w_{U,i}^\top B_U v_{U,j}, \quad \log \left(1 + \sum_{k \geq 1} a_k x^k \right) = \sum_{k \geq 1} f_k x^k,$$

then coefficient comparison after logarithmic differentiation gives

$$f_k = a_k - \frac{1}{k} \sum_{j=1}^{k-1} j f_j a_{k-j}. \quad (77)$$

Every vector has dimension at most 2^{p+1} . Even dense matrix arithmetic and enumeration of every $W \subseteq U$ cost only $\exp(O(p)) \text{poly}(p)$ operations per U . Connected subsets of size at most $p+1$ can be enumerated by traversals of their spanning trees, with deduplication, in $n \exp(O_d(p))$ time; a traversal has length at most $2p$. The case $d = 0$ consists just of singletons.

Appendix A.2 bounds all denominators and intermediate numerators by polynomial bit lengths, proving the asserted bit complexity. $\square$

Proof of Theorem 5.11. Use $c = 1/\mu$, ρ_s from (70), and the rational map parameters of Lemma 5.6. Given $0 < \tau < 1$, choose the least $p \geq 0$ for which

$$4nq^{p+1} \leq \tau(1 - q).$$

Rational multiplication and comparison find p ; (67) gives $p = O_{d,J,\mu}(\log(2 + n/\tau))$. Use Lemma 5.15, the triangular transform, and (76) to compute a Gaussian rational S_B with $|S_B - F_B(c)| \leq \tau$. Also $|S_B| \leq Kn + 1$. The preceding lemma and the rational transform give polynomial bit cost.

We make explicit that exponentiation does not introduce an unbounded-precision oracle. Let T_0 be an integer upper bound on $Kn + 1$, and choose an integer ℓ with $2^{-\ell} \leq \tau$. Truncate the Taylor series of e^{S_B} at degree

$$M = \lceil 10(2T_0 + \ell + 1) \rceil$$

by taking $\sum_{j=0}^M S_B^j / j!$. The remainder is at most $e^{T_0} T_0^{M+1} / (M+1)!$. Using $r! \geq (r/3)^r$, it is less than τe^{-T_0} . Since $|e^{S_B}| \geq e^{-T_0}$, this is relative error less than τ . All terms are Gaussian rational, $M = O_{d,J,\mu}(n + \log \tau^{-1})$, and the numerators and denominators have polynomial bit length. Furthermore $|e^{S_B} / G_B(c) - 1| \leq e^\tau - 1$. Choosing $\tau = \epsilon/32$ supplies the required margin. Repeat for $B = I$, and divide the two nonzero rational estimates to obtain the estimate of G_B/G_I . For errors at most $\epsilon/8$ in each factor, the ratio error is at most $(\epsilon/4)/(1 - \epsilon/8) < \epsilon$.

The nonvanishing assertions follow from Lemma 5.13; I is a stable product test because its local kernel is $1 + zw$. For a simple eigenvalue the spectral projection is $vw^\top / (w^\top v)$, so this quotient is indeed $\text{Tr}(B\Pi)$. $\square$

5.10. Product-state queries and equatorial measurement sampling. For $s \in \mathbb{C}$ with $|s| \leq 1$, define

$$|s\rangle = \frac{|0\rangle + s|1\rangle}{\sqrt{1 + |s|^2}}.$$

These are the pure states in the closed northern hemisphere of the Bloch sphere in our fixed basis. The amplitude and probability queries extend to this whole hemisphere. Complete two-outcome projective measurement sampling is stated on

the equator, where the two states $|s\rangle, |-s\rangle$ are orthogonal and both satisfy the condition.

Theorem 5.16 (Classical access to Suzuki–Fisher ground states). *Fix d, J, μ as in Theorem 5.7. Let H be a rational Suzuki–Fisher Hamiltonian on $n \geq 1$ qubits satisfying that theorem’s hypotheses, and let $0 < \epsilon < 1$ be rational. Its unique unit ground vector ψ can be normalized by $\langle 0^n | \psi \rangle > 0$. The query algorithms in (1)–(2) have bit complexity $\text{poly}_{d,J,\mu}(n, \epsilon^{-1}, L_{\text{in}})$.*

- (1) *Given $s_1, \dots, s_n \in \mathbb{Q}(i)$ with $|s_i| \leq 1$, approximate the nonzero complex amplitude $\langle s_1 \cdots s_n | \psi \rangle$ with relative complex error at most ϵ .*
- (2) *Given $S \subseteq [n]$ and $s_i \in \mathbb{Q}(i)$, $|s_i| \leq 1$ for $i \in S$, approximate the positive probability*

$$\left\langle \psi \left| \bigotimes_{i \in S} |s_i\rangle \langle s_i| \otimes I_{S^c} \right| \psi \right\rangle$$

by a positive rational with relative error at most ϵ .

- (3) *Given a polynomial-time adaptive strategy that measures each site at most once, in a basis $\{|s\rangle, |-s\rangle\}$ with $s \in \mathbb{Q}(i)$, $|s| = 1$, sample its outcome record by a classical randomized algorithm with total variation error at most ϵ .*

The strategy in (3) has a finite description, takes previous outcomes as input, and outputs the next unmeasured site and a phase. More precisely, if its total evaluation time along any record is at most T_{strategy} , and all generated phases have bit length at most L_{phase} , the sampler uses at most

$$T_{\text{strategy}} + \text{poly}_{d,J,\mu}(n, \epsilon^{-1}, L_{\text{in}} + L_{\text{phase}})$$

bit operations. In particular the sampler is polynomial-time for each fixed polynomial-time strategy.

Proof. Use $A_0 = -(H + \mu \sum_i Z_i)$ as in (68). Its local norm bound is a fixed rational J' depending on the Pauli bound J and on μ ; all radii in the preceding lemmas are evaluated with this bound, not with an unadjusted J . At positive real x , $H(x)$ is Hermitian, so its vacuum-normalized transpose eigenvector is $w = \bar{v}$. Consequently $G_I = \|v\|^2 > 0$ and $\psi = v/\sqrt{G_I}$. Along the positive real path the normalized logarithm F_I is the real logarithm, because it starts at zero and G_I stays positive.

For amplitudes take

$$B_i^{\text{amp}} = \begin{pmatrix} 1 & \bar{s}_i \\ 0 & 0 \end{pmatrix}.$$

Its kernel $1 + \bar{s}_i w$ is disk-stable, and

$$\langle s_1 \cdots s_n | \psi \rangle = \left(\prod_i (1 + |s_i|^2)^{-1/2} \right) \exp(F_{B^{\text{amp}}}(c) - F_I(c)/2).$$

For the event in (2), take $B_i = I$ off S and

$$B_i^{\text{prob}} = \begin{pmatrix} 1 & \bar{s}_i \\ s_i & |s_i|^2 \end{pmatrix} = (1 + |s_i|^2) |s_i\rangle \langle s_i| \quad (i \in S).$$

Its kernel factors as $(1 + s_i z)(1 + \bar{s}_i w)$. Thus

$$P_S = \left(\prod_{i \in S} (1 + |s_i|^2)^{-1} \right) \exp(F_B(c) - F_I(c)).$$

Nonvanishing follows from Lemma 5.13; positivity of the projector expectation then gives $P_S > 0$.

Compute the logarithms with a fixed fraction of the requested error, using the constructive proof of Theorem 5.11. For (2), take the real part of the logarithmic difference before exponentiating; its error does not increase because the exact difference is real on the positive path. For either logarithmic difference use $T_0 = \lceil 2Kn + 2 \rceil$ in that finite Taylor procedure. In case (2), the resulting rational estimate is real and positive when its relative error is less than one. The probability normalization is rational and exact. For (1), the product $R_s = \prod_i (1 + |s_i|^2)$ is a positive rational between 1 and 2^n . Rational bisection for $\sqrt{R_s}$, followed by inversion, computes its reciprocal square root to any required relative error with $O(n + \log \epsilon^{-1} + L_{\text{in}})$ bits and polynomial arithmetic cost. Splitting the error budget between the two logarithms, exponentiation and this factor proves (1) and (2), including the phase assertion.

For (3), fix an already generated history. Its two extensions have true probabilities $p_+, p_- > 0$. Part (2) computes positive estimates $\hat{p}_\pm$ with relative error at most $\eta = \epsilon/(16n)$. Normalizing their sum gives a conditional probability with error at most $2\eta/(1 - \eta) < 4\eta$, regardless of the probability of the parent history. Round it to a dyadic probability within $\epsilon/(4n)$ and draw the corresponding Bernoulli variable using $O(\log(n/\epsilon))$ unbiased random bits. On a fixed history, projectors on distinct sites commute, so its probability is exactly the product event in (2), even though the bases were chosen adaptively. Coupling the two processes at each step bounds their total variation distance by the sum of the at most n conditional errors, which is less than ϵ . At most $2n$ queries are made. Their accuracy and phase bit lengths give the displayed polynomial overhead; evaluating the strategy contributes T_{strategy} . $\square$

The novelty of this conclusion is its uniform local input and its fixed-parameter polynomial complexity on the full positive-field Suzuki–Fisher family. For general Lee–Yang states with a fixed spatial radius $r > 1$, [41, Section 4, Theorems 5–6] already provides state-dependent quasipolynomial circuits for relative X -basis amplitudes, with marginal probabilities and phases in the state-preparation proof; see also [40]. Our proof does not assume a size-independent spatial radius margin. It uses the common reciprocal-field domain instead. Connected coefficient computation and zero-free interpolation are established methods [25, 42, 38]; the new input to those methods is Lemma 5.13 for principal-state tests. In particular [42, Theorem 15] already treats tensorized measurements of thermal partition functions. Its thermal zero-free hypothesis does not itself supply the ground-state parameter domain used here.

The queries above do not include arbitrary product bases: the projector onto $|1\rangle$ has no nonzero vacuum entry and is not an allowed normalized test. Nor does the fixed-field theorem extend the zero-field energy-value scheme to state approximation or sampling. The broad quantum algorithms in [14, 30] have different parameter ranges and outputs; the present classical result does not subsume them.

6. ALL-TEMPERATURE LEE–YANG HAMILTONIANS

We now specialize the complex generator classification to Hermitian operators on qubits. The resulting inequalities are precisely the Suzuki–Fisher conditions, including their full transverse operator-norm form. Their sufficiency is classical;

the generator theorem supplies necessity and excludes additional interactions of arbitrary support. Positive longitudinal fields then improve the zero-free domain of the actual Gibbs tensor, at every positive temperature.

6.1. From matrix tensors to polynomial generators. Fix the computational basis $\{|a\rangle : a \in \{0, 1\}^n\}$, with $Z|0\rangle = |0\rangle$. Associate to a matrix M the polynomial

$$F_M(z, w) = \sum_{a, b \in \{0, 1\}^n} M_{ab} z^a w^b. \quad (78)$$

We say that M has the tensor Lee–Yang property at radius r if F_M is nonzero on $\mathbb{D}_r^n \times \mathbb{D}_r^n$. The two groups of variables in this definition are independent complex variables. In particular, (78) is not the sesquilinear form $\langle z | M | z \rangle$.

For a Hermitian Hamiltonian H , the Gibbs operator is $M_\beta = e^{-\beta H}$, and its normalized density matrix is $\rho_\beta = M_\beta / \text{Tr } M_\beta$. Since $\text{Tr } e^{-\beta H} > 0$ for real β , the two matrices have exactly the same polynomial zeros.

The following bridge fixes both the tensor and operator conventions. It is the binary instance, with a unitary scalar normalization, of the Cayley correspondence (8).

Lemma 6.1. *Let*

$$C_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} i & -i \\ 1 & 1 \end{pmatrix}, \quad C = C_1^{\otimes n}, \quad D_1 = \text{diag}(-1, 1).$$

For an invertible matrix M , its tensor has the unit-disk Lee–Yang property if and only if CMC^{-1} , acting on multiaffine coefficient vectors, preserves upper-half-plane stability. Consequently, the Gibbs family corresponds to the generator

$$A = -CHC^*. \quad (79)$$

Here operator conjugation is a similarity, whereas transforming both groups of tensor variables gives the congruence CMC^T .

Proof. The coefficient action of C is

$$(Cf)(s) = 2^{-n/2} \prod_i (s_i + i) f\left(\frac{s_1 - i}{s_1 + i}, \dots, \frac{s_n - i}{s_n + i}\right).$$

The fractions map the upper half-plane onto the unit disk, and the prefactors are nonzero there. Thus C identifies the two stability classes. By the binary form of the contraction theorem, or Lemma 4.1, the disk-stable tensor of an invertible M is equivalent to preservation of disk-stable coefficient vectors. This proves the equivalence by similarity.

For completeness, the reverse tensor conversion can also be checked directly. The unitary matrix C_1 satisfies $C_1 C_1^T = D_1$, so the coefficient matrix of

$$\prod_i (s_i t_i - 1)$$

is $D_1^{\otimes n} = CC^T$. Each factor is stable in the two upper-half-plane variables: if $s_i t_i = 1$, then $t_i = 1/s_i$ has negative imaginary part. Acting on its first group of variables by $N = CMC^{-1}$ gives coefficient matrix $NCC^T = CMC^T$. For

fixed $t \in \mathbb{H}^n$, invertibility excludes a zero output, so stability preservation gives nonvanishing jointly in $(s, t) \in \mathbb{H}^{2n}$. Finally,

$$F_{CMC^T}(s, t) = 2^{-n} \prod_i (s_i + i)(t_i + i) F_M\left(\frac{s-i}{s+i}, \frac{t-i}{t+i}\right).$$

This verifies the congruence and the sign of the half-plane kernel. As $C^{-1} = C^*$, exponential similarity yields $Ce^{-\beta H}C^{-1} = e^{\beta A}$. $\square$

6.2. The exact Hamiltonian classification. Write X, Y, Z for the Pauli matrices, with $Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$. For a real 2×2 matrix, $\|\cdot\|_{\text{op}}$ denotes its largest singular value.

Theorem 6.2 (All-temperature qubit classification). *For a Hermitian operator H on $(\mathbb{C}^2)^{\otimes n}$, the following conditions are equivalent:*

- (1) $e^{-\beta H}$ has the tensor Lee–Yang property at radius one for every $\beta \geq 0$.
- (2) The same property holds for every β in some interval $[0, \varepsilon)$, where $\varepsilon > 0$.
- (3) There are real coefficients such that

$$\begin{aligned} H = & h_0 I + \sum_i (h_i^x X_i + h_i^y Y_i + h_i^z Z_i) \\ & + \sum_{i < j} (a_{ij} X_i X_j + b_{ij} X_i Y_j + c_{ij} Y_i X_j + d_{ij} Y_i Y_j + e_{ij} Z_i Z_j), \end{aligned} \quad (80)$$

where

$$h_i^z \leq 0, \quad e_{ij} \leq -\left\| \begin{pmatrix} a_{ij} & b_{ij} \\ c_{ij} & d_{ij} \end{pmatrix} \right\|_{\text{op}}. \quad (81)$$

There is no restriction on h_0, h_i^x, h_i^y .

Equivalently, in the ferromagnetic convention

$$H = h_0 I - \sum_i \sum_{a=x,y,z} \mu_i^a \sigma_i^a - \sum_{i < j} \left(J_{ij}^{zz} Z_i Z_j + \sum_{a,b=x,y} J_{ij}^{ab} \sigma_i^a \sigma_j^b \right), \quad (82)$$

the conditions are $\mu_i^z \geq 0$ and $J_{ij}^{zz} \geq \|J_{ij}^\perp\|_{\text{op}}$, where the rows and columns of $J_{ij}^\perp$ are ordered x, y .

Proof. The first condition implies the second. By Lemma 6.1, the second says that e^{tA} preserves upper-half-plane stability for small $t \geq 0$. The semigroup identity extends preservation to all $t \geq 0$: for any t , choose an integer m with $t/m < \varepsilon$ and compose $e^{(t/m)A}$ m times. We may therefore apply Theorem 3.2 on the binary capacity box.

We first spell out how differential order becomes physical support. On one binary site a basis of endomorphisms is

$$I, \quad J^- = \partial_s, \quad J^0 = s\partial_s - \frac{1}{2}, \quad J^+ = s^2\partial_s - s.$$

The last three operators are traceless; their first-order coefficients are the linearly independent polynomials $1, s, s^2$. Expand A in tensor products of this basis and choose a support set S of maximal cardinality occurring with a nonzero coefficient. The canonical coefficient of $\partial_S = \prod_{i \in S} \partial_i$ is then the sum of its exact-support terms, because no strictly larger support occurs. Those terms have distinct principal monomials $\prod_{i \in S} s_i^{r_i}$, with $r_i \in \{0, 1, 2\}$. The order-two conclusion of Theorem 3.2 forces this coefficient to vanish if $|S| \geq 3$. Linear independence then eliminates

every term of that support. Descending in support eliminates all interactions on three or more sites. Local similarity preserves the identity and traceless subspaces at each site, so the same support conclusion holds for H .

Next write $A = R + iS$ with entrywise real matrices R, S . The imaginary-part conclusion of Theorem 3.2 has the form

$$S = c_0 I + \sum_i J_i(q_i), \quad q_i(s) = \alpha_i + \eta_i s + \gamma_i s^2 \geq 0 \quad (s \in \mathbb{R}), \quad (83)$$

where $J(q) = q\partial - q'/2$. On coefficients $(1, s)$,

$$J(q) = \begin{pmatrix} -\eta/2 & \alpha \\ -\gamma & \eta/2 \end{pmatrix}.$$

Since A is Hermitian, $R^T = R$ and $S^T = -S$. Taking the trace in (83) first gives $c_0 = 0$. Independence of the remaining one-site traceless components shows that each $J(q_i)$ is skew-symmetric. Thus $\eta_i = 0$ and $\gamma_i = \alpha_i$; nonnegativity gives $\alpha_i \geq 0$. In particular,

$$q_i(s) = \alpha_i(1 + s^2), \quad iS = -\sum_i \alpha_i Y_i. \quad (84)$$

The local Pauli conjugations are

$$C_1 X C_1^* = -Z, \quad C_1 Y C_1^* = X, \quad C_1 Z C_1^* = -Y. \quad (85)$$

Therefore the physical field $h_i^x X_i + h_i^y Y_i + h_i^z Z_i$ contributes $h_i^x Z_i - h_i^y X_i + h_i^z Y_i$ to A . Comparison with (84) gives $h_i^z = -\alpha_i \leq 0$. A two-site physical Pauli term containing exactly one Z becomes a term containing exactly one Y , and is entrywise imaginary. The four such terms on each pair are linearly independent. Since S has no two-site component, all physical XZ, YZ, ZX, ZY coefficients vanish. This leaves exactly the form (80).

It remains to determine the pair inequality. Suppress the pair indices. A physical term $aXX + bXY + cYX + dYY + eZZ$ contributes

$$-aZZ + bZX + cXZ - dXX - eYY$$

to A . The first-order coefficients of the coefficient operators X, Z, Y , respectively, are

$$p_X(s) = 1 - s^2, \quad p_Z(s) = -2s, \quad p_Y(s) = -i(1 + s^2).$$

Hence the full coefficient of $\partial_i \partial_j$ is

$$q_{ij}(s, t) = e(1 + s^2)(1 + t^2) - 4ast - 2bs(1 - t^2) - 2ct(1 - s^2) - d(1 - s^2)(1 - t^2). \quad (86)$$

There is no factor of two in this expression: it is the full mixed coefficient in Theorem 3.2. Define

$$u(s) = \left(\frac{2s}{1 + s^2}, \frac{1 - s^2}{1 + s^2} \right), \quad J = \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

Then

$$\frac{q_{ij}(s, t)}{(1 + s^2)(1 + t^2)} = e - u(s)^T J u(t). \quad (87)$$

The set $u(\mathbb{R})$ is dense in the unit circle. Consequently the required inequality $q_{ij}(s, t) \leq 0$ for all real s, t is equivalent to

$$e \leq \min_{\|u\|_2 = \|v\|_2 = 1} u^T J v = -\|J\|_{\text{op}}.$$

This proves necessity of (81).

Conversely, suppose (80)–(81) hold. The displayed conjugations show that A has canonical differential order at most two. Its mixed coefficients are real and nonpositive by (87), and its imaginary part is (83) with $q_i(s) = -h_i^z(1 + s^2) \geq 0$. Its real part therefore satisfies the real criterion in Theorem 3.2, and its imaginary part satisfies that theorem’s one-site condition. The theorem proves preservation for all $\beta \geq 0$; the bridge then gives the tensor Lee–Yang property of $e^{-\beta H}$. Finally, changing the signs of the coefficients converts (81) into (82). $\square$

Remark 6.3. The sufficient Suzuki–Fisher family originates in [35, 4, 3]. The full transverse operator-norm condition is already contained, in equivalent algebraic form, in [35, equation (2.47)] and [20, Theorem 2, equation (8)]; see also the modern notation in [14, Definition 1 and Fact 3]. Indeed, for a real matrix $J = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$,

$$2\|J\|_{\text{op}} = \sqrt{(a-d)^2 + (b+c)^2} + \sqrt{(a+d)^2 + (b-c)^2}.$$

To verify the identity, the squared radicals are $\|J\|_F^2 - 2\det J$ and $\|J\|_F^2 + 2\det J$. If the singular values are $s_1 \geq s_2 \geq 0$, the two radicals are therefore $s_1 + s_2$ and $s_1 - s_2$, in some order. Full matrix generating functions and their composition are also explicit in Asano’s contemporaneous account for the XXZ subfamily [4, equations (1.15), (3.7)–(3.10)]. Thus a distinction between classical traces and modern full tensors alone would not identify the contribution. Theorem 6.2 adds the converse among all finite Hermitian qubit Hamiltonians, including the exclusion of higher-support interactions. The proof also shows that testing only a sufficiently small interval of inverse temperatures is enough.

Corollary 6.4. *Let H be an arbitrary complex qubit matrix. If $e^{-\beta H}$ has the unit-disk tensor Lee–Yang property for all sufficiently small real $\beta \geq 0$, then its Pauli expansion contains no term supported on three or more sites.*

Proof. The Cayley bridge and the semigroup identity still apply. The complex generator theorem bounds the canonical order by two. The maximal-support basis argument in the proof of Theorem 6.2 does not use Hermiticity and therefore gives the assertion. $\square$

Remark 6.5. For a complex Hamiltonian, tensor stability does not justify trace normalization. For example, $H = \text{diag}(0, i\pi)$ has Gibbs polynomial $1 + e^{-i\pi\beta}zw$, which is nonzero on the open unit bidisk for every real β , but $\text{Tr } e^{-H} = 0$. The support conclusion in Corollary 6.4 applies to this complex setting; the normalized Gibbs-state statements use Hermiticity.

6.3. Strict positive fields and the actual Gibbs kernel.

Theorem 6.6 (Strict-field Gibbs radius). *Let H have the form (82), with $J_{ij}^{zz} \geq \|J_{ij}^\perp\|_{\text{op}}$ and $\mu_i^z > 0$ at every site. Then, for every $\beta > 0$, $F_{e^{-\beta H}}$ is nonzero on the closed unit polydisk in its full $2n$ variables. In particular, it has some radius $r_\beta > 1$.*

The ground eigenspace of H is one-dimensional. Its nonzero coefficient vector ψ defines a polynomial $f_\psi(z) = \sum_a \psi_a z^a$ of radius greater than one. For every $\beta_0 > 0$, one radius $r > 1$ works for all Gibbs kernels at $\beta \geq \beta_0$, and also for the image under each $e^{-\beta H}$ of every nonzero unit-disk-stable coefficient vector.

Proof. Let $\mu = \min_i \mu_i^z > 0$, and write $H = H_0 - \mu \sum_i Z_i$. The Hamiltonian H_0 satisfies the nonstrict Suzuki–Fisher conditions. By Theorem 6.2 and the disk

version of Lemma 6.1, its coefficient operator $A_0 = -H_0$ generates a disk-stability preserver. In the multiaffine coefficient basis, $Z_i = I - 2E_i$. Thus

$$-H = A_0 - 2\mu E + \mu n I.$$

The harmless scalar factor $e^{\beta\mu n}$ reduces every kernel and evolution assertion to Theorem 4.3.

Proposition 4.5 supplies an algebraically simple dominant eigenvalue of $-H$. Since H is Hermitian, this is exactly the negative of its ground energy, so the ground space is one-dimensional. The same proposition gives the strict radius of its right eigenvector. Equivalently, one may fix a Gibbs kernel radius $r > 1$ and normalize its powers by their largest eigenvalue. These powers retain radius r by contraction, and converge to $|\psi\rangle\langle\psi|$. Their nonzero limiting polynomial is

$$f_\psi(z)\overline{f_\psi(\bar{w})}.$$

Hurwitz gives nonvanishing on $\mathbb{D}_r^{2n}$, and therefore nonvanishing of each factor on $\mathbb{D}_r^n$. $\square$

Remark 6.7. The spectral conclusion

$$E_1(H) - E_0(H) \geq 2 \min_i \mu_i^z$$

is the Hermitian specialization of Theorem 4.8 and was already proved for this family in [14, Theorem 1]. Here the additional conclusion is strict nonvanishing of the actual Gibbs tensor at every positive inverse temperature, and its transfer to the ground-state polynomial. No numerical lower bound for $r_\beta - 1$, uniform in system size, is asserted. This distinction also separates the actual kernel from the strict kernels used in a product-formula approximation.

6.4. A qualitative radius for deformed EPR ground states. Consider a finite graph without isolated vertices, with positive edge weights w_e . On an edge $e = \{i, j\}$, fix $0 \leq s_e < 1$ and $\theta_e \in \mathbb{R}$, and define

$$|\eta_e\rangle = |00\rangle + s_e e^{i\theta_e} |11\rangle, \quad H_{\text{EPR}} = - \sum_e w_e |\eta_e\rangle\langle\eta_e|_{ij}.$$

The edge vectors here are unnormalized; replacing them by normalized vectors simply rescales the positive weights.

Corollary 6.8. *The ground space of H_{EPR} is one-dimensional, and its ground-state polynomial is zero-free on a polydisk of some radius $r > 1$.*

Proof. On one edge, direct expansion in Pauli matrices gives

$$\begin{aligned} |\eta_e\rangle\langle\eta_e| &= \frac{1+s_e^2}{4}(I + Z_i Z_j) + \frac{1-s_e^2}{4}(Z_i + Z_j) \\ &\quad + \frac{s_e}{2} [\cos \theta_e (X_i X_j - Y_i Y_j) + \sin \theta_e (X_i Y_j + Y_i X_j)]. \end{aligned}$$

Its ferromagnetic coefficients are therefore

$$J_e^{zz} = \frac{w_e(1+s_e^2)}{4}, \quad J_e^\perp = \frac{w_e s_e}{2} \begin{pmatrix} \cos \theta_e & \sin \theta_e \\ \sin \theta_e & -\cos \theta_e \end{pmatrix}.$$

The displayed real matrix is orthogonal, so $\|J_e^\perp\|_{\text{op}} = w_e s_e/2$ and $J_e^{zz} - \|J_e^\perp\|_{\text{op}} = w_e(1 - s_e)^2/4 \geq 0$. The longitudinal field at site i is

$$\mu_i^z = \sum_{e \ni i} \frac{w_e(1 - s_e^2)}{4} > 0.$$

All hypotheses of Theorem 6.6 follow, including strictness because every vertex has an incident edge. Apply that theorem to obtain the stated uniqueness and radius. $\square$

For a common deformation parameter $s < 1$, the corollary gives the qualitative strict-radius conclusion asked for in [41, Conjecture 1]. The radius may depend on the graph and its parameters. Edge phases may be chosen independently; no simultaneous gauge removal is used in the proof.

For the unweighted common-parameter model, the field calculation above and [14, Theorem 1] give

$$\Delta(H_s) \geq \frac{d_{\min}(G)}{2}(1 - s^2), \quad d_{\min}(G) = \min_i \deg_G(i),$$

where Δ denotes the gap between the two lowest eigenvalues. Thus the lower bound proposed in [41, Conjecture 3] already follows when $d_{\min}(G) \geq 2$. For graphs with leaves this estimate gives $(1 - s^2)/2$; attaining the conjectured lower bound requires additional information. For stars with at least three vertices, equality holds in the conjectured bound by [41, Lemma 13]; the single-edge gap is $1 + s^2$ by its rank-one spectrum.

7. ALL-TEMPERATURE HAMILTONIANS AT HIGHER SPINS

The finite-box generator classification also determines the coherent Gibbs kernels at arbitrary finite local spins. We retain the binomial normalization throughout.

7.1. Coherent normalization and the Hamiltonian cone. At site i , fix a positive integer κ_i and put $j_i = \kappa_i/2$. Let $\mathcal{H}_{\kappa_i} = \mathbb{C}^{\kappa_i+1}$ have orthonormal basis $|a_i\rangle$, $0 \leq a_i \leq \kappa_i$. For multi-indices a set

$$b_a = \prod_i \binom{\kappa_i}{a_i}, \quad |a\rangle \longleftrightarrow \sqrt{b_a} z^a. \quad (88)$$

Thus the monomial inner product is $\langle z^a, z^b \rangle = \delta_{ab}/b_a$. All adjoints in this section refer to the physical orthonormal basis, or equivalently to this weighted polynomial inner product. The coherent kernel of a matrix M is

$$F_M(z, w) = \sum_{a,b} \sqrt{b_a b_b} M_{ab} z^a w^b. \quad (89)$$

In particular, $F_I(z, w) = \prod_i (1 + z_i w_i)^{\kappa_i}$. We call M coherently Lee–Yang if F_M has no zero when all its variables belong to $\mathbb{D}$. Both weights in (89) are essential.

The spin matrices have the following differential realization:

$$S_i^x = \frac{(1 - z_i^2)\partial_i + \kappa_i z_i}{2}, \quad S_i^y = \frac{(1 + z_i^2)\partial_i - \kappa_i z_i}{2i}, \quad S_i^z = \frac{\kappa_i}{2} - z_i \partial_i. \quad (90)$$

They satisfy $(S_i^x)^2 + (S_i^y)^2 + (S_i^z)^2 = j_i(j_i + 1)I$. Write $S_i^\perp = (S_i^x, S_i^y)^\top$.

Theorem 7.1 (Higher-spin Hamiltonians). *For a Hermitian operator H on $\bigotimes_i \mathcal{H}_{\kappa_i}$, the following conditions are equivalent:*

- (1) $e^{-\beta H}$ is coherently Lee–Yang for every $\beta \geq 0$;
- (2) this holds for every $0 \leq \beta < \varepsilon$, for some $\varepsilon > 0$;
- (3) H admits the expression

$$H = h_0 I + \sum_i (h_i^x S_i^x + h_i^y S_i^y + h_i^z S_i^z) + \sum_{i: \kappa_i \geq 2} (S_i^\perp)^\top B_i S_i^\perp + \sum_{i < j} ((S_i^\perp)^\top J_{ij} S_j^\perp + e_{ij} S_i^z S_j^z), \quad (91)$$

where all coefficients are real, $B_i = B_i^\top$, and

$$h_i^z \leq 0, \quad B_i \succeq 0, \quad e_{ij} \leq -\|J_{ij}\|_{\text{op}}. \quad (92)$$

The coefficient $(B_i)_{12}$ multiplies $S_i^x S_i^y + S_i^y S_i^x$. At a capacity-one site all single-site quadrupoles reduce to scalars or zero and are omitted from (91).

We give the normalization and coordinate computations in the proof, so that the finite-capacity infinitesimal theorem can be applied without changing the physical adjoint.

Lemma 7.2 (Polarization and the coherent Cayley transform). *Define the Dicke isometry*

$$W_i |a\rangle = \binom{\kappa_i}{a}^{-1/2} \sum_{|x|=a} |x\rangle, \quad W = \bigotimes_i W_i. \quad (93)$$

The binary matrix kernel of WMW^* is the normalized polarization of F_M , separately in each row and column block. Moreover, let

$$C_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} i & -i \\ 1 & 1 \end{pmatrix}, \quad C_i = W_i^* C_1^{\otimes \kappa_i} W_i, \quad C = \bigotimes_i C_i. \quad (94)$$

For Hermitian H , coherent stability of e^{-tH} for all sufficiently small $t \geq 0$ is equivalent to the condition that $\exp(-tCHC^*)$ is a complete upper-half-plane stability preserver for every $t \geq 0$.

Proof. At binary words of weights a, b , the matrix entry of WMW^* is $M_{ab}/\sqrt{b_a b_b}$. Identifying the variables in their respective blocks multiplies this coefficient by $b_a b_b$, giving exactly (89). The disk polarization theorem therefore identifies stability of the two kernels; see [8, Theorem 2.1]. Polarization, binary tensor contraction, and diagonal specialization show that a coherently stable matrix sends each stable coherent vector to a stable vector or zero. An invertible matrix cannot produce the latter alternative. This argument does not require WMW^* to be invertible outside the symmetric subspace.

The matrix C_i is unitary. On polynomials it acts by

$$f(z_i) \mapsto \left(\frac{s_i + i}{\sqrt{2}} \right)^{\kappa_i} f\left(\frac{s_i - i}{s_i + i} \right), \quad (95)$$

which identifies disk stability with upper-half-plane stability. Consequently coherent stability of e^{-tH} implies ordinary stability preservation by e^{-tCHC^*} . Preservation for a sufficiently small time interval implies preservation for all times by composition. Theorem 3.2 then gives complete preservation.

For the converse, put $D_i = \text{diag}_{0 \leq a \leq \kappa_i}((-1)^{\kappa_i - a})$ and $D = \bigotimes_i D_i$. Symmetric powers of $C_1 C_1^\top$ give $CC^\top = D$, and

$$F_D(s, t) = \prod_i (s_i t_i - 1)^{\kappa_i} \quad (96)$$

is upper-half-plane stable. Indeed $st = 1$ is impossible for two points of the upper half-plane. A complete preserver $N = e^{-tCHC^*} = Ce^{-tH}C^*$ applied in the s variables to this kernel produces the coherent coefficient matrix $ND = Ce^{-tH}C^\top$. This is precisely the separate Cayley transform of both groups of variables in $F_{e^{-tH}}$. It is nonzero because N is invertible. Applying the inverse transforms proves the required disk stability. $\square$

Proof of Theorem 7.1. Put $A = -CHC^*$. By Lemma 7.2, Theorem 3.2 applies to A . We recall the parts of its conclusion used here. The generator belongs to the degree-at-most-two part of the enveloping algebra of the local $\mathfrak{sl}_2$ actions. Its second-order differential coefficients are real and nonpositive on the appropriate real lines or planes. If $A = R + iS$ in the real monomial basis, then its imaginary part has the form

$$S = cI + \sum_i J_{i, \kappa_i}(q_i), \quad J_{i, \kappa_i}(q) = q\partial_i - \frac{\kappa_i}{2}q', \quad q_i \in \mathbb{R}[z_i]_{\leq 2}, \quad q_i \geq 0 \text{ on } \mathbb{R}. \quad (97)$$

Conversely these conditions, together with preservation of the finite degree box, characterize the generators.

The degree-two operator space of Proposition 3.4 decomposes as the direct sum of the scalar space, the local spin spaces, the local traceless symmetric quadrupoles when $\kappa_i \geq 2$, and the spaces spanned by $S_i^a S_j^b$ for $i < j$. Its dimension is

$$1 + 3n + 5\#\{i : \kappa_i \geq 2\} + 9\binom{n}{2}. \quad (98)$$

For clarity, the local assertion follows from the usual decomposition of the square of the three-dimensional spin representation into its scalar, antisymmetric, and traceless symmetric parts: the antisymmetric part is the spin commutator, the scalar part is the Casimir, and the rank-two part is nonzero precisely for $\kappa_i \geq 2$. Distinct sites give independent tensor factors. This also proves the stated directness. Since the Cayley action is a local spin rotation, it preserves this operator space. Thus an arbitrary Hermitian H satisfying condition (2) already has total spin degree at most two, with the stated capacity-one exceptions.

Hermiticity of A and the real diagonal monomial inner product give $R^* = R$ and $S^* = -S$. For one site write

$$J^- = \partial, \quad J^0 = z\partial - \kappa/2, \quad J^+ = z^2\partial - \kappa z.$$

In the weighted inner product, $(J^-)^* = -J^+$ and $(J^0)^* = J^0$. Consequently skew-adjointness of $aJ^- + bJ^0 + cJ^+$, for real a, b, c , forces $b = 0$ and $a = c$. The scalar in (97) vanishes, and hence

$$q_i(s) = \alpha_i(1 + s^2), \quad \alpha_i \geq 0, \quad iJ_{i, \kappa_i}(q_i) = -2\alpha_i S_i^y. \quad (99)$$

The Cayley rotations are

$$CS_i^x C^* = -S_i^z, \quad CS_i^y C^* = S_i^x, \quad CS_i^z C^* = -S_i^y. \quad (100)$$

Thus the original longitudinal field contributes $h_i^z S_i^y$ to A , and (99) gives $h_i^z = -2\alpha_i \leq 0$. Each mixed longitudinal–transverse quadratic term in H gives an imaginary second-order coefficient in A . The direct decomposition above, or the linearly independent principal polynomials at each site, prevents cancellation between distinct such terms. They all vanish.

It remains to compute the real second-order inequalities. The principal coefficients of (S^x, S^y, S^z) in (90) are $((1-s^2)/2, -i(1+s^2)/2, -s)$. Put

$$u(s) = \left(\frac{2s}{1+s^2}, \frac{1-s^2}{1+s^2} \right)^\top;$$

its range is dense in the real unit circle. For the pair term $(S_i^\perp)^\top J_{ij} S_j^\perp + e_{ij} S_i^z S_j^z$, the coefficient q_{ij} of $\partial_i \partial_j$ in A satisfies

$$\frac{4q_{ij}(s, t)}{(1+s^2)(1+t^2)} = e_{ij} - u(s)^\top J_{ij} u(t). \quad (101)$$

Its nonpositivity is therefore equivalent to $e_{ij} \leq -\|J_{ij}\|_{\text{op}}$. Before fixing the local Casimir gauge, write a surviving local quadratic term as $(S_i^\perp)^\top K_i S_i^\perp + c_i (S_i^z)^2$, with K_i real symmetric. For $\kappa_i \geq 2$ its coefficient q_i of ∂_i^2 satisfies

$$\frac{4q_i(s)}{(1+s^2)^2} = c_i - u(s)^\top K_i u(s). \quad (102)$$

Thus $c_i \leq \lambda_{\min}(K_i)$. The Casimir identity rewrites this term as $(S_i^\perp)^\top B_i S_i^\perp$ plus a scalar, where $B_i = K_i - c_i I_2 \succeq 0$. This proves necessity.

Conversely, (92) makes (101) and (102) real and nonpositive and makes the imaginary part satisfy (97). The expression preserves the degree box because it is built from the spin operators. Theorem 3.2 and Lemma 7.2 give coherent stability at every time. This proves (3) $\Rightarrow$ (1); (1) $\Rightarrow$ (2) is immediate. $\square$

Corollary 7.3 (Exact symmetric-qubit realization). *Every Hamiltonian in Theorem 7.1 has a Hermitian qubit extension $\hat{H}$, invariant under permutations within the capacity blocks, whose pair coefficients satisfy the qubit Lee–Yang cone and such that*

$$\hat{H}W = WH, \quad e^{-\beta \hat{H}}W = We^{-\beta H} \quad (\beta \in \mathbb{R}). \quad (103)$$

Proof. For each $B_i \succeq 0$, let $\lambda_i = \lambda_{\max}(B_i)$ and use the exact identity

$$(S_i^\perp)^\top B_i S_i^\perp = (S_i^\perp)^\top (B_i - \lambda_i I_2) S_i^\perp - \lambda_i (S_i^z)^2 + \lambda_i j_i(j_i + 1)I. \quad (104)$$

Replace each spin in this expression and in (91) by $\hat{S}_i^a = \frac{1}{2} \sum_{r=1}^{\kappa_i} \sigma_{ir}^a$. Retain the last term of (104) as the indicated scalar. Interblock pairs have transverse matrix $J_{ij}/4$ and longitudinal coefficient $e_{ij}/4$. Intrablock pairs have transverse matrix $(B_i - \lambda_i I_2)/2$ and longitudinal coefficient $-\lambda_i/2$. If $b_i = \lambda_{\min}(B_i)$, then

$$\|B_i - \lambda_i I_2\|_{\text{op}} = \lambda_i - b_i \leq \lambda_i.$$

All pairs therefore satisfy the qubit cone. Identical-slot symmetric off-diagonal products vanish by Pauli anticommutation; the remaining identical-slot terms are scalars. Each collective spin preserves the symmetric sector and intertwines the physical spin through W_i . This proves both identities in (103). $\square$

The reduction of higher spins to spin one-half and the corresponding anisotropic sufficient Lee–Yang theorems are classical; see, in particular, [35, Theorem 2 and equations (2.38)–(2.44)]. The corollary supplies explicit coefficients for every Hamiltonian in the necessary-and-sufficient cone, including single-ion quadrupoles. It is an invariant-sector identity and does not assert a spectral comparison with the other qubit spin sectors.

APPENDIX A. EXACT ARITHMETIC AND BIT BOUNDS

The algorithms in Section 5 use rational recursions. We supply the denominator and intermediate-magnitude bounds that convert their arithmetic-operation counts into bit-complexity bounds.

A.1. Eigenvalue continuation. Use the notation of Section 5 and the coefficient order p chosen in the proof of Theorem 5.1. Let B_0 bound the bit length of each numerator and denominator in the original local matrices. For $p > 0$, the interpolation nodes $s_\ell = -1 + 2\ell/p$ have $O(\log(p+1))$ bits. Taking local Hermitian parts and forming $B + s_\ell C$ therefore produces Gaussian-rational inputs with bit lengths polynomial in B_0 and $\log(p+1)$.

Consider one Hermitian coefficient computation at these nodes. Let Q_0 be the product of the positive denominators of the real and imaginary parts of its local entries. There are $O_D(N)$ such entries, so $\log Q_0$ is polynomial in the input length. Set

$$K_p = 48(p+1)!, \quad D_j = (Q_0 K_p)^j \quad (0 \leq j \leq p). \quad (105)$$

We recall the properties of the coefficient recursion used in Sections 3–5 of [12]. An order- j creation coefficient $C_j(M)$ is zero unless M is contained in a connected set of at most $j+1$ vertices. Each nonzero term in its recursion contains one local matrix entry, previous creation coefficients of total order $j-1$, and a divisor

$$k! E_0(M), \quad k \leq 4, \quad E_0(M) = 2|M|.$$

The basis creation matrices $\prod_{u \in M} |1\rangle\langle 0|_u$ have entries zero or one; expanding their nested commutators therefore produces signed sums of local perturbation entries. The corresponding energy recursion has the same form without the outer divisor $E_0(M)$. These are the explicit recursions used to implement the published coefficient algorithm, rather than an assumption about arbitrary symbolic perturbation expressions.

For $j \leq p$, every product $k! E_0(M)$ just displayed divides K_p . Induction in j consequently shows that

$$D_j C_j(M) \in \mathbb{Z}[i], \quad D_j e_j(B + s_\ell C) \in \mathbb{Z}[i]. \quad (106)$$

Indeed, the previous coefficients contribute denominator dividing $(Q_0 K_p)^{j-1}$, the new local entry contributes a factor dividing Q_0 , and the remaining divisor divides K_p . All terms at a fixed order may be accumulated over this common denominator. Its bit length is

$$O(p \log Q_0 + p^2 \log(p+1)).$$

Intermediate numerators have polynomial bit length as well. Here a deliberately loose bound suffices. Fully unfold a term of perturbation order at most p . It has at most p local insertions and bounded-arity recursion nodes. Choose its local edges in at most $(ND)^{O(p)}$ ways. Their union has $O(p)$ vertices, and all intermediate

creation sets lie in this union. Choosing these sets at $O(p)$ nodes, together with the tree shape, order partitions, and bounded-depth commutator terms, gives at most

$$\exp(O_D(p^2 + p \log(N+1)))$$

fully unfolded summands. Each summand contains at most p bounded local entries; the positive integer energy and factorial divisors cannot increase its absolute value. Its magnitude is therefore at most $\max(1, 2J)^{O(p)}$ times a fixed exponential factor. After introducing (105), this bounds the logarithmic magnitude of every intermediate numerator by a polynomial in p, N, B_0 . This argument concerns bit sizes only. The algorithm does not unfold these trees: its sharper operation count is the inherited $N \exp(O_D(p))$ count.

Exact interpolation also has polynomial bit cost. For the nodes $s_\ell = -1 + 2\ell/p$, its weights at i can be written

$$L_\ell(i) = \frac{\prod_{\substack{0 \leq r \leq p \\ r \neq \ell}} (pi + p - 2r)}{2^p (-1)^{p-\ell} \ell! (p-\ell)!}. \quad (107)$$

Both numerator and denominator have $O(p \log(p+1))$ bits. The coefficients $e_k(V)$ are obtained by multiplying these weights by the computed node values and adding $p+1$ terms. The rational powers and binomial coefficients in (63), and the powers in (65)–(66), increase bit length only polynomially. Thus a safe total bit bound is

$$N \exp(O_D(p)) \text{poly}(N, B_0, p, L_{\text{in}}),$$

where L_{in} is the total input bit length, including the requested accuracy. This proves the polynomial bit overhead used in Theorem 5.1.

A.2. Connected principal-state tests. We use the subsystem coefficients and notation from the proof of Lemma 5.15. Let Q_0 be a common positive denominator for all local entries of V , let Q_B be the product of the denominators of the entries of the B_i , and put

$$D_p = 2(p+1)!, \quad T = Q_0 D_p.$$

In a subsystem of size at most $p+1$, each nonzero diagonal entry of the reduced inverse S_U is $1/(2r)$, $1 \leq r \leq p+1$. Induction in the vector recursion gives denominators dividing T^{2k} for $v_{U,k}, w_{U,k}$ and T^{2k-1} for the energy coefficient. The denominator of a_k divides $Q_B T^{2k}$. Equation (77) then bounds the denominator of f_k by $k! Q_B^k T^{2k}$. At each order, accumulate the sums over these common denominators; their logarithms are polynomial in p and the total input length.

Intermediate numerators also have polynomial bit length. For a subsystem U , $\|V_U\| \leq m(d+1)J$. On $|x| \leq r_U = [16(1+m(d+1)J)]^{-1}$, the resolvent Neumann series on the circle $|z|=1$ bounds the spectral projection at the vacuum and its difference from the unperturbed projection. In particular, the projection has norm at most $16/15$, its vacuum entry differs from one by at most $1/15$, and both vacuum-normalized eigenvectors have norm less than two. Cauchy's estimate gives $\|v_{U,k}\|, \|w_{U,k}\| \leq 2r_U^{-k}$. As $\|B_U\| \leq 2^m$,

$$|a_k| \leq 4(k+1)2^m r_U^{-k}.$$

For $k \leq p$, choose $A_p = 4(p+1)2^{p+1}$. The finite formula $\log(1+a) = \sum_{\ell=1}^p (-1)^{\ell+1} a^\ell / \ell \pmod{x^{p+1}}$ bounds $|f_k|$ by $(2A_p)^k r_U^{-k}$. Matrix products, vector sums and the partial sums in (77) have the same type of bound, up to

factors $2^{O(p)}$ $\text{poly}(p)$. Subset sums add at most 2^{p+1} terms, and the final connected sum has at most $n \exp(O_d(p))$ terms. Combining these magnitude bounds with the common denominators controls every stored numerator, including before cancellation. This proves the bit complexity.

APPENDIX B. CONNECTED PRESERVERS AND THE OBSTRUCTION TO FLOW GENERATION

The exact infinitesimal cone does not generate every connected static preserver, even after taking limits of products. We prove this on the fixed multiaffine space $V_n = \mathbb{C}[z_1, \dots, z_n]_{\text{ma}} \simeq (\mathbb{C}^2)^{\otimes n}$. Let S_n be the semigroup of invertible complex-linear operators that preserve disk stability. For its ordinary monomial coefficient matrix M , write

$$G_M(z, w) = \sum_{a, b \in \{0, 1\}^n} M_{ab} z^a w^b. \quad (108)$$

For $M \in \text{GL}(V_n)$, the symbol criterion is

$$M \in S_n \iff G_M \text{ is stable on } \mathbb{D}^{2n}. \quad (109)$$

For necessity, fix $w \in \mathbb{D}^n$ and apply M to the stable polynomial $\prod_i (1 + w_i z_i)$. Invertibility excludes the zero output. Conversely, tensor contraction of G_M with a stable input proves preservation, again excluding zero by invertibility. This is the multiaffine disk-symbol convention of the preceding sections. Hurwitz's theorem shows that S_n is closed relative to $\text{GL}(V_n)$: an invertible limit cannot have identically zero symbol. Nonzero scalar multiplication preserves S_n .

Define

$$\begin{aligned} W_n &= \{A : e^{tA} \in S_n \text{ for every } t \geq 0\}, \\ E_n &= \langle e^A : A \in W_n \rangle, \quad T_n = \overline{E_n}^{\text{GL}(V_n)}, \quad H_n = S_n \cap S_n^{-1}. \end{aligned} \quad (110)$$

Here E_n consists of finite products. The closure T_n is a different semigroup. The group H_n is the group of reversible preservers, and H_n^0 denotes its identity component.

Theorem B.1 (Global connectedness and a strict flow obstruction). *For every $n \geq 1$, S_n is path connected. Nevertheless,*

$$E_n \cap H_n = H_n^0, \quad T_n \cap T_n^{-1} = H_n^0. \quad (111)$$

For $n \geq 2$, every nontrivial coordinate permutation π satisfies

$$\pi \in S_n \setminus T_n. \quad (112)$$

Thus a positive-radius matrix neighborhood of π contains no finite product of admissible one-parameter flows, even with unbounded product lengths and generator norms.

Path connectedness. Let $D_r z^a = r^{|a|} z^a$, $0 < r \leq 1$. It is an invertible preserver, and

$$G_{D_r M D_r}(z, w) = G_M(rz, rw). \quad (113)$$

For $M \in S_n$, its constant coefficient M_{00} is nonzero. A path of nonzero scalars normalizes it to one. With this normalization, $D_r M D_r$ converges as $r \downarrow 0$ to the vacuum projector $|0\rangle\langle 0|$.

Consider the convex open set in the complex affine hyperplane $X_{00} = 1$ given by

$$\mathcal{B} = \left\{ X : X_{00} = 1, \sum_{(a,b) \neq (0,0)} |X_{ab}| < 1 \right\}. \quad (114)$$

Every member has a kernel nonvanishing on the closed polydisk, by strict constant-term domination. For small positive r, s , both $D_r M D_r$ and D_s belong to $\mathcal{B}$ and are invertible. Indeed the nonconstant coefficient sum for D_s is $(1+s)^n - 1$.

The invertible part of $\mathcal{B}$ is path connected. To see this, take invertible $X, Y \in \mathcal{B}$ and set $L(\zeta) = (1-\zeta)X + \zeta Y$. Its determinant is a nonzero polynomial in ζ , since it is nonzero at zero. The real segment $L([0, 1])$ is compactly contained in $\mathcal{B}$, so a sufficiently thin complex neighborhood of $[0, 1]$ still maps into $\mathcal{B}$. A path from 0 to 1 in this neighborhood avoids the finitely many determinant zeros. Its image connects X to Y through invertible members of $\mathcal{B}$. Now connect M to $D_r M D_r$ using (113), connect this point to D_s inside $\mathcal{B}$, and connect D_s to I by increasing s to one. Every point is in S_n . The singular vacuum projector is used only as a limit to enter $\mathcal{B}$; it is never a point of the path. $\square$

For reference, there is also an explicit two-variable path. On $\text{span}(z_1, z_2)$ let $P_{\pm}$ be the symmetric and antisymmetric projections, and let U_{θ} act as $P_+ + e^{i\theta} P_-$ there and as the identity on $\text{span}(1, z_1 z_2)$. For $r = 1/4$,

$$\begin{aligned} G_{D_r U_{\theta}} &= 1 + r a_{\theta}(z_1 w_1 + z_2 w_2) + r b_{\theta}(z_1 w_2 + z_2 w_1) + r^2 z_1 z_2 w_1 w_2, \\ a_{\theta} &= (1 + e^{i\theta})/2, \quad b_{\theta} = (1 - e^{i\theta})/2. \end{aligned} \quad (115)$$

Its nonconstant coefficient sum is at most $2\sqrt{2}r + r^2 < 1$, and its determinant is $r^4 e^{i\theta} \neq 0$. The paths $I \rightarrow D_r, D_r U_{\theta}$ for $0 \leq \theta \leq \pi$, and $D_s \pi$ for $r \leq s \leq 1$ connect the identity to the coordinate exchange. These paths are not asserted to have preserving forward increments.

Lemma B.2 (Reversible factors). *If $A, B \in S_n$ and $AB \in H_n$, then $A, B \in H_n$. Moreover every element of H_n^0 is a nonzero scalar times a tensor product of one-site invertible matrices, whereas no nontrivial coordinate permutation has this form.*

Proof. The identities $A^{-1} = B(AB)^{-1}$ and $B^{-1} = (AB)^{-1}A$ prove the first assertion. The group H_n is a closed subgroup of $\text{GL}(V_n)$ and hence a real Lie group. Its Lie algebra is $W_n \cap (-W_n)$. By Corollary 3.11, in half-plane coordinates this algebra is

$$\mathbb{C}I + \bigoplus_i \mathfrak{sl}_2(\mathbb{R})_i. \quad (116)$$

The fixed tensor Cayley transformation preserves products and coordinate permutations. A connected Lie group is generated by the exponentials of its Lie algebra. Exponentiating (116) and taking finite products therefore gives only scalar multiples of local tensor products. A nontrivial coordinate permutation cannot have this form: varying one input factor varies a different output factor, whereas an invertible local tensor product varies only the corresponding output factor. Equivalently, for a two-factor exchange, an identity $Ax \otimes By = cy \otimes x$ with fixed nonzero x and varying independent y would force the fixed vector Ax to be proportional to every y . $\square$

The exact-product assertion in Theorem B.1. If $e^{A_m} \dots e^{A_1} \in H_n$, with every $A_j \in W_n$, Lemma B.2 makes every exponential factor a unit. For $0 \leq s \leq 1$, factor $e^{A_j} = e^{(1-s)A_j} e^{sA_j}$. Both factors preserve stability, so the same lemma gives $e^{sA_j} \in H_n$. Thus each factor belongs to H_n^0 , and so does their product. Conversely H_n^0 is generated by exponentials of its Lie algebra, which is contained in W_n . This proves $E_n \cap H_n = H_n^0$. $\square$

For the closure assertion we use Neeb's classical connected-unit theorem: if G is a connected Lie group and $T \subseteq G$ is a closed submonoid satisfying

$$T = \overline{\langle \exp L(T) \rangle}^G, \quad L(T) = \{A : \exp(tA) \in T \text{ for every } t \geq 0\},$$

then $T \cap T^{-1}$ is connected. This is [24, Definition II.1(3) and Proposition III.2, pp. 171, 178]; see also [34, Section 6.1].

The closure assertion in Theorem B.1. The ambient real Lie group $G = \mathrm{GL}_{2n}(\mathbb{C})$ is connected. By definition T_n is a closed submonoid and $T_n \subseteq S_n$. Every $A \in W_n$ generates a one-parameter semigroup in E_n , while $L(T_n) \subseteq L(S_n) = W_n$. Hence

$$L(T_n) = W_n, \quad T_n = \overline{\langle \exp L(T_n) \rangle}^G.$$

Neeb's theorem therefore makes $T_n \cap T_n^{-1}$ a connected subgroup of H_n , so it is contained in H_n^0 . The reverse inclusion follows because the exponentials of both signs of (116) belong to T_n and generate H_n^0 .

If a nontrivial coordinate permutation π belonged to T_n , its finite order would imply $\pi^{-1} \in T_n$. It would therefore be a unit of T_n , and hence belong to H_n^0 , contrary to Lemma B.2. This proves (112). Relative closure in the general linear group and ordinary matrix-norm closure agree at the invertible point π , proving the neighborhood statement. $\square$

Corollary B.3 (Exact local divisibility). *Call $M \in S_n$ exactly locally divisible if, for every identity neighborhood U in $\mathrm{GL}(V_n)$, it is a finite product of members of $S_n \cap U$. The exactly locally divisible units of S_n are precisely H_n^0 . In addition, if a continuous path $M : [0, 1] \rightarrow S_n$ starts at I and satisfies $M(t)M(s)^{-1} \in S_n$ for $s \leq t$, then a unit endpoint must belong to H_n^0 .*

Proof. The identity component of a Lie group is open, so choose U with $H_n \cap U \subset H_n^0$. Any exact factorization of a unit into $S_n \cap U$ consists entirely of units by Lemma B.2, hence has endpoint in H_n^0 . Conversely, subdividing a finite exponential representation of a member of H_n^0 puts all its factors in any prescribed identity neighborhood. For the path assertion, write $M(1) = [M(1)M(t)^{-1}]M(t)$. A unit endpoint makes $M(t)$ a unit for every t , so the continuous path stays in H_n and ends in H_n^0 . $\square$

Remark B.4. The identity for the closure in (111) is $T_n \cap T_n^{-1} = H_n^0$. This equality concerns elements whose inverses also belong to T_n ; finite order supplies this inverse inclusion for permutations. The result concerns flows of the infinitesimal cone. Static apolar maps may lie on other branches: for $F(s, t) = s - t$, $B_F = \pi - I$, so $I + B_F = \pi$, whereas

$$e^{tB_F} = \frac{1 + e^{-2t}}{2}I + \frac{1 - e^{-2t}}{2}\pi.$$

Allowing such a static permutation as an additional primitive changes the factorization question.

APPENDIX C. AN EXACT COUNTEREXAMPLE TO A STRONGER GROUND-STATE RADIUS

The qualitative strict radius in Theorem 6.6 cannot in general be replaced by the proposed explicit radius for deformed EPR ground states. We address the stronger clause of Conjecture 1 in [41, Section 5.4], with its stated unweighted-graph hypothesis.

The computations in [41, Numerical Study 1, equation (70)] test the signed diagonal specializations $z_i = (-1)^{y_i} z$, $y \in \{0, 1\}^n$, of the ground-state polynomial. The zero certified below lies outside every such slice: its two fixed coordinate values are neither equal nor opposite. The certificate therefore concerns points outside the family tested in that numerical study.

Theorem C.1. *Let G have vertices $\{0, 1, 2, 3, 4, 5\}$ and edges $\{01, 02, 03, 14, 15\}$.*

Put

$$H = - \sum_{ij \in E(G)} (|00\rangle + \tfrac{1}{2}|11\rangle)(\langle 00| + \tfrac{1}{2}\langle 11|)_{ij}.$$

The ground state ψ is unique up to scalar. Its multiaffine polynomial $f_\psi(z) = \sum_b \psi_b z^b$ has a zero satisfying

$$|z_i| < \frac{499}{500}\sqrt{2} \quad (0 \leq i \leq 5).$$

Thus the proposed radius $s^{-1/2}$ fails at $s = 1/2$ on an unweighted six-vertex tree.

Proof. We give rational spectral and zero certificates. All inequalities below are verified by the supplementary finite procedure described at the end of this appendix, using only integer arithmetic and exact fractions.

Write $K = -H$. Index the computational basis by integers $b \in \{0, \dots, 63\}$, with bit b_i at vertex i . For each edge ij and each input with $b_i = b_j = a$, changing both bits to $c \in \{0, 1\}$ contributes $2^{-(a+c)}$ to the corresponding entry of K . All other entries from that edge are zero. This specifies the symmetric rational matrix K completely.

Consider $M = 37I - 8K$. The charge

$$Q(b) = b_0 + b_4 + b_5 - b_1 - b_2 - b_3$$

is conserved. In charge order $-3, -2, \dots, 3$, the block dimensions are

$$1, 6, 15, 20, 15, 6, 1.$$

Order each block by increasing integer b . Successive rational symmetric elimination has pivot signs as follows:

Q	-3	-2	-1	0	1	2	3
negative pivots	0	0	0	1	0	0	0
positive pivots	1	6	15	19	15	6	1.

No pivot is zero; the negative pivot is the first in the charge-zero block. For clarity, the recurrence at step i is

$$p_i = M_{ii}^{(i)}, \quad M_{jk}^{(i+1)} = M_{jk}^{(i)} - M_{ji}^{(i)} M_{ik}^{(i)} / p_i \quad (j, k > i).$$

It proves the stated inertia by exact rational comparison, rather than taking the pivot signs as external data. Sylvester's law of inertia shows that K has exactly

one eigenvalue above $37/8$. Since $K_{00} = 5$, its largest eigenvalue is at least 5 and is simple. Every other eigenvalue is below $37/8$.

Define a rational vector v as follows. For a bit string use the key

$$(a, b, i, j) = (b_0, b_1, b_2 + b_3, b_4 + b_5).$$

The amplitude for each individual bit string is the following numerator divided by 10^{15} ; a missing key has amplitude zero.

Key	Numerator
(0, 0, 0, 0)	1000000000000000
(0, 0, 1, 1)	8942179087130
(0, 0, 2, 2)	136748346620
(0, 1, 0, 1)	171232824560490
(0, 1, 1, 2)	2452435031309
(1, 0, 1, 0)	171232824560490
(1, 0, 2, 1)	2452435031309
(1, 1, 0, 0)	97255575326164
(1, 1, 1, 1)	37858301388951
(1, 1, 2, 2)	1200949534669

These are computational-basis amplitudes, not normalized orbit-basis coordinates. Put

$$\hat{\lambda} = \frac{539109343678406}{10^{14}}.$$

Direct rational summation gives

$$\|(K - \hat{\lambda}I)v\|_2^2 < 10^{-24}, \quad \hat{\lambda} - 37/8 > 1/2.$$

Let v_g be the orthogonal projection onto the largest eigenspace of K . Expanding in an orthonormal eigenbasis and using the preceding separation gives

$$\|v - v_g\|_2 \leq 2\|(K - \hat{\lambda}I)v\|_2 < 2 \cdot 10^{-12}.$$

Since $v_{000000} = 1$, v_g is nonzero, hence is a scalar multiple of the true ground state. Both v and v_g have charge zero; in particular they have even parity.

For an even-parity vector w put

$$\tilde{f}_w(x) = f_w(\sqrt{2}x) = \sum_b 2^{|b|/2} w_b x^b.$$

The coefficients of $\tilde{f}_v$ are rational. Set

$$c = \frac{175105 + 981180i}{10^6}, \quad \ell = \frac{-379834 + 921467i}{10^6}.$$

Exact squaring gives $|c|, |\ell| < 499/500$. Fix $x_0 = x_1 = c$ and $x_2 = x_3 = x_4 = \ell$. Multiaffinity yields

$$\tilde{f}_v(c, c, \ell, \ell, \ell, x_5) = A_v + B_v x_5, \quad |A_v|^2 < (85/1000)^2, \quad |B_v|^2 > (852/10000)^2.$$

The displayed inequalities are finite sums over the amplitude table. Each coefficient A or B contains at most 32 terms. A fixed-variable monomial has modulus at most one, and its even tilt factor is at most eight. Therefore

$$|A_{v_g} - A_v|, |B_{v_g} - B_v| < 32 \cdot 8 \cdot 2 \cdot 10^{-12} < 10^{-8}.$$

In particular $B_{v_g} \neq 0$, and

$$x_5 = -A_{v_g}/B_{v_g}, \quad |x_5| < \frac{85/1000 + 10^{-8}}{852/10000 - 10^{-8}} < 499/500.$$

The last inequality is rational. This gives a zero of $\tilde{f}_{v_g}$ with all coordinates of modulus below $499/500$. Rescaling by $\sqrt{2}$ proves the theorem. $\square$

The certificate locates a zero of the exact ground-state polynomial. It does not treat a rounded eigenvector as an exact eigenvector. The existence of some radius greater than one, proved above by positive-field strictification, is compatible with this example.

Reproducible rational verification. The supplementary Python procedure `epr_six_vertex_certificate.py` reproduces the inertia, residual, and zero inequalities used above from the displayed rational data. It uses only standard-library integer and fraction arithmetic; Gaussian-rational operations are written explicitly. The supplementary procedure `epr_independent_certificate.py` provides a second exact verification. Its floating-point diagnostics are separate from the rational assertions used in the proof. Both procedures are supplied with execution instructions and are available at the fixed repository version identified below.

DATA AND CODE AVAILABILITY

All rational certificate data are included in Appendix C. The two verification procedures are supplied as supplementary material and available in the `certificates` directory of a fixed GitHub version (certificate snapshot 8552777). The procedures use Python 3 and its standard library.

DISCLOSURE OF ARTIFICIAL INTELLIGENCE USE

Most of the mathematical content, proofs, computational certificates, and text in this article were generated with substantial assistance from OpenAI GPT-6 Pro in ChatGPT chat mode and GPT-6 Astra in Codex, both using the ultra reasoning setting. Claude Fable 5.1 in chat mode also provided several rounds of revision suggestions. The author organized and directed the research through repeated interactions, including questions, choices of research directions, requests for extensions and critical review, and selection and organization of the material. AI assistance was also used for literature searches, proof audits, exact computational checks, and manuscript revision. These checks do not constitute complete proof-assistant verification, which is not claimed. The author takes full responsibility for the accuracy, originality, attribution, and conclusions of the article.

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